

**GigaDevice Semiconductor Inc.**

**GD32L233xx**

**Arm® Cortex®-M23 32-bit MCU**

**Datasheet**

Revision 1.5

(Jul. 2023)

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## 1. General description

The GD32L233xx device belongs to the value line of GD32 MCU family. It is a new 32-bit general-purpose microcontroller based on the ARM® Cortex®-M23 core. The Cortex-M23 processor is an energy-efficient processor with a very low gate count. It is intended to be used for microcontroller and deeply embedded applications that require an area-optimized processor. The processor delivers high energy efficiency through a small but powerful instruction set and extensively optimized design, providing high-end processing hardware including a single-cycle multiplier and a 17-cycle divider.

The GD32L233xx device incorporates the ARM® Cortex®-M23 32-bit processor core operating at up to 64 MHz frequency with Flash accesses 0~3 wait states to obtain maximum efficiency. It provides up to 256 KB embedded Flash memory and up to 32 KB SRAM memory. An extensive range of enhanced I/Os and peripherals connected to two APB buses. The devices offer one 12-bit ADC, one DAC, two comparators, up to four general 16-bit timers, two basic timers, and a 32-bit low power timer, as well as standard and advanced communication interfaces: up to two SPIs, three I2Cs, two USARTs, two UARTs, an I2S, and an LPUART. Additional peripherals as segment LCD controller (SLCD), cryptographic acceleration unit (CAU) are included.

The device operates from a 1.71 to 3.63 V power supply and available in -40 to +85 °C temperature range for grade 6 devices, and -40°C to +105°C temperature range for grade 7 devices. Several power saving modes provide the flexibility for maximum optimization between wakeup latency and power consumption, an especially important consideration in low power applications.

The above features make the GD32L233xx devices suitable for a wide range of applications, especially in areas such as industrial control, motor drives, user interface, power monitor and alarm systems, consumer and handheld equipment, gaming and GPS, E-bike and so on.



## 2. Device overview

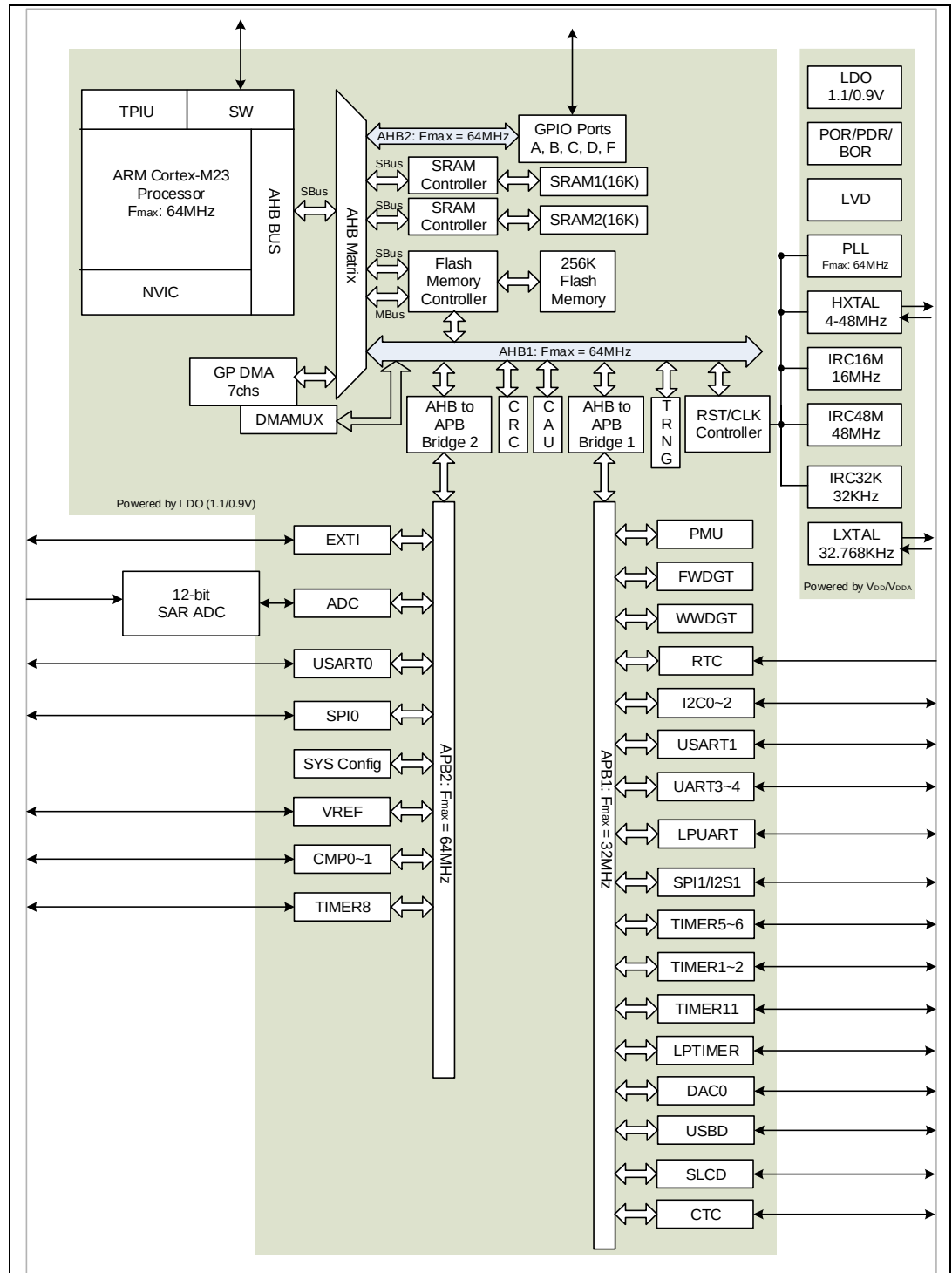
### 2.1. Device information

Table 2-1. GD32L233xx devices features and peripheral list

| Part Number  |                         | GD32L233xx                      |                                 |                                 |                                 |                                 |                                   |                                   |                                 |                                   |                                   |                                   |
|--------------|-------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|-----------------------------------|-----------------------------------|---------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
|              |                         | K8Q6                            | KBQ6                            | K8T6                            | KBT6                            | C8T6                            | CBT6                              | CCT6                              | R8T6                            | RBT6                              | RCT6                              | RCT7                              |
| FLASH (KB)   |                         | 64                              | 128                             | 64                              | 128                             | 64                              | 128                               | 256                               | 64                              | 128                               | 256                               | 256                               |
| SRAM (KB)    |                         | 16                              | 24                              | 16                              | 24                              | 16                              | 24                                | 32                                | 16                              | 24                                | 32                                | 32                                |
| Timers       | General timer(16-bit)   | 3<br><small>(1, 2, 8)</small>   | 3<br><small>(1, 2, 8)</small>   | 3<br><small>(1, 2, 8)</small>   | 3<br><small>(1, 2, 8)</small>   | 3<br><small>(1, 2, 8)</small>   | 4<br><small>(1, 2, 8, 11)</small> | 4<br><small>(1, 2, 8, 11)</small> | 3<br><small>(1, 2, 8)</small>   | 4<br><small>(1, 2, 8, 11)</small> | 4<br><small>(1, 2, 8, 11)</small> | 4<br><small>(1, 2, 8, 11)</small> |
|              | Low power timer(32-bit) | 1                               | 1                               | 1                               | 1                               | 1                               | 1                                 | 1                                 | 1                               | 1                                 | 1                                 | 1                                 |
|              | SysTick                 | 1                               | 1                               | 1                               | 1                               | 1                               | 1                                 | 1                                 | 1                               | 1                                 | 1                                 | 1                                 |
|              | Basic timer(16-bit)     | 2<br><small>(5, 6)</small>      | 2<br><small>(5, 6)</small>      | 2<br><small>(5, 6)</small>      | 2<br><small>(5, 6)</small>      | 2<br><small>(5, 6)</small>      | 2<br><small>(5, 6)</small>        | 2<br><small>(5, 6)</small>        | 2<br><small>(5, 6)</small>      | 2<br><small>(5, 6)</small>        | 2<br><small>(5, 6)</small>        | 2<br><small>(5, 6)</small>        |
|              | Watchdog                | 2                               | 2                               | 2                               | 2                               | 2                               | 2                                 | 2                                 | 2                               | 2                                 | 2                                 | 2                                 |
|              | RTC                     | 1                               | 1                               | 1                               | 1                               | 1                               | 1                                 | 1                                 | 1                               | 1                                 | 1                                 | 1                                 |
| Connectivity | UART                    | 1<br><small>(3)</small>         | 1<br><small>(3)</small>         | 1<br><small>(3)</small>         | 1<br><small>(3)</small>         | 1<br><small>(3)</small>         | 2<br><small>(3, 4)</small>        | 2<br><small>(3, 4)</small>        | 1<br><small>(3)</small>         | 2<br><small>(3, 4)</small>        | 2<br><small>(3, 4)</small>        | 2<br><small>(3, 4)</small>        |
|              | USART                   | 2<br><small>(0, 1)</small>      | 2<br><small>(0, 1)</small>      | 2<br><small>(0, 1)</small>      | 2<br><small>(0, 1)</small>      | 2<br><small>(0, 1)</small>      | 2<br><small>(0, 1)</small>        | 2<br><small>(0, 1)</small>        | 2<br><small>(0, 1)</small>      | 2<br><small>(0, 1)</small>        | 2<br><small>(0, 1)</small>        | 2<br><small>(0, 1)</small>        |
|              | LPUART                  | 1                               | 1                               | 1                               | 1                               | 1                               | 1                                 | 1                                 | 1                               | 1                                 | 1                                 | 1                                 |
|              | I2C                     | 2<br><small>(0-1)</small>       | 2<br><small>(0-1)</small>       | 2<br><small>(0-1)</small>       | 2<br><small>(0-1)</small>       | 2<br><small>(0-1)</small>       | 2<br><small>(0-1)</small>         | 2<br><small>(0-1)</small>         | 3<br><small>(0-2)</small>       | 3<br><small>(0-2)</small>         | 3<br><small>(0-2)</small>         | 3<br><small>(0-2)</small>         |
|              | SPI/I2S                 | 2/1<br><small>(0-1)/(1)</small> | 2/1<br><small>(0-1)/(1)</small> | 2/1<br><small>(0-1)/(1)</small> | 2/1<br><small>(0-1)/(1)</small> | 2/1<br><small>(0-1)/(1)</small> | 2/1<br><small>(0-1)/(1)</small>   | 2/1<br><small>(0-1)/(1)</small>   | 2/1<br><small>(0-1)/(1)</small> | 2/1<br><small>(0-1)/(1)</small>   | 2/1<br><small>(0-1)/(1)</small>   | 2/1<br><small>(0-1)/(1)</small>   |
|              | USBD                    | 1                               | 1                               | 1                               | 1                               | 1                               | 1                                 | 1                                 | 1                               | 1                                 | 1                                 | 1                                 |
| GPIO         |                         | 29                              | 29                              | 27                              | 27                              | 43                              | 43                                | 43                                | 59                              | 59                                | 59                                | 59                                |
| ADC          | Units                   | 1                               | 1                               | 1                               | 1                               | 1                               | 1                                 | 1                                 | 1                               | 1                                 | 1                                 | 1                                 |
|              | Channels (External)     | 10                              | 10                              | 10                              | 10                              | 10                              | 10                                | 10                                | 16                              | 16                                | 16                                | 16                                |
|              | Channels (Internal)     | 4                               | 4                               | 4                               | 4                               | 4                               | 4                                 | 4                                 | 4                               | 4                                 | 4                                 | 4                                 |
| DAC          |                         | 1                               | 1                               | 1                               | 1                               | 1                               | 1                                 | 1                                 | 1                               | 1                                 | 1                                 | 1                                 |
| CMP          |                         | 2                               | 2                               | 2                               | 2                               | 2                               | 2                                 | 2                                 | 2                               | 2                                 | 2                                 | 2                                 |
| SLCD         |                         | 0                               | 0                               | 0                               | 0                               | 0                               | 0                                 | 0                                 | 1                               | 1                                 | 1                                 | 1                                 |
| Package      |                         | QFN32                           |                                 | LQFP32                          |                                 | LQFP48                          |                                   |                                   | LQFP64                          |                                   |                                   |                                   |

## 2.2. Block diagram

Figure 2-1. GD32L233xx block diagram



## 2.3. Pinouts and pin assignment

Figure 2-2. GD32L233Rx LQFP64 pinouts

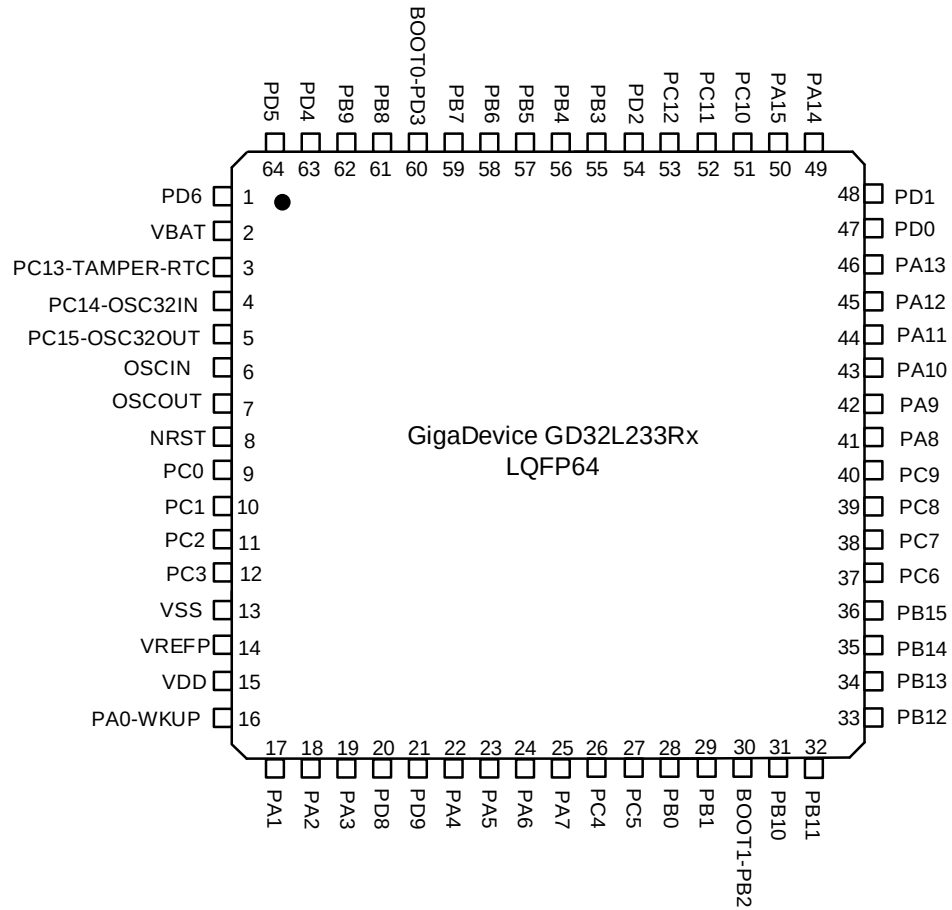


Figure 2-3. GD32L233Cx LQFP48 pinouts

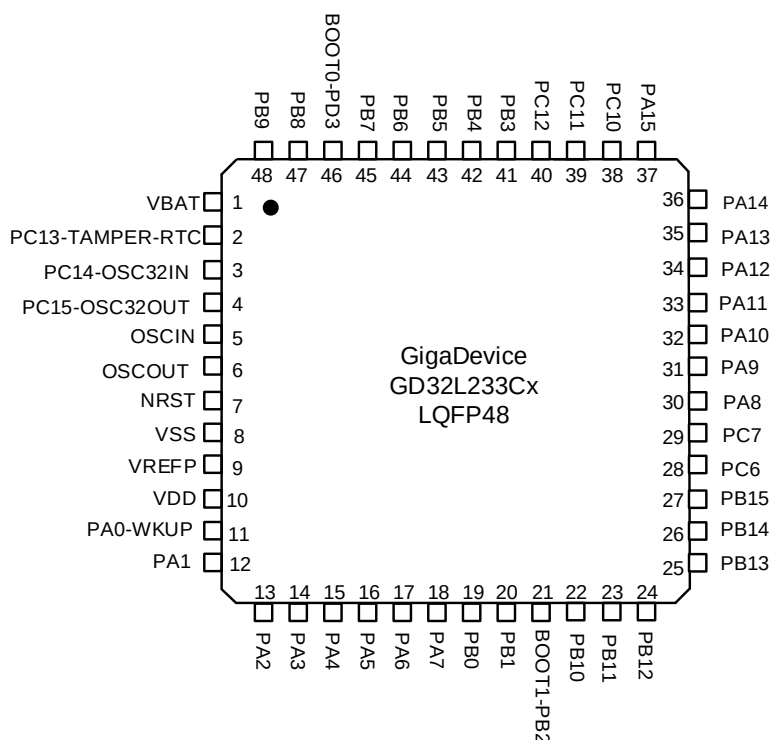


Figure 2-4. GD32L233Kx LQFP32 pinouts

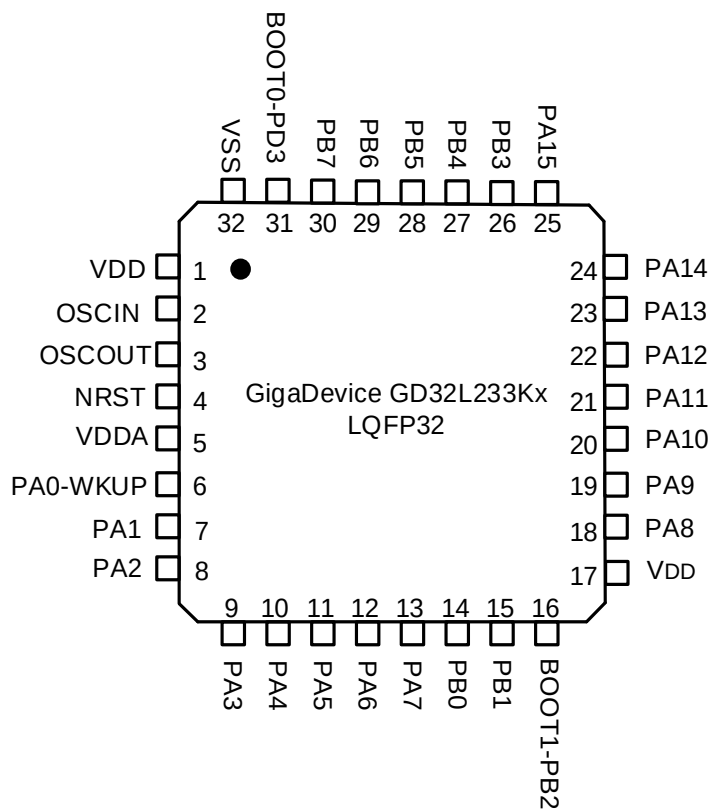
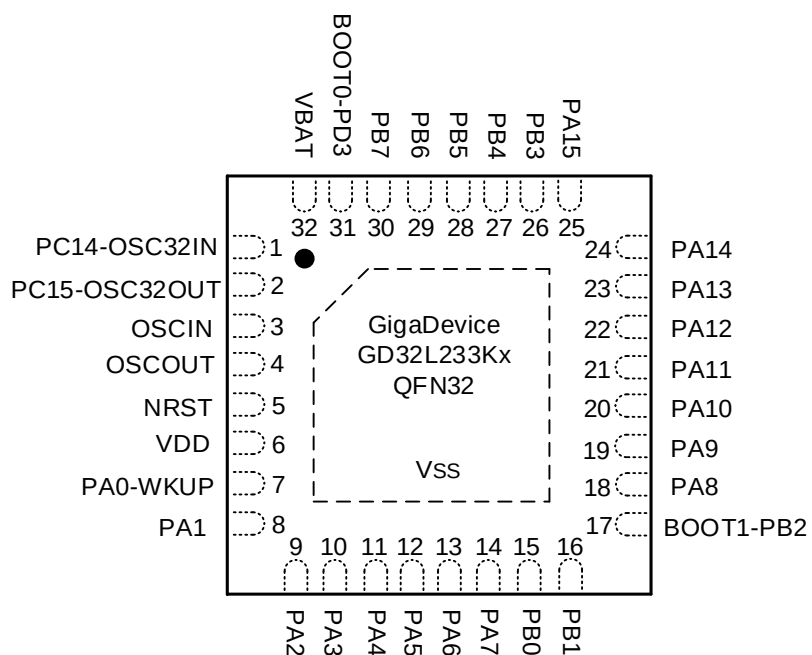


Figure 2-5. GD32L233Kx QFN32 pinouts



## 2.4. Memory map

Table 2-2. GD32L233xx memory map

| Pre-defined Regions | Bus  | ADDRESS                   | Peripherals                      |
|---------------------|------|---------------------------|----------------------------------|
|                     |      | 0xE000 0000 - 0xE00F FFFF | Cortex®-M23 internal peripherals |
| External Device     |      | 0xA000 0000 - 0xDFFF FFFF | Reserved                         |
| External RAM        |      | 0x60000000 - 0x9FFFFFFF   | Reserved                         |
| Peripherals         | AHB1 | 0x5006 1000 - 0x5FFF FFFF | Reserved                         |
|                     |      | 0x5006 0C00 - 0x5006 0FFF | Reserved                         |
|                     |      | 0x5006 0800 - 0x5006 0BFF | TRNG                             |
|                     |      | 0x5006 0400 - 0x5006 07FF | Reserved                         |
|                     |      | 0x5006 0000 - 0x5006 03FF | CAU                              |
|                     |      | 0x5005 0400 - 0x5005 FFFF | Reserved                         |
|                     |      | 0x5005 0000 - 0x5005 03FF | Reserved                         |
|                     |      | 0x5004 0000 - 0x5004 FFFF | Reserved                         |
|                     |      | 0x5000 0000 - 0x5003 FFFF | Reserved                         |
|                     | AHB2 | 0x4800 1800 - 0x4FFF FFFF | Reserved                         |
|                     |      | 0x4800 1400 - 0x4800 17FF | GPIOF                            |
|                     |      | 0x4800 1000 - 0x4800 13FF | Reserved                         |
|                     |      | 0x4800 0C00 - 0x4800 0FFF | GPIOD                            |
|                     |      | 0x4800 0800 - 0x4800 0BFF | GPIOC                            |
|                     |      | 0x4800 0400 - 0x4800 07FF | GPIOB                            |
|                     |      | 0x4800 0000 - 0x4800 03FF | GPIOA                            |
|                     | AHB1 | 0x4002 4400 - 0x47FF FFFF | Reserved                         |
|                     |      | 0x4002 4000 - 0x4002 43FF | Reserved                         |
|                     |      | 0x4002 3400 - 0x4002 3FFF | Reserved                         |
|                     |      | 0x4002 3000 - 0x4002 33FF | CRC                              |
|                     |      | 0x4002 2400 - 0x4002 2FFF | Reserved                         |
|                     |      | 0x4002 2000 - 0x4002 23FF | FMC                              |
|                     |      | 0x4002 1400 - 0x4002 1FFF | Reserved                         |
|                     |      | 0x4002 1000 - 0x4002 13FF | RCU                              |
|                     |      | 0x4002 0C00 - 0x4002 0FFF | Reserved                         |
|                     |      | 0x4002 0800 - 0x4002 0BFF | DMAMUX                           |
|                     |      | 0x4002 0400 - 0x4002 07FF | Reserved                         |
|                     |      | 0x4002 0000 - 0x4002 03FF | DMA                              |
|                     | APB2 | 0x4001 8000 - 0x4001 FFFF | Reserved                         |
|                     |      | 0x4001 7C00 - 0x4001 7FFF | CMP                              |
|                     |      | 0x4001 5C00 - 0x4001 7BFF | Reserved                         |
|                     |      | 0x4001 5800 - 0x4001 5BFF | DBG                              |
|                     |      | 0x4001 5000 - 0x4001 57FF | Reserved                         |
|                     |      | 0x4001 4C00 - 0x4001 4FFF | TIMER8                           |

| Pre-defined Regions | Bus  | ADDRESS                   | Peripherals          |
|---------------------|------|---------------------------|----------------------|
|                     |      | 0x4001 3C00 - 0x4001 4BFF | Reserved             |
|                     |      | 0x4001 3800 - 0x4001 3BFF | USART0               |
|                     |      | 0x4001 3400 - 0x4001 37FF | Reserved             |
|                     |      | 0x4001 3000 - 0x4001 33FF | SPI0                 |
|                     |      | 0x4001 2C00 - 0x4001 2FFF | Reserved             |
|                     |      | 0x4001 2800 - 0x4001 2BFF | Reserved             |
|                     |      | 0x4001 2400 - 0x4001 27FF | ADC                  |
|                     |      | 0x4001 0800 - 0x4001 23FF | Reserved             |
|                     |      | 0x4001 0400 - 0x4001 07FF | EXTI                 |
|                     |      | 0x4001 0000 - 0x4001 03FF | SYSCFG + VREF        |
|                     | APB1 | 0x4000 CC00 - 0x4000 FFFF | Reserved             |
|                     |      | 0x4000 C800 - 0x4000 CBFF | CTC                  |
|                     |      | 0x4000 C400 - 0x4000 C7FF | Reserved             |
|                     |      | 0x4000 C000 - 0x4000 C3FF | I2C2                 |
|                     |      | 0x4000 9800 - 0x4000 BFFF | Reserved             |
|                     |      | 0x4000 9400 - 0x4000 97FF | LPTIMER              |
|                     |      | 0x4000 8400 - 0x4000 93FF | Reserved             |
|                     |      | 0x4000 8000 - 0x4000 83FF | LPUART               |
|                     |      | 0x4000 7C00 - 0x4000 7FFF | Reserved             |
|                     |      | 0x4000 7800 - 0x4000 7BFF | Reserved             |
|                     |      | 0x4000 7400 - 0x4000 77FF | DAC0                 |
|                     |      | 0x4000 7000 - 0x4000 73FF | PMU                  |
|                     |      | 0x4000 6400 - 0x4000 6FFF | Reserved             |
|                     |      | 0x4000 6000 - 0x4000 63FF | USBD RAM (512 bytes) |
|                     |      | 0x4000 5C00 - 0x4000 5FFF | USBD                 |
|                     |      | 0x4000 5800 - 0x4000 5BFF | I2C1                 |
|                     |      | 0x4000 5400 - 0x4000 57FF | I2C0                 |
|                     |      | 0x4000 5000 - 0x4000 53FF | UART4                |
|                     |      | 0x4000 4C00 - 0x4000 4FFF | UART3                |
|                     |      | 0x4000 4800 - 0x4000 4BFF | Reserved             |
|                     |      | 0x4000 4400 - 0x4000 47FF | USART1               |
|                     |      | 0x4000 4000 - 0x4000 43FF | Reserved             |
|                     |      | 0x4000 3C00 - 0x4000 3FFF | Reserved             |
|                     |      | 0x4000 3800 - 0x4000 3BFF | SPI1/I2S1            |
|                     |      | 0x4000 3400 - 0x4000 37FF | Reserved             |
|                     |      | 0x4000 3000 - 0x4000 33FF | FWDGT                |
|                     |      | 0x4000 2C00 - 0x4000 2FFF | WWDGT                |
|                     |      | 0x4000 2800 - 0x4000 2BFF | RTC                  |
|                     |      | 0x4000 2400 - 0x4000 27FF | SLCD                 |
|                     |      | 0x4000 2000 - 0x4000 23FF | Reserved             |

| Pre-defined Regions | Bus | ADDRESS                   | Peripherals                       |
|---------------------|-----|---------------------------|-----------------------------------|
|                     |     | 0x4000 1C00 - 0x4000 1FFF | Reserved                          |
|                     |     | 0x4000 1800 - 0x4000 1BFF | TIMER11                           |
|                     |     | 0x4000 1400 - 0x4000 17FF | TIMER6                            |
|                     |     | 0x4000 1000 - 0x4000 13FF | TIMER5                            |
|                     |     | 0x4000 0800 - 0x4000 0FFF | Reserved                          |
|                     |     | 0x4000 0400 - 0x4000 07FF | TIMER2                            |
|                     |     | 0x4000 0000 - 0x4000 03FF | TIMER1                            |
|                     |     | 0x4000 0000 - 0x4000 03FF | Reserved                          |
| SRAM                |     | 0x2000 8000 - 0x3FFF FFFF | Reserved                          |
|                     |     | 0x2000 5000 - 0x2000 7FFF | SRAM1(16KB)                       |
|                     |     | 0x2000 4000 - 0x2000 4FFF |                                   |
|                     |     | 0x2000 2000 - 0x2000 3FFF | SRAM0(16KB)                       |
|                     |     | 0x2000 1000 - 0x2000 1FFF |                                   |
|                     |     | 0x2000 0000 - 0x2000 0FFF |                                   |
| Code                |     | 0x1FFF F810 - 0x1FFF FFFF | Reserved                          |
|                     |     | 0x1FFF F800 - 0x1FFF F80F | Option bytes(16B)                 |
|                     |     | 0x1FFF D000- 0x1FFF F7FF  | System memory(10KB)               |
|                     |     | 0x1FFF 7200 - 0x1FFF CFFF | Reserved                          |
|                     |     | 0x1FFF 7000 - 0x1FFF 71FF | OTP(512B)                         |
|                     |     | 0x1000 0000 - 0x1FFF 6FFF | Reserved                          |
|                     |     | 0x0804 0000 - 0x0FFF FFFF | Reserved                          |
|                     |     | 0x0802 0000 - 0x0803 FFFF | Main Flash memory(256KB)          |
|                     |     | 0x0801 0000 - 0x0801 FFFF |                                   |
|                     |     | 0x0800 0000 - 0x0800 FFFF |                                   |
|                     |     | 0x0001 0000 - 0x07FF FFFF | Reserved                          |
|                     |     | 0x0000 0000 - 0x0000 FFFF | Aliased to Flash or system memory |

IRC32K: Internal 32K RC oscillator

## 2.6. Pin definitions

### 2.6.1. GD32L233Rx LQFP64 pin definitions

Table 2-3. GD32L233Rx LQFP64 pin definitions

| GD32L233Rx LQFP64 |      |                         |                          |                                                                                                      |
|-------------------|------|-------------------------|--------------------------|------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                |
| PD6               | 1    | I/O                     |                          | Default: PD6<br>Alternate: USART1_RX, EVENTOUT, SPI0_MOSI, LPTIMER_IN1<br>Additional: VSLCD          |
| VBAT              | 2    | P                       |                          | Default: VBAT                                                                                        |
| PC13-TAMPER-RTC   | 3    | I/O                     |                          | Default: PC13<br>Alternate: EVENTOUT<br>Additional: RTC_TAMP0, RTC_OUT, RTC_TS, WKUP1                |
| PC14-OSC32IN      | 4    | I/O                     |                          | Default: PC14<br>Alternate: EVENTOUT<br>Additional: OSC32IN                                          |
| PC15-OSC32OUT     | 5    | I/O                     |                          | Default: PC15<br>Alternate: EVENTOUT<br>Additional: OSC32OUT                                         |
| OSCIIN            | 6    | I/O                     |                          | Default: OSCIIN<br>Alternate: EVENTOUT, SPI1_NSS, I2S1_WS<br>Additional: PF0                         |
| OSCOOUT           | 7    | I/O                     |                          | Default: OSCOUT<br>Alternate: EVENTOUT, SPI1_SCK, I2S1_CK<br>Additional: PF1                         |
| NRST              | 8    | I/O                     |                          | Default: NRST                                                                                        |
| PC0               | 9    | I/O                     | 5VT                      | Default: PC0<br>Alternate: SEG18, I2C2_SCL, LPUART_RX, LPTIMER_IN0, EVENTOUT<br>Additional: ADC_IN10 |
| PC1               | 10   | I/O                     | 5VT                      | Default: PC1<br>Alternate: SEG19, I2C2_SDA, LPUART_TX, LPTIMER_OUT, EVENTOUT<br>Additional: ADC_IN11 |
| PC2               | 11   | I/O                     | 5VT                      | Default: PC2<br>Alternate: SPI1_MISO, I2S1_MCK, SEG20, EVENTOUT, LPTIMER_IN1<br>Additional: ADC_IN12 |
| PC3               | 12   | I/O                     | 5VT                      | Default: PC3<br>Alternate: SPI1_MOSI, I2S1_SD, SEG21, LPTIMER_ETIO, EVENTOUT<br>Additional: ADC_IN13 |

| GD32L233Rx LQFP64 |      |                         |                          |                                                                                                                                                                |
|-------------------|------|-------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                                                          |
| VSS               | 13   | P                       |                          | Default: VSS                                                                                                                                                   |
| VREFP             | 14   | P                       |                          | Default: VREFP                                                                                                                                                 |
| VDD               | 15   | P                       |                          | Default: VDD                                                                                                                                                   |
| PA0-WKUP          | 16   | I/O                     | 5VT                      | Default: PA0<br>Alternate: USART1_CTS, TIMER1_CH0_ETI, CMP0_OUT, EVENTOUT, UART3_TX<br>Additional: WKUP0, ADC_IN0, RTC_TAMP1, CMP0_IM4                         |
| PA1               | 17   | I/O                     | 5VT                      | Default: PA1<br>Alternate: USART1_RTS/USART1_DE, TIMER1_CH1, I2C0_SMBA, SPI0_SCK, SEG0, EVENTOUT, UART3_RX<br>Additional: ADC_IN1, CMP0_IP                     |
| PA2               | 18   | I/O                     | 5VT                      | Default: PA2<br>Alternate: USART1_TX, TIMER8_CH0, TIMER1_CH2, SPI0_IO2, CMP1_OUT, LPUART_TX, SEG1, EVENTOUT<br>Additional: ADC_IN2, CMP1_IM4, RTC_TAMP2, WKUP2 |
| PA3               | 19   | I/O                     | 5VT                      | Default: PA3<br>Alternate: USART1_RX, TIMER8_CH1, TIMER1_CH3, SPI0_IO3, LPUART_RX, SEG2, EVENTOUT<br>Additional: ADC_IN3, CMP1_IP0                             |
| PD8               | 20   | I/O                     | 5VT                      | Default: PD8<br>Alternate: LPTIMER_ETI0, LPUART_TX, EVENTOUT, SEG30                                                                                            |
| PD9               | 21   | I/O                     | 5VT                      | Default: PD9<br>Alternate: LPTIMER_IN0, LPUART_RX, EVENTOUT, SEG31                                                                                             |
| PA4               | 22   | I/O                     |                          | Default: PA4<br>Alternate: SPI0_NSS, USART1_CK, SPI1_NSS, I2S1_WS, LPTIMER_OUT, EVENTOUT<br>Additional: ADC_IN4, DAC_OUT                                       |
| PA5               | 23   | I/O                     | 5VT                      | Default: PA5<br>Alternate: SPI0_SCK, TIMER1_CH0_ETI, LPTIMER_ETI0, EVENTOUT<br>Additional: ADC_IN5                                                             |
| PA6               | 24   | I/O                     | 5VT                      | Default: PA6<br>Alternate: SPI0_MISO, TIMER2_CH0, LPTIMER_IN0, CMP0_OUT, LPUART_CTS, SEG3, EVENTOUT<br>Additional: ADC_IN6                                     |
| PA7               | 25   | I/O                     | 5VT                      | Default: PA7<br>Alternate: SPI0_MOSI, TIMER2_CH1, LPTIMER_ETI0, I2C2_SCL, CMP1_OUT, SEG4, EVENTOUT<br>Additional: ADC_IN7                                      |

| GD32L233Rx LQFP64 |      |                         |                          |                                                                                                                       |
|-------------------|------|-------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                 |
| PC4               | 26   | I/O                     | 5VT                      | Default: PC4<br>Alternate: LPUART_TX, USART0_TX, TIMER1_CH0_ETI, SEG22, EVENTOUT<br>Additional: ADC_IN14              |
| PC5               | 27   | I/O                     | 5VT                      | Default: PC5<br>Alternate: LPUART_RX, USART0_RX, TIMER1_CH1, SEG23, EVENTOUT<br>Additional: ADC_IN15                  |
| PB0               | 28   | I/O                     | 5VT                      | Default: PB0<br>Alternate: TIMER2_CH2, LPTIMER_OUT, SPI0_NSS, CMP0_OUT, SEG5, EVENTOUT<br>Additional: ADC_IN8         |
| PB1               | 29   | I/O                     | 5VT                      | Default: PB1<br>Alternate: TIMER2_CH3, LPUART_RTS, LPTIMER_IN0, SEG6, EVENTOUT<br>Additional: ADC_IN9                 |
| BOOT1-PB2         | 30   | I/O                     | 5VT                      | Default: BOOT1<br>Alternate: LPTIMER_OUT, EVENTOUT, RTC_OUT<br>Additional: PB2, WKUP3                                 |
| PB10              | 31   | I/O                     | 5VT                      | Default: PB10<br>Alternate: SPI1_SCK, I2S1_CK, LPUART_TX, I2C1_SCL, LPUART_RX, TIMER1_CH2, CMP0_OUT, SEG10, EVENTOUT  |
| PB11              | 32   | I/O                     | 5VT                      | Default: PB11<br>Alternate: LPUART_RX, I2C1_SDA, LPUART_TX, TIMER1_CH3, CMP1_OUT, SEG11, EVENTOUT                     |
| PB12              | 33   | I/O                     | 5VT                      | Default: PB12<br>Alternate: SPI1_NSS, I2S1_WS, I2C1_SMBA, LPUART_RTS, SEG12, EVENTOUT                                 |
| PB13              | 34   | I/O                     | 5VT                      | Default: PB13<br>Alternate: CK_OUT, SPI1_SCK, I2S1_CK, LPUART_CTS, I2C1_SCL, SEG13, EVENTOUT                          |
| PB14              | 35   | I/O                     | 5VT                      | Default: PB14<br>Alternate: SPI1_MISO, LPUART_RTS, I2C1_SDA, TIMER11_CH0 <sup>(3)</sup> , SEG14, EVENTOUT, RTC_OUT    |
| PB15              | 36   | I/O                     | 5VT                      | Default: PB15<br>Alternate: SPI1_MOSI, I2S1_SD, TIMER11_CH1 <sup>(3)</sup> , SEG15, EVENTOUT<br>Additional: RTC_REFIN |
| PC6               | 37   | I/O                     | 5VT                      | Default: PC6<br>Alternate: I2S1_MCK, TIMER2_CH0, SEG24, EVENTOUT<br>Additional: WKUP4                                 |
| PC7               | 38   | I/O                     | 5VT                      | Default: PC7<br>Alternate: TIMER2_CH1, SEG25, EVENTOUT                                                                |

| GD32L233Rx LQFP64 |      |                         |                          |                                                                                                                                   |
|-------------------|------|-------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                             |
| PC8               | 39   | I/O                     | 5VT                      | Default: PC8<br>Alternate: TIMER2_CH2, I2C2_SDA, SEG26, EVENTOUT                                                                  |
| PC9               | 40   | I/O                     | 5VT                      | Default: PC9<br>Alternate: TIMER2_CH3, I2C2_SCL, SEG27, EVENTOUT                                                                  |
| PA8               | 41   | I/O                     | 5VT                      | Default: PA8<br>Alternate: USART0_CK, CK_OUT, LPTIMER_OUT, I2C2_SMBA, COM0, EVENTOUT, CTC_SYNC<br>Additional: VCORE               |
| PA9               | 42   | I/O                     | 5VT                      | Default: PA9<br>Alternate: CK_OUT, USART0_TX, I2C0_SCL, COM1, EVENTOUT, LPTIMER_IN1                                               |
| PA10              | 43   | I/O                     | 5VT                      | Default: PA10<br>Alternate: USART0_RX, I2C0_SDA, COM2, EVENTOUT                                                                   |
| PA11              | 44   | I/O                     | 5VT                      | Default: PA11<br>Alternate: CMP0_OUT, USART0_CTS, SPI0_MISO, EVENTOUT<br>Additional: USBDM                                        |
| PA12              | 45   | I/O                     | 5VT                      | Default: PA12<br>Alternate: CMP1_OUT, USART0_RTS/USART0_DE, SPI0_MOSI, EVENTOUT<br>Additional: USBDP                              |
| PA13              | 46   | I/O                     | 5VT                      | Default: SWDIO<br>Alternate: LPUART_RX, I2C0_SCL, USART0_TX, SPI0_IO2, SPI0_NSS, EVENTOUT<br>Additional: PA13                     |
| PD0               | 47   | I/O                     | 5VT                      | Default: PD0<br>Alternate: SPI1_NSS, I2S1_WS, LPTIMER_OUT, USART1_CK, EVENTOUT, CTC_SYNC                                          |
| PD1               | 48   | I/O                     | 5VT                      | Default: PD1<br>Alternate: SPI1_SCK, I2S1_CK, SPI1_MISO, USART1_CTS, EVENTOUT                                                     |
| PA14              | 49   | I/O                     | 5VT                      | Default: SWCLK<br>Alternate: LPUART_TX, USART1_TX, I2C0_SDA, USART0_RX, SPI0_IO3, SPI1_NSS, I2S1_WS, EVENTOUT<br>Additional: PA14 |
| PA15              | 50   | I/O                     | 5VT                      | Default: PA15<br>Alternate: SPI1_NSS, I2S1_WS, TIMER1_CH0_ET1, SPI0_NSS, USART1_RX, SEG17, EVENTOUT                               |
| PC10              | 51   | I/O                     | 5VT                      | Default: PC10<br>Alternate: UART3_TX, LPUART_TX, SPI1_SCK, I2S1_CK, SEG28, COM4, EVENTOUT                                         |

| GD32L233Rx LQFP64 |      |                         |                          |                                                                                                                                                                         |
|-------------------|------|-------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                                                                   |
| PC11              | 52   | I/O                     | 5VT                      | Default: PC11<br>Alternate: UART3_RX, LPUART_RX, SPI1_MISO, SEG29, COM5, EVENTOUT                                                                                       |
| PC12              | 53   | I/O                     | 5VT                      | Default: PC12<br>Alternate: UART4_TX <sup>(3)</sup> , SPI1_MOSI, I2S1_SD, SEG30, COM6, EVENTOUT                                                                         |
| PD2               | 54   | I/O                     | 5VT                      | Default: PD2<br>Alternate: LPUART_RTS, TIMER2_ETI, UART4_RX <sup>(3)</sup> , SEG31, COM7, EVENTOUT                                                                      |
| PB3               | 55   | I/O                     | 5VT                      | Default: PB3<br>Alternate: UART4_TX <sup>(3)</sup> , SPI1_SCK, I2S1_CK, TIMER1_CH1, SPI0_SCK, USART0_RTS/USART0_DE, SEG7, EVENTOUT, LPTIMER_IN1<br>Additional: CMP1_IM6 |
| PB4               | 56   | I/O                     | 5VT                      | Default: PB4<br>Alternate: UART4_RX <sup>(3)</sup> , SPI1_MISO, TIMER2_CH0, SPI0_MISO, USART0_CTS, SEG8, EVENTOUT<br>Additional: CMP1_IP1                               |
| PB5               | 57   | I/O                     | 5VT                      | Default: PB5<br>Alternate: LPTIMER_IN0, I2C0_SMBA, SPI1_MOSI, I2S1_SD, TIMER2_CH1, SPI0_MOSI, USART0_CK, CMP1_OUT, SEG9, EVENTOUT<br>Additional: CMP1_IP2               |
| PB6               | 58   | I/O                     | 5VT                      | Default: PB6<br>Alternate: LPTIMER_ETI0, I2C1_SCL, I2C0_SCL, USART0_TX, SPI0_IO2, EVENTOUT<br>Additional: CMP1_IP3                                                      |
| PB7               | 59   | I/O                     | 5VT                      | Default: PB7<br>Alternate: I2C1_SDA, I2C0_SDA, USART0_RX, SPI0_IO3, EVENTOUT<br>Additional: CMP1_IP4                                                                    |
| BOOT0-PD3         | 60   | I/O                     | 5VT                      | Default: BOOT0<br>Alternate: USART1_CTS, SPI1_MISO, I2S1_MCK<br>Additional: PD3                                                                                         |
| PB8               | 61   | I/O                     | 5VT                      | Default: PB8<br>Alternate: I2C1_SCL, I2C0_SCL, CMP0_OUT, SEG16, EVENTOUT                                                                                                |
| PB9               | 62   | I/O                     | 5VT                      | Default: PB9<br>Alternate: I2C1_SDA, SPI1_NSS, I2S1_WS, I2C0_SDA, CMP1_OUT, COM3, EVENTOUT                                                                              |
| PD4               | 63   | I/O                     | 5VT                      | Default: PD4<br>Alternate: SPI1_MOSI, I2S1_SD, USART1_RTS/USART1_DE, EVENTOUT, SEG28                                                                                    |
| PD5               | 64   | I/O                     | 5VT                      | Default: PD5                                                                                                                                                            |

| GD32L233Rx LQFP64 |      |                         |                          |                                                  |
|-------------------|------|-------------------------|--------------------------|--------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                            |
|                   |      |                         |                          | Alternate: USART1_TX, EVENTOUT, SPI0_MISO, SEG29 |

**Note:**

- (1) Type: I = input, O = output, A = analog, P = power.
- (2) I/O Level: 5VT = 5 V tolerant.
- (3) Functions are available on GD32L233RB/C devices only.

## 2.6.2. GD32L233Cx LQFP48 pin definitions

Table 2-4. GD32L233Cx LQFP48 pin definitions

| GD32L233Cx LQFP48 |      |                         |                          |                                                                                                                                                          |
|-------------------|------|-------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                                                    |
| VBAT              | 1    | P                       |                          | Default: VBAT                                                                                                                                            |
| PC13-TAMPER-RTC   | 2    | I/O                     |                          | Default: PC13<br>Alternate: EVENTOUT<br>Additional: RTC_TAMP0, RTC_OUT, RTC_TS, WKUP1                                                                    |
| PC14-OSC32IN      | 3    | I/O                     |                          | Default: PC14<br>Alternate: EVENTOUT<br>Additional: OSC32IN                                                                                              |
| PC15-OSC32OUT     | 4    | I/O                     |                          | Default: PC15<br>Alternate: EVENTOUT<br>Additional: OSC32OUT                                                                                             |
| OSCIN             | 5    | I/O                     |                          | Default: OSCIN<br>Alternate: EVENTOUT, SPI1_NSS, I2S1_WS<br>Additional: PF0                                                                              |
| OSCOUT            | 6    | I/O                     |                          | Default: OSCOUT<br>Alternate: EVENTOUT, SPI1_SCK, I2S1_CK<br>Additional: PF1                                                                             |
| NRST              | 7    | I/O                     |                          | Default: NRST                                                                                                                                            |
| VSS               | 8    | P                       |                          | Default: VSS                                                                                                                                             |
| VREFP             | 9    | P                       |                          | Default: VREFP                                                                                                                                           |
| VDD               | 10   | P                       |                          | Default: VDD                                                                                                                                             |
| PA0-WKUP          | 11   | I/O                     | 5VT                      | Default: PA0<br>Alternate: USART1_CTS, TIMER1_CH0_ETI, CMP0_OUT, EVENTOUT, UART3_TX<br>Additional: WKUP0, ADC_IN0, RTC_TAMP1, CMP0_IM4                   |
| PA1               | 12   | I/O                     | 5VT                      | Default: PA1<br>Alternate: USART1_RTS/USART1_DE, TIMER1_CH1, I2C0_SMBA, SPI0_SCK, EVENTOUT, UART3_RX<br>Additional: ADC_IN1, CMP0_IP                     |
| PA2               | 13   | I/O                     | 5VT                      | Default: PA2<br>Alternate: USART1_TX, TIMER8_CH0, TIMER1_CH2, SPI0_IO2, CMP1_OUT, LPUART_TX, EVENTOUT<br>Additional: ADC_IN2, CMP1_IM4, RTC_TAMP2, WKUP2 |
| PA3               | 14   | I/O                     | 5VT                      | Default: PA3<br>Alternate: USART1_RX, TIMER8_CH1, TIMER1_CH3, SPI0_IO3, LPUART_RX, EVENTOUT<br>Additional: ADC_IN3, CMP1_IP0                             |

| GD32L233Cx LQFP48 |      |                         |                          |                                                                                                                          |
|-------------------|------|-------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                    |
| PA4               | 15   | I/O                     |                          | Default: PA4<br>Alternate: SPI0_NSS, USART1_CK, SPI1_NSS, I2S1_WS, LPTIMER_OUT, EVENTOUT<br>Additional: ADC_IN4, DAC_OUT |
| PA5               | 16   | I/O                     | 5VT                      | Default: PA5<br>Alternate: SPI0_SCK, TIMER1_CH0_ETI, LPTIMER_ETIO, EVENTOUT<br>Additional: ADC_IN5                       |
| PA6               | 17   | I/O                     | 5VT                      | Default: PA6<br>Alternate: SPI0_MISO, TIMER2_CH0, LPTIMER_IN0, CMP0_OUT, LPUART_CTS, EVENTOUT<br>Additional: ADC_IN6     |
| PA7               | 18   | I/O                     | 5VT                      | Default: PA7<br>Alternate: SPI0_MOSI, TIMER2_CH1, LPTIMER_ETIO, CMP1_OUT, EVENTOUT<br>Additional: ADC_IN7                |
| PB0               | 19   | I/O                     | 5VT                      | Default: PB0<br>Alternate: TIMER2_CH2, LPTIMER_OUT, SPI0_NSS, CMP0_OUT, EVENTOUT<br>Additional: ADC_IN8                  |
| PB1               | 20   | I/O                     | 5VT                      | Default: PB1<br>Alternate: TIMER2_CH3, LPUART_RTS, LPTIMER_IN0, EVENTOUT<br>Additional: ADC_IN9                          |
| BOOT1-PB2         | 21   | I/O                     | 5VT                      | Default: BOOT1<br>Alternate: LPTIMER_OUT, EVENTOUT, RTC_OUT<br>Additional: PB2, WKUP3                                    |
| PB10              | 22   | I/O                     | 5VT                      | Default: PB10<br>Alternate: SPI1_SCK, I2S1_CK, LPUART_TX, I2C1_SCL, LPUART_RX, TIMER1_CH2, CMP0_OUT, EVENTOUT            |
| PB11              | 23   | I/O                     | 5VT                      | Default: PB11<br>Alternate: LPUART_RX, I2C1_SDA, LPUART_TX, TIMER1_CH3, CMP1_OUT, EVENTOUT                               |
| PB12              | 24   | I/O                     | 5VT                      | Default: PB12<br>Alternate: SPI1_NSS, I2S1_WS, I2C1_SMBA, LPUART_RTS, EVENTOUT                                           |
| PB13              | 25   | I/O                     | 5VT                      | Default: PB13<br>Alternate: CK_OUT, SPI1_SCK, I2S1_CK, LPUART_CTS, I2C1_SCL, EVENTOUT                                    |
| PB14              | 26   | I/O                     | 5VT                      | Default: PB14<br>Alternate: SPI1_MISO, LPUART_RTS, I2C1_SDA, TIMER11_CH0 <sup>(3)</sup> , EVENTOUT, RTC_OUT              |
| PB15              | 27   | I/O                     | 5VT                      | Default: PB15<br>Alternate: SPI1_MOSI, I2S1_SD, TIMER11_CH1 <sup>(3)</sup> ,                                             |

| GD32L233Cx LQFP48 |      |                         |                          |                                                                                                                                   |
|-------------------|------|-------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                             |
|                   |      |                         |                          | EVENTOUT<br>Additional: RTC_REFIN                                                                                                 |
| PC6               | 28   | I/O                     | 5VT                      | Default: PC6<br>Alternate: I2S1_MCK, TIMER2_CH0, EVENTOUT<br>Additional: WKUP4                                                    |
| PC7               | 29   | I/O                     | 5VT                      | Default: PC7<br>Alternate: TIMER2_CH1, EVENTOUT                                                                                   |
| PA8               | 30   | I/O                     | 5VT                      | Default: PA8<br>Alternate: USART0_CK, CK_OUT, LPTIMER_OUT, EVENTOUT, CTC_SYNC<br>Additional: VCORE                                |
| PA9               | 31   | I/O                     | 5VT                      | Default: PA9<br>Alternate: CK_OUT, USART0_TX, I2C0_SCL, EVENTOUT, LPTIMER_IN1                                                     |
| PA10              | 32   | I/O                     | 5VT                      | Default: PA10<br>Alternate: USART0_RX, I2C0_SDA, EVENTOUT                                                                         |
| PA11              | 33   | I/O                     | 5VT                      | Default: PA11<br>Alternate: CMP0_OUT, USART0_CTS, SPI0_MISO, EVENTOUT<br>Additional: USBDM                                        |
| PA12              | 34   | I/O                     | 5VT                      | Default: PA12<br>Alternate: CMP1_OUT, USART0_RTS/USART0_DE, SPI0_MOSI, EVENTOUT<br>Additional: USBDP                              |
| PA13              | 35   | I/O                     | 5VT                      | Default: SWDIO<br>Alternate: LPUART_RX, I2C0_SCL, USART0_TX, SPI0_IO2, SPI0_NSS, EVENTOUT<br>Additional: PA13                     |
| PA14              | 36   | I/O                     | 5VT                      | Default: SWCLK<br>Alternate: LPUART_TX, USART1_TX, I2C0_SDA, USART0_RX, SPI0_IO3, SPI1_NSS, I2S1_WS, EVENTOUT<br>Additional: PA14 |
| PA15              | 37   | I/O                     | 5VT                      | Default: PA15<br>Alternate: SPI1_NSS, I2S1_WS, TIMER1_CH0_ETI, SPI0_NSS, USART1_RX, EVENTOUT                                      |
| PC10              | 38   | I/O                     | 5VT                      | Default: PC10<br>Alternate: UART3_TX, LPUART_TX, SPI1_SCK, I2S1_CK, EVENTOUT                                                      |
| PC11              | 39   | I/O                     | 5VT                      | Default: PC11<br>Alternate: UART3_RX, LPUART_RX, SPI1_MISO, EVENTOUT                                                              |
| PC12              | 40   | I/O                     | 5VT                      | Default: PC12<br>Alternate: UART4_TX <sup>(3)</sup> , SPI1_MOSI, I2S1_SD, EVENTOUT                                                |

| GD32L233Cx LQFP48 |      |                         |                          |                                                                                                                                                                    |
|-------------------|------|-------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                                                              |
| PB3               | 41   | I/O                     | 5VT                      | Default: PB3<br>Alternate: UART4_TX <sup>(3)</sup> , SPI1_SCK, I2S1_CK, TIMERR1_CH1, SPI0_SCK, USART0_RTS/USART0_DE, EVENTOUT, LPTIMER_IN1<br>Additional: CMP1_IM6 |
| PB4               | 42   | I/O                     | 5VT                      | Default: PB4<br>Alternate: UART4_RX <sup>(3)</sup> , SPI1_MISO, TIMER2_CH0, SPI0_MISO, USART0_CTS, EVENTOUT<br>Additional: CMP1_IP1                                |
| PB5               | 43   | I/O                     | 5VT                      | Default: PB5<br>Alternate: LPTIMER_IN0, I2C0_SMBA, SPI1_MOSI, I2S1_SD, TIMER2_CH1, SPI0_MOSI, USART0_CK, CMP1_OUT, EVENTOUT<br>Additional: CMP1_IP2                |
| PB6               | 44   | I/O                     | 5VT                      | Default: PB6<br>Alternate: LPTIMER_ETIO, I2C1_SCL, I2C0_SCL, USART0_TX, SPI0_IO2, EVENTOUT<br>Additional: CMP1_IP3                                                 |
| PB7               | 45   | I/O                     | 5VT                      | Default: PB7<br>Alternate: I2C1_SDA, I2C0_SDA, USART0_RX, SPI0_IO3, EVENTOUT<br>Additional: CMP1_IP4                                                               |
| BOOT0-PD3         | 46   | I/O                     | 5VT                      | Default: BOOT0<br>Alternate: USART1_CTS, SPI1_MISO, I2S1_MCK<br>Additional: PD3                                                                                    |
| PB8               | 47   | I/O                     | 5VT                      | Default: PB8<br>Alternate: I2C1_SCL, I2C0_SCL, CMP0_OUT, EVENTOUT                                                                                                  |
| PB9               | 48   | I/O                     | 5VT                      | Default: PB9<br>Alternate: I2C1_SDA, SPI1_NSS, I2S1_WS, I2C0_SDA, CMP1_OUT, EVENTOUT                                                                               |

**Note:**

(1) Type: I = input, O = output, A = analog, P = power.

(2) I/O Level: 5VT = 5 V tolerant.

(3) Functions are available on GD32L233CB/C devices only.

### 2.6.3. GD32L233Kx LQFP32 pin definitions

Table 2-5. GD32L233Kx LQFP32 pin definitions

| GD32L233Kx LQFP32 |      |                         |                          |                                                                                                                                                          |
|-------------------|------|-------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                                                    |
| VDD               | 1    | P                       |                          | Default: VDD                                                                                                                                             |
| OSCIN             | 2    | I/O                     |                          | Default: OSCIN<br>Alternate: EVENTOUT, SPI1_NSS, I2S1_WS<br>Additional: PF0                                                                              |
| OSCOUT            | 3    | I/O                     |                          | Default: OSCOUT<br>Alternate: EVENTOUT, SPI1_SCK, I2S1_CK<br>Additional: PF1                                                                             |
| NRST              | 4    | I/O                     |                          | Default: NRST                                                                                                                                            |
| VDDA              | 5    | P                       |                          | Default: VDDA                                                                                                                                            |
| PA0-WKUP          | 6    | I/O                     | 5VT                      | Default: PA0<br>Alternate: USART1_CTS, TIMER1_CH0_ETI, CMP0_OUT, EVENTOUT, UART3_TX<br>Additional: WKUP0, ADC_IN0, RTC_TAMP1, CMP0_IM4                   |
| PA1               | 7    | I/O                     | 5VT                      | Default: PA1<br>Alternate: USART1_RTS/USART1_DE, TIMER1_CH1, I2C0_SMBA, SPI0_SCK, EVENTOUT, UART3_RX<br>Additional: ADC_IN1, CMP0_IP                     |
| PA2               | 8    | I/O                     | 5VT                      | Default: PA2<br>Alternate: USART1_TX, TIMER8_CH0, TIMER1_CH2, SPI0_IO2, CMP1_OUT, LPUART_TX, EVENTOUT<br>Additional: ADC_IN2, CMP1_IM4, RTC_TAMP2, WKUP2 |
| PA3               | 9    | I/O                     | 5VT                      | Default: PA3<br>Alternate: USART1_RX, TIMER8_CH1, TIMER1_CH3, SPI0_IO3, LPUART_RX, EVENTOUT<br>Additional: ADC_IN3, CMP1_IP0                             |
| PA4               | 10   | I/O                     |                          | Default: PA4<br>Alternate: SPI0_NSS, USART1_CK, SPI1_NSS, I2S1_WS, LPTIMER_OUT, EVENTOUT<br>Additional: ADC_IN4, DAC_OUT                                 |
| PA5               | 11   | I/O                     | 5VT                      | Default: PA5<br>Alternate: SPI0_SCK, TIMER1_CH0_ETI, LPTIMER_ETIO, EVENTOUT<br>Additional: ADC_IN5                                                       |
| PA6               | 12   | I/O                     | 5VT                      | Default: PA6<br>Alternate: SPI0_MISO, TIMER2_CH0, LPTIMER_IN0, CMP0_OUT, LPUART_CTS, EVENTOUT<br>Additional: ADC_IN6                                     |

| GD32L233Kx LQFP32 |      |                         |                          |                                                                                                                                           |
|-------------------|------|-------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                                     |
| PA7               | 13   | I/O                     | 5VT                      | Default: PA7<br>Alternate: SPI0_MOSI, TIMER2_CH1, LPTIMER_ETI<br>0, CMP1_OUT, EVENTOUT<br>Additional: ADC_IN7                             |
| PB0               | 14   | I/O                     | 5VT                      | Default: PB0<br>Alternate: TIMER2_CH2, LPTIMER_OUT, SPI0_NSS,<br>CMP0_OUT, EVENTOUT<br>Additional: ADC_IN8                                |
| PB1               | 15   | I/O                     | 5VT                      | Default: PB1<br>Alternate: TIMER2_CH3, LPUART_RTS, LPTIMER_I<br>N0, EVENTOUT<br>Additional: ADC_IN9                                       |
| BOOT1-PB2         | 16   | I/O                     | 5VT                      | Default: BOOT1<br>Alternate: LPTIMER_OUT, EVENTOUT, RTC_OUT<br>Additional: PB2, WKUP3                                                     |
| VDD               | 17   | P                       |                          | Default: VDD                                                                                                                              |
| PA8               | 18   | I/O                     | 5VT                      | Default: PA8<br>Alternate: USART0_CK, CK_OUT, LPTIMER_OUT,<br>EVENTOUT, CTC_SYNC<br>Additional: VCORE                                     |
| PA9               | 19   | I/O                     | 5VT                      | Default: PA9<br>Alternate: CK_OUT, USART0_TX, I2C0_SCL, EVEN<br>TOUT, LPTIMER_IN1                                                         |
| PA10              | 20   | I/O                     | 5VT                      | Default: PA10<br>Alternate: USART0_RX, I2C0_SDA, EVENTOUT                                                                                 |
| PA11              | 21   | I/O                     | 5VT                      | Default: PA11<br>Alternate: CMP0_OUT, USART0_CTS, SPI0_MISO,<br>EVENTOUT<br>Additional: USBDM                                             |
| PA12              | 22   | I/O                     | 5VT                      | Default: PA12<br>Alternate: CMP1_OUT, USART0_RTS/USART0_DE,<br>SPI0_MOSI, EVENTOUT<br>Additional: USBDP                                   |
| PA13              | 23   | I/O                     | 5VT                      | Default: SWDIO<br>Alternate: LPUART_RX, I2C0_SCL, USART0_TX, S<br>PI0_IO2, SPI0_NSS, EVENTOUT<br>Additional: PA13                         |
| PA14              | 24   | I/O                     | 5VT                      | Default: SWCLK<br>Alternate: LPUART_TX, USART1_TX, I2C0_SDA, U<br>SART0_RX, SPI0_IO3, SPI1_NSS, I2S1_WS, EVEN<br>TOUT<br>Additional: PA14 |
| PA15              | 25   | I/O                     | 5VT                      | Default: PA15<br>Alternate: SPI1_NSS, I2S1_WS, TIMER1_CH0_ETI,<br>SPI0_NSS, USART1_RX, EVENTOUT                                           |

| GD32L233Kx LQFP32 |      |                         |                          |                                                                                                                                                     |
|-------------------|------|-------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Pin Name          | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                                               |
| PB3               | 26   | I/O                     | 5VT                      | Default: PB3<br>Alternate: SPI1_SCK, I2S1_CK, TIMER1_CH1, SPI0_SCK, USART0_RTS/USART0_DE, EVENTOUT, LPTIMER_IN1<br>Additional: CMP1_IM6             |
| PB4               | 27   | I/O                     | 5VT                      | Default: PB4<br>Alternate: SPI1_MISO, TIMER2_CH0, SPI0_MISO, USART0_CTS, EVENTOUT<br>Additional: CMP1_IP1                                           |
| PB5               | 28   | I/O                     | 5VT                      | Default: PB5<br>Alternate: LPTIMER_IN0, I2C0_SMBA, SPI1_MOSI, I2S1_SD, TIMER2_CH1, SPI0_MOSI, USART0_CK, CMP1_OUT, EVENTOUT<br>Additional: CMP1_IP2 |
| PB6               | 29   | I/O                     | 5VT                      | Default: PB6<br>Alternate: LPTIMER_ETIO, I2C1_SCL, I2C0_SCL, USART0_TX, SPI0_IO2, EVENTOUT<br>Additional: CMP1_IP3                                  |
| PB7               | 30   | I/O                     | 5VT                      | Default: PB7<br>Alternate: I2C1_SDA, I2C0_SDA, USART0_RX, SPI0_IO3, EVENTOUT<br>Additional: CMP1_IP4                                                |
| BOOT0-PD3         | 31   | I/O                     | 5VT                      | Default: BOOT0<br>Alternate: USART1_CTS, SPI1_MISO, I2S1_MCK<br>Additional: PD3                                                                     |
| VSS               | 32   | P                       |                          | Default: VSS                                                                                                                                        |

**Note:**

(1) Type: I = input, O = output, A = analog, P = power.

(2) I/O Level: 5VT = 5 V tolerant.

## 2.6.4. GD32L233Kx QFN32 pin definitions

Table 2-6. GD32L233Kx QFN32 pin definitions

| GD32L233Kx QFN32 |      |                         |                          |                                                                                                                                                          |
|------------------|------|-------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pin Name         | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                                                    |
| PC14-OSC32IN     | 1    | I/O                     |                          | Default: PC14<br>Alternate: EVENTOUT<br>Additional: OSC32IN                                                                                              |
| PC15-OSC32OUT    | 2    | I/O                     |                          | Default: PC15<br>Alternate: EVENTOUT<br>Additional: OSC32OUT                                                                                             |
| OSCIN            | 3    | I/O                     |                          | Default: OSCIN<br>Alternate: EVENTOUT, SPI1_NSS, I2S1_WS<br>Additional: PF0                                                                              |
| OSCOUT           | 4    | I/O                     |                          | Default: OSCOUT<br>Alternate: EVENTOUT, SPI1_SCK, I2S1_CK<br>Additional: PF1                                                                             |
| NRST             | 5    | I/O                     |                          | Default: NRST                                                                                                                                            |
| VDD              | 6    | P                       |                          | Default: VDD                                                                                                                                             |
| PA0-WKUP         | 7    | I/O                     | 5VT                      | Default: PA0<br>Alternate: USART1_CTS, TIMER1_CH0_ETI, CMP0_OUT, EVENTOUT, UART3_TX<br>Additional: WKUP0, ADC_IN0, RTC_TAMP1, CMP0_IM4                   |
| PA1              | 8    | I/O                     | 5VT                      | Default: PA1<br>Alternate: USART1_RTS/USART1_DE, TIMER1_CH1, I2C0_SMBA, SPI0_SCK, EVENTOUT, UART3_RX<br>Additional: ADC_IN1, CMP0_IP                     |
| PA2              | 9    | I/O                     | 5VT                      | Default: PA2<br>Alternate: USART1_TX, TIMER8_CH0, TIMER1_CH2, SPI0_IO2, CMP1_OUT, LPUART_TX, EVENTOUT<br>Additional: ADC_IN2, CMP1_IM4, RTC_TAMP2, WKUP2 |
| PA3              | 10   | I/O                     | 5VT                      | Default: PA3<br>Alternate: USART1_RX, TIMER8_CH1, TIMER1_CH3, SPI0_IO3, LPUART_RX, EVENTOUT<br>Additional: ADC_IN3, CMP1_IP0                             |
| PA4              | 11   | I/O                     |                          | Default: PA4<br>Alternate: SPI0_NSS, USART1_CK, SPI1_NSS, I2S1_WS, LPTIMER_OUT, EVENTOUT<br>Additional: ADC_IN4, DAC_OUT                                 |
| PA5              | 12   | I/O                     | 5VT                      | Default: PA5<br>Alternate: SPI0_SCK, TIMER1_CH0_ETI, LPTIMER_ETIO, EVENTOUT                                                                              |

| GD32L233Kx QFN32 |      |                         |                          |                                                                                                                      |
|------------------|------|-------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------|
| Pin Name         | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                |
|                  |      |                         |                          | Additional: ADC_IN5                                                                                                  |
| PA6              | 13   | I/O                     | 5VT                      | Default: PA6<br>Alternate: SPI0_MISO, TIMER2_CH0, LPTIMER_IN0, CMP0_OUT, LPUART_CTS, EVENTOUT<br>Additional: ADC_IN6 |
| PA7              | 14   | I/O                     | 5VT                      | Default: PA7<br>Alternate: SPI0_MOSI, TIMER2_CH1, LPTIMER_ETI0, CMP1_OUT, EVENTOUT<br>Additional: ADC_IN7            |
| PB0              | 15   | I/O                     | 5VT                      | Default: PB0<br>Alternate: TIMER2_CH2, LPTIMER_OUT, SPI0_NSS, CMP0_OUT, EVENTOUT<br>Additional: ADC_IN8              |
| PB1              | 16   | I/O                     | 5VT                      | Default: PB1<br>Alternate: TIMER2_CH3, LPUART_RTS, LPTIMER_I1N0, EVENTOUT<br>Additional: ADC_IN9                     |
| BOOT1-PB2        | 17   | I/O                     | 5VT                      | Default: BOOT1<br>Alternate: LPTIMER_OUT, EVENTOUT, RTC_OUT<br>Additional: PB2, WKUP3                                |
| PA8              | 18   | I/O                     | 5VT                      | Default: PA8<br>Alternate: USART0_CK, CK_OUT, LPTIMER_OUT, EVENTOUT, CTC_SYNC<br>Additional: VCORE                   |
| PA9              | 19   | I/O                     | 5VT                      | Default: PA9<br>Alternate: CK_OUT, USART0_TX, I2C0_SCL, EVENTOUT, LPTIMER_IN1                                        |
| PA10             | 20   | I/O                     | 5VT                      | Default: PA10<br>Alternate: USART0_RX, I2C0_SDA, EVENTOUT                                                            |
| PA11             | 21   | I/O                     | 5VT                      | Default: PA11<br>Alternate: CMP0_OUT, USART0_CTS, SPI0_MISO, EVENTOUT<br>Additional: USBDM                           |
| PA12             | 22   | I/O                     | 5VT                      | Default: PA12<br>Alternate: CMP1_OUT, USART0_RTS/USART0_DE, SPI0_MOSI, EVENTOUT<br>Additional: USBDP                 |
| PA13             | 23   | I/O                     | 5VT                      | Default: SWDIO<br>Alternate: LPUART_RX, I2C0_SCL, USART0_TX, SPI0_IO2, SPI0_NSS, EVENTOUT<br>Additional: PA13        |
| PA14             | 24   | I/O                     | 5VT                      | Default: SWCLK<br>Alternate: LPUART_TX, USART1_TX, I2C0_SDA, USART0_RX, SPI0_IO3, SPI1_NSS, I2S1_WS, EVENTOUT        |

| GD32L233Kx QFN32 |      |                         |                          |                                                                                                                                                     |
|------------------|------|-------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Pin Name         | Pins | Pin Type <sup>(1)</sup> | I/O Level <sup>(2)</sup> | Functions description                                                                                                                               |
|                  |      |                         |                          | Additional: PA14                                                                                                                                    |
| PA15             | 25   | I/O                     | 5VT                      | Default: PA15<br>Alternate: SPI1_NSS, I2S1_WS, TIMER1_CH0_ETI, SPI0_NSS, USART1_RX, EVENTOUT                                                        |
| PB3              | 26   | I/O                     | 5VT                      | Default: PB3<br>Alternate: SPI1_SCK, I2S1_CK, TIMER1_CH1, SPI0_SCK, USART0_RTS/USART0_DE, EVENTOUT, LP_TIMER_IN1<br>Additional: CMP1_IM6            |
| PB4              | 27   | I/O                     | 5VT                      | Default: PB4<br>Alternate: SPI1_MISO, TIMER2_CH0, SPI0_MISO, USART0_CTS, EVENTOUT<br>Additional: CMP1_IP1                                           |
| PB5              | 28   | I/O                     | 5VT                      | Default: PB5<br>Alternate: LPTIMER_IN0, I2C0_SMBA, SPI1_MOSI, I2S1_SD, TIMER2_CH1, SPI0_MOSI, USART0_CK, CMP1_OUT, EVENTOUT<br>Additional: CMP1_IP2 |
| PB6              | 29   | I/O                     | 5VT                      | Default: PB6<br>Alternate: LPTIMER_ETI0, I2C1_SCL, I2C0_SCL, USART0_TX, SPI0_IO2, EVENTOUT<br>Additional: CMP1_IP3                                  |
| PB7              | 30   | I/O                     | 5VT                      | Default: PB7<br>Alternate: I2C1_SDA, I2C0_SDA, USART0_RX, SPI0_IO3, EVENTOUT<br>Additional: CMP1_IP4                                                |
| BOOT0-PD3        | 31   | I/O                     | 5VT                      | Default: BOOT0<br>Alternate: USART1_CTS, SPI1_MISO, I2S1_MCK<br>Additional: PD3                                                                     |
| VBAT             | 32   | P                       |                          | Default: VBAT                                                                                                                                       |

**Note:**

(1) Type: I = input, O = output, A = analog, P = power.

(2) I/O Level: 5VT = 5 V tolerant.

## 2.6.5. GD32L233xx pin alternate functions

Table 2-7. Port A alternate functions summary

| Pin Name | AF0    | AF1            | AF2          | AF3                  | AF4                     | AF5       | AF6              | AF7                  | AF8        | AF9       |
|----------|--------|----------------|--------------|----------------------|-------------------------|-----------|------------------|----------------------|------------|-----------|
| PA0      |        | TIMER1_CH0_ETI |              |                      |                         |           | CMP0_OUT         | USART1_CTS           | UART3_TX   | EVENTO UT |
| PA1      |        | TIMER1_CH1     |              | SEG0 <sup>(2)</sup>  | I2C0_SMB                | SPI0_SCK  |                  | USART1_RTS/USART1_DE | UART3_RX   | EVENTO UT |
| PA2      |        | TIMER1_CH2     | TIMER8_CH0   | SEG1 <sup>(2)</sup>  |                         | SPI0_IO2  | CMP1_OUT         | USART1_TX            | LPUART_TX  | EVENTO UT |
| PA3      |        | TIMER1_CH3     | TIMER8_CH1   | SEG2 <sup>(2)</sup>  |                         | SPI0_IO3  |                  | USART1_RX            | LPUART_RX  | EVENTO UT |
| PA4      |        |                | LPTIMER_OUT  |                      |                         | SPI0_NSS  | SPI1_NSS/I2S1_WS | USART1_CK            |            | EVENTO UT |
| PA5      |        | TIMER1_CH0_ETI | LPTIMER_ETI0 |                      |                         | SPI0_SCK  |                  |                      |            | EVENTO UT |
| PA6      |        | TIMER2_CH0     | LPTIMER_IN0  | SEG3 <sup>(2)</sup>  |                         | SPI0_MISO | CMP0_OUT         |                      | LPUART_CTS | EVENTO UT |
| PA7      |        | TIMER2_CH1     | LPTIMER_ETI0 | SEG4 <sup>(2)</sup>  | I2C2_SCL <sup>(2)</sup> | SPI0_MOSI | CMP1_OUT         |                      |            | EVENTO UT |
| PA8      | CK_OUT |                | LPTIMER_OUT  | COM0 <sup>(2)</sup>  | I2C2_SMB <sup>(2)</sup> |           |                  | USART0_CK            | CTC_SYNC   | EVENTO UT |
| PA9      | CK_OUT |                | LPTIMER_IN1  | COM1 <sup>(2)</sup>  | I2C0_SCL                |           |                  | USART0_TX            |            | EVENTO UT |
| PA10     |        |                |              | COM2 <sup>(2)</sup>  | I2C0_SDA                |           |                  | USART0_RX            |            | EVENTO UT |
| PA11     |        |                |              |                      |                         | SPI0_MISO | CMP0_OUT         | USART0_CTS           |            | EVENTO UT |
| PA12     |        |                |              |                      |                         | SPI0_MOSI | CMP1_OUT         | USART0_RTS/USART0_DE |            | EVENTO UT |
| PA13     | SWDIO  |                | LPUART_RX    |                      | I2C0_SCL                | SPI0_IO2  | SPI0_NSS         | USART0_TX            |            | EVENTO UT |
| PA14     | SWCLK  |                | LPUART_TX    |                      | I2C0_SDA                | SPI0_IO3  | SPI1_NSS/I2S1_WS | USART0_RX            | USART1_TX  | EVENTO UT |
| PA15     |        | TIMER1_CH0_ETI |              | SEG17 <sup>(2)</sup> |                         | SPI0_NSS  | SPI1_NSS/I2S1_WS | USART1_RX            |            | EVENTO UT |

Table 2-8. Port B alternate functions summary

| Pin Name | AF0     | AF1        | AF2          | AF3                 | AF4      | AF5       | AF6               | AF7                  | AF8                     | AF9       |
|----------|---------|------------|--------------|---------------------|----------|-----------|-------------------|----------------------|-------------------------|-----------|
| PB0      |         | TIMER2_CH2 | LPTIMER_OUT  | SEG5 <sup>(2)</sup> |          | SPI0_NSS  | CMP0_OUT          |                      |                         | EVENTO UT |
| PB1      |         | TIMER2_CH3 | LPTIMER_IN0  | SEG6 <sup>(2)</sup> |          |           |                   |                      | LPUART_RTS              | EVENTO UT |
| PB2      | RTC_OUT |            | LPTIMER_OUT  |                     |          |           |                   |                      |                         | EVENTO UT |
| PB3      |         | TIMER1_CH1 | LPTIMER_IN1  | SEG7 <sup>(2)</sup> |          | SPI0_SCK  | SPI1_SCK/I2S1_CK  | USART0_RTS/USART0_DE | UART4_TX <sup>(1)</sup> | EVENTO UT |
| PB4      |         | TIMER2_CH0 |              | SEG8 <sup>(2)</sup> |          | SPI0_MISO | SPI1_MISO         | USART0_CTS           | UART4_RX <sup>(1)</sup> | EVENTO UT |
| PB5      |         | TIMER2_CH1 | LPTIMER_IN0  | SEG9 <sup>(2)</sup> | I2C0_SMB | SPI0_MOSI | SPI1_MOSI/I2S1_SD | USART0_CK            | CMP1_OUT                | EVENTO UT |
| PB6      |         |            | LPTIMER_ETI0 |                     | I2C0_SCL | SPI0_IO2  |                   | USART0_TX            | I2C1_SCL                | EVENTO UT |

| Pin Name | AF0         | AF1            | AF2                            | AF3                  | AF4           | AF5                          | AF6                          | AF7           | AF8            | AF9          |
|----------|-------------|----------------|--------------------------------|----------------------|---------------|------------------------------|------------------------------|---------------|----------------|--------------|
| PB7      |             |                |                                |                      | I2C0_SD<br>A  | SPI0_IO3                     |                              | USART0<br>_RX | I2C1_SD<br>A   | EVENTO<br>UT |
| PB8      |             |                |                                | SEG16 <sup>(2)</sup> | I2C0_SC<br>L  |                              | CMP0_O<br>UT                 |               | I2C1_SC<br>L   | EVENTO<br>UT |
| PB9      |             |                |                                | COM3 <sup>(2)</sup>  | I2C0_SD<br>A  | SPI1_NS<br>S/I2S1_<br>WS     | CMP1_O<br>UT                 |               | I2C1_SD<br>A   | EVENTO<br>UT |
| PB10     |             | TIMER1_<br>CH2 |                                | SEG10 <sup>(2)</sup> | I2C1_SC<br>L  | SPI1_SC<br>K/I2S1_<br>C<br>K | CMP0_O<br>UT                 | LPUART<br>_TX | LPUART<br>_RX  | EVENTO<br>UT |
| PB11     |             | TIMER1_<br>CH3 |                                | SEG11 <sup>(2)</sup> | I2C1_SD<br>A  |                              | CMP1_O<br>UT                 | LPUART<br>_RX | LPUART<br>_TX  | EVENTO<br>UT |
| PB12     |             |                |                                | SEG12 <sup>(2)</sup> | I2C1_SM<br>BA |                              | SPI1_NS<br>S/I2S1_<br>WS     |               | LPUART<br>_RTS | EVENTO<br>UT |
| PB13     | CK_OUT      |                |                                | SEG13 <sup>(2)</sup> | I2C1_SC<br>L  |                              | SPI1_SC<br>K/I2S1_<br>C<br>K |               | LPUART<br>_CTS | EVENTO<br>UT |
| PB14     | RTC_OU<br>T |                | TIMER11_<br>CH0 <sup>(1)</sup> | SEG14 <sup>(2)</sup> | I2C1_SD<br>A  |                              | SPI1_MI<br>SO                |               | LPUART<br>_RTS | EVENTO<br>UT |
| PB15     |             |                | TIMER11_<br>CH1 <sup>(1)</sup> | SEG15 <sup>(2)</sup> |               |                              | SPI1_MO<br>SI/I2S1_<br>SD    |               |                | EVENTO<br>UT |

Table 2-9. Port C alternate functions summary

| Pin Name | AF0 | AF1                | AF2              | AF3                                          | AF4                         | AF5                          | AF6          | AF7                         | AF8           | AF9          |
|----------|-----|--------------------|------------------|----------------------------------------------|-----------------------------|------------------------------|--------------|-----------------------------|---------------|--------------|
| PC0      |     |                    | LPTIMER<br>IN0   | SEG18 <sup>(2)</sup>                         | I2C2_SC<br>L <sup>(2)</sup> |                              |              |                             | LPUART<br>_RX | EVENTO<br>UT |
| PC1      |     |                    | LPTIMER<br>OUT   | SEG19 <sup>(2)</sup>                         | I2C2_SD<br>A <sup>(2)</sup> |                              |              |                             | LPUART<br>_TX | EVENTO<br>UT |
| PC2      |     |                    | LPTIMER<br>IN1   | SEG20 <sup>(2)</sup>                         |                             | SPI1_MI<br>SO                | I2S1_MC<br>K |                             |               | EVENTO<br>UT |
| PC3      |     |                    | LPTIMER<br>_ETI0 | SEG21 <sup>(2)</sup>                         |                             | SPI1_MO<br>SI/I2S1_<br>SD    |              |                             |               | EVENTO<br>UT |
| PC4      |     | TIMER1_<br>CH0_ETI |                  | SEG22 <sup>(2)</sup>                         |                             |                              |              | USART0<br>_TX               | LPUART<br>_TX | EVENTO<br>UT |
| PC5      |     | TIMER1_<br>CH1     |                  | SEG23 <sup>(2)</sup>                         |                             |                              |              | USART0<br>_RX               | LPUART<br>_RX | EVENTO<br>UT |
| PC6      |     | TIMER2_<br>CH0     |                  | SEG24 <sup>(2)</sup>                         |                             | I2S1_MC<br>K                 |              |                             |               | EVENTO<br>UT |
| PC7      |     | TIMER2_<br>CH1     |                  | SEG25 <sup>(2)</sup>                         |                             |                              |              |                             |               | EVENTO<br>UT |
| PC8      |     | TIMER2_<br>CH2     |                  | SEG26 <sup>(2)</sup>                         | I2C2_SD<br>A <sup>(2)</sup> |                              |              |                             |               | EVENTO<br>UT |
| PC9      |     | TIMER2_<br>CH3     |                  | SEG27 <sup>(2)</sup>                         | I2C2_SC<br>L <sup>(2)</sup> |                              |              |                             |               | EVENTO<br>UT |
| PC10     |     |                    |                  | SEG28 <sup>(2)</sup><br>/COM4 <sup>(2)</sup> |                             | SPI1_SC<br>K/I2S1_<br>C<br>K |              | UART3_T<br>X                | LPUART<br>_TX | EVENTO<br>UT |
| PC11     |     |                    |                  | SEG29 <sup>(2)</sup><br>/COM5 <sup>(2)</sup> |                             | SPI1_MI<br>SO                |              | UART3_<br>RX                | LPUART<br>_RX | EVENTO<br>UT |
| PC12     |     |                    |                  | SEG30 <sup>(2)</sup><br>/COM6 <sup>(2)</sup> |                             | SPI1_MO<br>SI/I2S1_<br>SD    |              | UART4_T<br>X <sup>(1)</sup> |               | EVENTO<br>UT |
| PC13     |     |                    |                  |                                              |                             |                              |              |                             |               | EVENTO<br>UT |
| PC14     |     |                    |                  |                                              |                             |                              |              |                             |               | EVENTO<br>UT |
| PC15     |     |                    |                  |                                              |                             |                              |              |                             |               | EVENTO<br>UT |

Table 2-10. Port D alternate functions summary

| Pin Name | AF0 | AF1        | AF2          | AF3                                       | AF4 | AF5               | AF6              | AF7                     | AF8        | AF9      |
|----------|-----|------------|--------------|-------------------------------------------|-----|-------------------|------------------|-------------------------|------------|----------|
| PD0      |     |            | LPTIMER_OUT  |                                           |     |                   | SPI1_NSS/I2S1_WS | USART1_CK               | CTC_SYNC   | EVENTOUT |
| PD1      |     |            |              |                                           |     | SPI1_MISO         | SPI1_SCK/I2S1_CK | USART1_CTS              |            | EVENTOUT |
| PD2      |     | TIMER2_ETI |              | SEG31 <sup>(2)</sup> /COM7 <sup>(2)</sup> |     |                   |                  | UART4_RX <sup>(1)</sup> | LPUART_RTS | EVENTOUT |
| PD3      |     |            |              |                                           |     | SPI1_MISO         | I2S1_MCK         | USART1_CTS              |            |          |
| PD4      |     |            |              | SEG28 <sup>(2)</sup>                      |     | SPI1_MOSI/I2S1_SD |                  | USART1_RTS/USART1_DE    |            | EVENTOUT |
| PD5      |     |            |              | SEG29 <sup>(2)</sup>                      |     | SPI0_MISO         |                  | USART1_TX               |            | EVENTOUT |
| PD6      |     |            | LPTIMER_IN1  |                                           |     | SPI0_MOSI         |                  | USART1_RX               |            | EVENTOUT |
| PD8      |     |            | LPTIMER_ETIO | SEG30 <sup>(2)</sup>                      |     |                   |                  |                         | LPUART_TX  | EVENTOUT |
| PD9      |     |            | LPTIMER_IN0  | SEG31 <sup>(2)</sup>                      |     |                   |                  |                         | LPUART_RX  | EVENTOUT |

Table 2-11. Port F alternate functions summary

| Pin Name | AF0 | AF1 | AF2 | AF3 | AF4 | AF5              | AF6 | AF7 | AF8 | AF9      |
|----------|-----|-----|-----|-----|-----|------------------|-----|-----|-----|----------|
| PF0      |     |     |     |     |     | SPI1_NSS/I2S1_WS |     |     |     | EVENTOUT |
| PF1      |     |     |     |     |     | SPI1_SCK/I2S1_CK |     |     |     | EVENTOUT |

**Note:**

- (1) Functions are available on GD32L233RC/RB/CC/CB devices only.
- (2) Functions are available on GD32L233Rx devices only.

### 3. Functional description

#### 3.1. Arm® Cortex®-M23 core

The Cortex-M23 processor is an energy-efficient processor with a very low gate count. It is intended to be used for microcontroller and deeply embedded applications that require an area-optimized processor. The processor is highly configurable enabling a wide range of implementations from those requiring memory protection and powerful trace technology to cost sensitive devices requiring minimal area, while delivering outstanding computational performance and an advanced system response to interrupts.

32-bit ARM® Cortex®-M23 processor core

- Up to 64 MHz operation frequency.
- Single-cycle multiplication and hardware divider.
- Ultra-low power, energy-efficient operation.
- Excellent code density.
- Integrated Nested Vectored Interrupt Controller (NVIC).
- 24-bit SysTick timer.

The Cortex®-M23 processor is based on the ARMv8-M architecture and supports both Thumb and Thumb-2 instruction sets. Some system peripherals listed below are also provided by Cortex®-M23:

- Internal Bus Matrix connected with AHB master, Serial Wire Debug Port and Single-cycle IO port.
- Nested Vectored Interrupt Controller (NVIC).
- Breakpoint Unit(BPU).
- Data Watchpoint and Trace (DWT).
- Serial Wire Debug Port.

#### 3.2. Embedded memory

- Up to 256 Kbytes of Flash memory.
- Up to 32 Kbytes of SRAM with hardware parity checking.

256 Kbytes of inner Flash memory, and 32 Kbytes of inner SRAM at most is available for storing programs and data, and Flash is accessed (read) at CPU clock speed with 0~3 wait states. [Table 2-2. GD32L233xx memory map](#) shows the memory map of the GD32L233xx series of devices, including code, SRAM, peripheral, and other pre-defined regions.

### 3.3. Clock, reset and supply management

- Internal 16 MHz factory-trimmed RC and external 4 to 48 MHz crystal oscillator.
- Internal 48 MHz factory-trimmed RC.
- Internal 32 KHz RC calibrated oscillator and external 32.768 KHz crystal oscillator.
- Integrated system clock PLL.
- 1.71 to 3.63 V application supply and I/Os.
- Supply Supervisor: POR (Power On Reset), PDR (Power Down Reset), and low voltage detector (LVD).

The Clock Control Unit (CCTL) provides a range of oscillator and clock functions. These include speed internal RC oscillator and external crystal oscillator, high speed and low speed two types. Several prescalers allow the frequency configuration of the AHB and two APB domains. The maximum frequency of the AHB, APB2 and APB1 domains is 64 MHz/64 MHz/32 MHz. See [Figure 2-6. GD32L233xx clock tree](#) for details on the clock tree.

The Reset Control Unit (RCU) controls three kinds of reset: system reset resets the processor core and peripheral IP components. Power-on reset (POR) and power-down reset (PDR) are always active, and ensures proper operation starting from 1.60 V and down to 1.56V. The device remains in reset mode when  $V_{DD}$  is below a specified threshold. The embedded low voltage detector (LVD) monitors the power supply, compares it to the voltage threshold and generates an interrupt as a warning message for leading the MCU into security.

Power supply schemes:

- $V_{DD}$  range: 1.71 to 3.63 V, external power supply for I/Os and the internal regulator. Provided externally through  $V_{DD}$  pins.
- $V_{SS}$  is 0 V.
- $V_{DDA}$  range: 1.71 to 3.63 V, external analog power supplies for ADC, reset blocks, RCs and PLL.
- $V_{BAT}$  range: 1.71 to 3.63 V, power supply for RTC unit, LXTAL oscillator, BPOR, and two pads, including PC13 to PC15 when  $V_{DD}$  is not present.

### 3.4. Boot modes

At startup, boot pins are used to select one of three boot options:

- Boot from main Flash memory (default).
- Boot from system memory.
- Boot from on-chip SRAM.

In default condition, boot from main Flash memory is selected. The boot loader is located in the internal boot ROM memory (system memory). It is used to reprogram the Flash

memory by using USART0 (PA9 and PA10) or USART1 (PA2 and PA3) or USB0 (PA11 and PA12).

### 3.5. Power saving modes

The MCU supports ten kinds of power saving modes to achieve even lower power consumption. They are Run, Run1, Run2, Sleep, Sleep1, Sleep2, Deep-sleep, Deep-sleep 1, Deep-sleep 2 and Standby mode. These operating modes reduce the power consumption and allow the application to achieve the best balance between the CPU operating time, speed and power consumption.

#### ■ Run mode

After system reset/ power reset or wakeup from standby mode, the MCU enters Run mode. And the NPLDO (normal power LDO) works in 1.1V mode.

#### ■ Run1 mode

When in Run mode, the NPLDO should be selected as 0.9V by configuring the LDOVS bits in PMU\_CTL0. In this mode, the system clock frequency should not exceed 16MHz.

#### ■ Run2 mode

When in Run mode or Run1 mode, the NPLDO can be selected as 0.9V by configuring the LDOVS bits in PMU\_CTL0. The LDNP in PMU\_CTL0 register should be configured to select the low-dirver mode. In this mode, the system clock frequency should not exceed 2MHz.

#### ■ Sleep mode

The Sleep mode is corresponding to the SLEEPING mode of the Cortex®-M23. In Sleep mode, only clock of Cortex®-M23 is off. To enter the Sleep mode, it is only necessary to clear the SLEEPDEEP bit in the Cortex®-M23 System Control Register, and execute a WFI or WFE instruction. If the Sleep mode is entered by executing a WFI instruction, any interrupt can wake up the system. If it is entered by executing a WFE instruction, any wakeup event can wake up the system (If SEVONPEND is 1, any interrupt can wake up the system, refer to Cortex®-M23 Technical Reference Manual). The mode offers the lowest wakeup time as no time is wasted in interrupt entry or exit.

#### ■ Sleep1 mode

The Sleep1 mode is corresponding to the SLEEPING mode of the Cortex®-M23. When in Run1 mode. The NPLDO should be selected as 0.9V by configuring the LDOVS bits in PMU\_CTL0.

#### ■ Sleep2 mode

The Sleep2 mode is corresponding to the SLEEPING mode of the Cortex®-M23. When in Run2 mode. The NPLDO should be selected as 0.9V by configuring the LDOVS bits in PMU\_CTL0. The LDNP in PMU\_CTL0 should be configured to select the low-dirver mode.

#### ■ Deep-sleep mode

The Deep-sleep mode is based on the SLEEPDEEP mode of the Cortex®-M23. In

Deep-sleep mode, all clocks in the 1.1V domain are off, and all of IRC16M, IRC48M, HXTAL and PLLs are disabled. The contents of SRAM and registers are preserved. The NPLDO can operate normally or in low driver mode depending on the LDNPDSB bit in the PMU\_CTL0 register. Before entering the Deep-sleep mode, it is necessary to set the SLEEPDEEP bit in the Cortex®-M23 System Control Register, and set LPMOD bits to “00” in the PMU\_CTL0 register. Then, the device enters the Deep-sleep mode after a WFI or WFE instruction is executed. If the Deep-sleep mode is entered by executing a WFI instruction, any interrupt from EXTI lines can wake up the system. If it is entered by executing a WFE instruction, any wakeup event from EXTI lines can wake up the system (If SEVONPEND is 1, any interrupt from EXTI lines can wake up the system, refer to Cortex®-M23 Technical Reference Manual). When exiting the Deep-sleep mode, the IRC16M is selected as the system clock. Notice that an additional wakeup delay will be incurred if the LDO operates in low driver mode.

#### ■ Deep-sleep 1 mode

The Deep-sleep 1 mode is based on the SLEEPDEEP mode of the Cortex®-M23. In Deep-sleep 1 mode, all clocks in the 1.1V domain are off, and all of IRC16M, IRC48M, HXTAL and PLLs are disabled. The LPLDO (low power LDO) can operate normally instead of NPLDO. Before entering the Deep-sleep 1 mode, it is necessary to set the SLEEPDEEP bit in the Cortex®-M23 System Control Register, set LPMOD bits to “01” in the PMU\_CTL0 register. Then, the device enters the Deep-sleep 1 mode after a WFI or WFE instruction is executed. If the Deep-sleep 1 mode is entered by executing a WFI instruction, any interrupt from EXTI lines can wake up the system. If it is entered by executing a WFE instruction, any wakeup event from EXTI lines can wake up the system (If SEVONPEND is 1, any interrupt from EXTI lines can wake up the system, refer to Cortex®-M23 Technical Reference Manual). When exiting the Deep-sleep 1 mode, the IRC16M is selected as the system clock. Waking up from Deep-sleep 1 mode needs an additional delay to wakeup NPLDO.

#### ■ Deep-sleep 2 mode

The Deep-sleep 2 mode is based on the SLEEPDEEP mode of the Cortex®-M23. In Deep-sleep 2 mode, all clocks in the 1.1V domain are off, and all of IRC16M, IRC48M, HXTAL and PLLs are disabled. The power of COREOFF0/SRAM1/COREOFF1 domain is cut off. The contents of COREOFF0/SRAM1/COREOFF1 domain are lost. The LPLDO can operate normally instead of NPLDO. Before entering the Deep-sleep 2 mode, it is necessary to set the SLEEPDEEP bit in the Cortex®-M23 System Control Register, set LPMOD bits to “10” in the PMU\_CTL0 register. Then, the device enters the Deep-sleep 2 mode after a WFI or WFE instruction is executed. If the Deep-sleep 2 mode is entered by executing a WFI instruction, any interrupt from EXTI lines can wake up the system. If it is entered by executing a WFE instruction, any wakeup event from EXTI lines can wake up the system (If SEVONPEND is 1, any interrupt from EXTI lines can wake up the system, refer to Cortex®-M23 Technical Reference Manual). When exiting the Deep-sleep 2 mode, the IRC16M is selected as the system clock. Waking up from Deep-sleep 2 mode needs an additional delay to wakeup NPLDO.

#### ■ Standby mode

The Standby mode is based on the SLEEPDEEP mode of the Cortex®-M23 too. In Standby mode, the whole 1.1V domain is power off, the NPLDO / LPLDO is shut down, and all of IRC16M, IRC48M, HXTAL and PLLs are disabled. Before entering the Standby mode, it is necessary to set the LPMOD bits to “11” in the PMU\_CTL0 register, and clear WUF bit in the PMU\_CS register, and set the SLEEPDEEP bit in the Cortex®-M23 System Control Register. Then, the device enters the Standby mode after a WFI or WFE instruction is executed, and the STBF status flag in the PMU\_CS register indicates that the MCU has been in Standby mode. There are four wakeup sources for the Standby mode, including the external reset from NRST pin, the RTC alarm / time stamp / tamper / auto wakeup events, the FWDGT reset, and the rising edge on WKUP pins. The Standby mode achieves the lowest power consumption, but spends longest time to wake up. Besides, the contents of SRAM and registers in 1.1V power domain are lost in Standby mode. When exiting from the Standby mode, a power-on reset occurs and the Cortex®-M23 will execute instruction code from the 0x00000000 address.

### 3.6. Clock trim controller (CTC)

- Three external reference signal source: GPIO, LXTAL clock, USBD\_SOF.
- Provide software reference sync pulse.
- Automatically trimmed by hardware without any software action.
- 16 bits trim counter with reference signal source capture and reload.
- 8 bits clock trim base value to frequency evaluation and automatically trim.

The Clock Trim Controller (CTC) is used to trim internal 48MHz RC oscillator (IRC48M) automatically by hardware. If using IRC48M clock to USBD, the IRC48M must be 48 MHz with 500ppm. The internal oscillator without such a high degree of accuracy needs to be trimmed. The CTC unit trim the frequency of the IRC48M based on an external accurate reference signal source. It can automatically adjust the trim value to provide a precise IRC48M clock.

### 3.7. General-purpose inputs/outputs (GPIOs)

- Up to 59 fast GPIOs, all mappable on 16 external interrupt lines.
- Analog input/output configurable.
- Alternate function input/output configurable.

There are up to 59 general purpose I/O pins (GPIO) in GD32L233xx, named PA0 ~ PA15, PB0 ~ PB15, PC0 ~ PC15, PD0~PD6, PD8~PD9, PF0~ PF1 to implement logic input/output functions. Each GPIO port has related control and configuration registers to satisfy the requirements of specific applications. The external interrupts on the GPIO pins of the device have related control and configuration registers in the Interrupt/Event

Controller Unit (EXTI). The GPIO ports are pin-shared with other alternative functions (AFs) to obtain maximum flexibility on the package pins.

Each of the GPIO pins can be configured by software as output (push-pull or open-drain), input, peripheral alternate function or analog mode. Most of the GPIO pins are shared with digital or analog alternate functions.

### 3.8. CRC calculation unit (CRC)

- Supports 7/8/16/32 bit data input.
- For 7(8)/16/32 bit input data length, the calculation cycles are 1/2/4 AHB clock cycles.
- Free 8-bit register is unrelated to calculation and can be used for any other goals by any other peripheral devices.
- User configurable polynomial value and size.

A cyclic redundancy check (CRC) is an error-detecting code commonly used in digital networks and storage devices to detect accidental changes to raw data. This CRC calculation unit can be used to calculate 7/8/16/32 bit CRC code within user configurable polynomial.

### 3.9. True Random number generator (TRNG)

- About 40 periods of TRNG\_CLK are needed between two consecutive random numbers.
- Disable TRNG module will reduce the chip power consumption.
- 32-bit random value seed is generated from analog noise, so the random number is a true random number.

The true random number generator (TRNG) module can generate a 32-bit random value by using continuous analog noise.

### 3.10. Direct memory access controller (DMA)

- 7 channels for DMA controller.
- DMA request from DMAMAX: peripherals (Timers, ADC, DAC, SPIs, I2S, I2Cs, USARTs, CAU and LPUART) and request generator.

The flexible general-purpose DMA controllers provide a hardware method of transferring data between peripherals and/or memory without intervention from the CPU, thereby increasing system performance by off-loading the MCU from copying large amounts of data and avoiding frequent interrupts to serve peripherals needing more data or having available data. Three types of access method are supported: peripheral to memory, memory to peripheral, memory to memory.

Each channel is connected to flexible hardware DMA requests. The priorities of DMA channel requests are determined by software configuration and hardware channel number. Transfer size of source and destination are independent and configurable.

### 3.11. DMA request multiplexer (DMAMUX)

- 7 channels for DMAMUX request multiplexer.
- 4 channels for DMAMUX request generator.
- Support 21 trigger inputs and 21 synchronization inputs.

DMAMUX is a transmission scheduler for DMA requests. The DMAMUX request multiplexer is used for routing a DMA request line between the peripherals / generated DMA request (from the DMAMUX request generator) and the DMA controller. Each DMAMUX request multiplexer channel selects a unique DMA request line, unconditionally or synchronously with events from its DMAMUX synchronization inputs. The DMA request is pending until it is served by the DMA controller which generates a DMA acknowledge signal (the DMA request signal is de-asserted).

### 3.12. Analog to digital converter (ADC)

- 12-bit SAR ADC's conversion rate is up to 1.07 MSPS.
- Hardware oversampling ratio adjustable from 2x to 256x improves resolution to 16-bit.
- Input voltage range:  $V_{SS}/V_{SSA}$  to  $V_{DD}/V_{DDA}$ .
- Temperature sensor.

A 12-bit multi-channel ADC is integrated in the device. It has a total of 20 multiplexed channels: up to 16 external channels, 1 channel for internal temperature sensor ( $V_{SENSE}$ ), 1 channel for internal reference voltage ( $V_{REFINT}$ ), 1 channel for external battery power supply ( $V_{BAT}$ ), and 1 channel for LCD voltage ( $V_{SLCD}$ ). The input voltage range is between  $V_{SS}/V_{SSA}$  and  $V_{DD}/V_{DDA}$ . An on-chip hardware oversampling scheme improves performance while off-loading the related computational burden from the CPU. The analog watchdog allows the application to detect whether the input voltage goes outside the user-defined higher or lower thresholds. A configurable channel management block can be used to perform conversions in single, continuous, scan or discontinuous mode to support more advanced use.

The ADC can be triggered from the events generated by the general level 0 timers (TIMERx, x=1, 2) and the general level 1 timers (TIMERx, x=8, 11) with internal connection. The temperature sensor can be used to generate a voltage that varies linearly with temperature. It is internally connected to the ADC\_IN16 input channel which is used to convert the sensor output voltage in a digital value.

To ensure a high accuracy on ADC, the independent power supply  $V_{DDA}$  is implemented

to achieve better performance of analog circuits.  $V_{DDA}$  can be externally connected to  $V_{DD}$  through the external filtering circuit that avoids noise on  $V_{DDA}$ .

### 3.13. Digital to analog converter (DAC)

- One 12-bit DAC with one output channel.
- 8-bit or 12-bit mode in conjunction with the DMA controller.
- Support references from internal 2.5 V precision reference or external VREFP pin.

The 12-bit buffered DAC is used to generate variable analog outputs. The DAC channels can be triggered by the timer or EXTI with DMA support. The maximum output value of the DAC is  $V_{REFP}$ .

### 3.14. Real time clock (RTC)

- Independent binary-coded decimal (BCD) format timer/counter with five 32-bit backup registers.
- Calendar with sub-second, second, minute, hour, week day, day, month and year automatically correction.
- Alarm function with wake up from deep-sleep and standby mode capability.
- On-the-fly correction for synchronization with master clock. Digital calibration with 0.95 ppm resolution for compensation of quartz crystal inaccuracy.

The real time clock is an independent timer which provides a set of continuously running counters in backup registers to provide a real calendar function, and provides an alarm interrupt or an expected interrupt. It is not reset by a system or power reset, or when the device wakes up from standby mode. In the RTC unit, there are two prescalers used for implementing the calendar and other functions. One prescaler is a 7-bit asynchronous prescaler and the other is a 15-bit synchronous prescaler.

### 3.15. Timers and PWM generation

- Up to four 16-bit general timers (TIMER1, TIMER2, TIMER8, TIMER11), two 16-bit basic timer (TIMER5, TIMER6), and one 32-bit low power timer (LPTIMER).
- Up to 4 independent channels of PWM, output compare or input capture for each general timer and external trigger input.
- Encoder interface controller with two inputs using quadrature decoder.
- 24-bit SysTick timers down counter.
- 2 watchdog timers (free watchdog timer and window watchdog timer).

The LPTIMER is a 32-bit timer and it is able to keep running in all power modes except for Standby mode with its diversity of clock sources. The LPTIMER provides one PWM out and also supports an encoder interface with two inputs using quadrature decoder.

The general timer can be used for a variety of purposes including general time, input signal pulse width measurement or output waveform generation such as a single pulse generation or PWM output, up to 4 independent channels for input capture/output compare. TIMER1 and TIMER2 are based on a 16-bit auto-reload up/down/center-aligned counter and a 16-bit prescaler. TIMER8 and TIMER11 is based on a 16-bit auto-reload up counter and a 16-bit prescaler. The general timer also supports an encoder interface with two inputs using quadrature decoder.

The basic timer TIMER5 and TIMER6, are mainly used as a simple 16-bit time base.

The GD32L233xx have two watchdog peripherals, free watchdog timer and window watchdog timer. They offer a combination of high safety level, flexibility of use and timing accuracy.

The free watchdog timer includes a 12-bit down-counting counter and an 8-stage prescaler. It is clocked from an independent 32 KHz internal RC and as it operates independently of the main clock, it can operate in deep-sleep and standby modes. It can be used either as a watchdog to reset the device when a problem occurs, or as a free-running timer for application timeout management.

The window watchdog is based on a 7-bit down counter that can be set as free-running. It can be used as a watchdog to reset the device when a problem occurs. It is clocked from the main clock. It has an early wakeup interrupt capability and the counter can be frozen in debug mode.

The SysTick timer is dedicated for OS, but could also be used as a standard down counter. The features are shown below:

- A 24-bit down counter.
- Auto reload capability.
- Maskable system interrupt generation when the counter reaches 0.
- Programmable clock source.

### 3.16. Universal synchronous/asynchronous receiver transmitter (USART/UART)

- Up to two USARTs and two UARTs with operating frequency up to 8 MBits/s.
- Supports both asynchronous and clocked synchronous serial communication modes.
- IrDA SIR encoder and decoder support.
- LIN break generation and detection.
- ISO 7816-3 compliant smart card interface.
- Dual clock domain.
- Wake up from Deep-sleep mode.

The USART (USART0, USART1) and UART (UART3, UART4) are used to translate data between parallel and serial interfaces, provides a flexible full duplex data exchange using

synchronous or asynchronous transfer. It is also commonly used for RS-232 standard communication. The USART/UART includes a programmable baud rate generator which is capable of dividing the system clock to produce a dedicated clock for the USART/UART transmitter and receiver. The USART/UART also supports DMA function for high speed data communication.

### 3.17. Universal asynchronous receiver transmitter (LPUART)

- Maximum speed up to 10 MBits/s.
- Supports asynchronous serial communication modes.
- Supports hardware modem operations (CTS/RTS) and RS485 drive.
- Dual clock domain.
- Wake up from Deep-sleep mode.

The Low-power universal Asynchronous Receiver/Transmitter (LPUART) provides a flexible serial data exchange interface with a limited power consumption. LPUART can perform asynchronous serial communication even with low power consumption. Data frames can be transferred in full duplex or half duplex mode, asynchronously through this interface. A programmable baud rate generator divides the clock to produce a dedicated wide range baudrate clock for the LPUART transmitter and receiver.

### 3.18. Inter-integrated circuit (I2C)

- Support both master and slave mode with a frequency up to 1 MHz (Fast mode plus).
- Provide arbitration function, optional PEC (packet error checking) generation and checking.
- Supports 7-bit and 10-bit addressing mode and general call addressing mode.
- Multiple 7-bit slave addresses (2 address, 1 with configurable mask).
- SMBus 3.0 and PMBus 1.3 compatible.
- Wakeup from Deep-sleep / Deep-sleep1 / Deep-sleep2 mode on I2C address match.

The I2C interface is an internal circuit allowing communication with an external I2C interface which is an industry standard two lines serial interface used for connection to external hardware. These two serial lines are known as a serial data line (SDA) and a serial clock line (SCL). The I2C module provides different data transfer rates: up to 100 KHz in standard mode, up to 400 KHz in the fast mode and up to 1 MHz in the fast mode plus. The I2C module also has an arbitration detect function to prevent the situation where more than one master attempts to transmit data to the I2C bus at the same time. A CRC-8 calculator is also provided in I2C interface to perform packet error checking for I2C data.

### 3.19. Serial peripheral interface (SPI)

- Up to two SPI interfaces with a frequency of up to 16 MHz.

- Support both master and slave mode.
- Hardware CRC calculation and transmit automatic CRC error checking.
- Separate transmit and receive 32-bit FIFO with DMA capability (only in SPI0).
- Data frame size can be 4 to 16 bits (only in SPI0).
- Quad-SPI configuration available in master mode (only in SPI0).

The SPI interface uses 4 pins, among which are the serial data input and output lines (MISO & MOSI), the clock line (SCK) and the slave select line (NSS). All SPIs can be served by the DMA controller. The SPI interface may be used for a variety of purposes, including simplex synchronous transfers on two lines with a possible bidirectional data line or reliable communication using CRC checking. Quad-SPI master mode is also supported in SPI0.

### 3.20. Inter-IC sound (I2S)

- Sampling frequency from 8 KHz to 192 KHz.
- Support either master or slave mode.

The Inter-IC sound (I2S) bus provides a standard communication interface for digital audio applications by 4-wire serial lines. GD32L233xx contain an I2S-bus interface that can be operated with 16/32 bit resolution in master or slave mode, pin multiplexed with SPI1. The audio sampling frequency from 8 KHz to 192 KHz is supported.

### 3.21. Cryptographic acceleration Unit (CAU)

- Supports DES, TDES or AES (128, 192, or 256) algorithms.
- DES/TDES supports Electronic codebook (ECB) or Cipher block chaining (CBC) mode.
- AES supports 128bits-key, 192bits-key or 256 bits-key.
- AES supports Electronic codebook (ECB), Cipher block chaining (CBC) mode, Counter mode (CTR) mode, Galois/counter mode (GCM), Galois message authentication code mode (GMAC), Counter with CBC-MAC (CCM), Cipher Feedback mode (CFB) and Output Feedback mode (OFB).
- DMA transfer for incoming and outgoing data is supported.

The Cryptographic Acceleration Unit supports acceleration of DES, TDES or AES (128, 192, or 256) algorithms. The DES/TDES supports Electronic codebook (ECB) or Cipher block chaining (CBC) mode. The AES supports Electronic codebook (ECB), Cipher block chaining (CBC) mode, Counter mode (CTR) mode, Galois/counter mode (GCM), Galois message authentication code mode (GMAC), Counter with CBC-MAC (CCM), Cipher Feedback mode (CFB) and Output Feedback mode (OFB).

### 3.22. Segment LCD controller (SLCD)

- Configurable frame frequency.
- Blinking of individual segments or all segments.
- Supports Static, 1/2, 1/3, 1/4, 1/6 and 1/8 duty.
- Supports 1/2, 1/3 and 1/4 bias.
- Double buffer up to 8x32 bits registers to store SLCD\_DATAx.
- The contrast can also be adjusted by configuring dead time.
- Optional voltage output driver for enhance SLCD driving capability.

The SLCD controller directly drives LCD displays by creating the AC segment and common voltage signals automatically. It can drive the monochrome passive liquid crystal display (LCD) which composed of a plurality of segments (pixels or complete symbols) that can be converted to visible or invisible. The SLCD controller can support up to 32 segments and 8 commons.

### 3.23. Comparators (CMP)

- Two fast rail-to-rail low-power comparators with software configurable.
- Programmable reference voltage (internal or external I/O).

Two Comparators (CMP) is implemented within the device. It can wake up from deep-sleep mode to generate interrupts and also can be combined as a window comparator. The internal voltage reference is also connected to ADC\_IN17 input channel of the ADC.

### 3.24. Universal serial bus full-speed device interface (USBD)

- USB 2.0 full-speed device controller.
- Support USB 2.0 Link Power Management.
- Dedicated 512-byte SRAM used for data packet buffer.
- Support embedded pull-up on the DP line.
- Integrated USB PHY.

The Universal Serial Bus full-speed device interface (USBD) module contains a full-speed internal USB PHY and no more external PHY chip is needed. USBBD supports all the four types of transfer (control, bulk, interrupt and isochronous) defined in USB 2.0 protocol. USBBD supports 8 USB bidirectional endpoints that can be individually configured.

### 3.25. Debug mode

- Serial wire debug port (SW-DP).

Debug capabilities can be accessed by a debug tool via Serial Wire (SW - Debug Port).

### 3.26. Package and operation temperature

- LQFP64 (GD32L233RxTx), LQFP48 (GD32L233CxTx), LQFP32 (GD32L233KxTx) and QFN32 (GD32L233KxQx).
- Operation temperature range: -40°C to +85°C (industrial level) for grade 6 devices, and -40°C to +105°C (industrial level) for grade 7 devices.

## 4 Electrical characteristics

### 4.1 Absolute maximum ratings

The maximum ratings are the limits to which the device can be subjected without permanently damaging the device. Note that the device is not guaranteed to operate properly at the maximum ratings. Exposure to the absolute maximum rating conditions for extended periods may affect device reliability.

**Table 4-1. Absolute maximum ratings<sup>(1)(4)</sup>**

| Symbol                             | Parameter                                                            | Min                    | Max                     | Unit |
|------------------------------------|----------------------------------------------------------------------|------------------------|-------------------------|------|
| V <sub>DD</sub>                    | External voltage range <sup>(2)</sup>                                | V <sub>SS</sub> - 0.3  | V <sub>SS</sub> + 3.63  | V    |
| V <sub>DDA</sub>                   | External analog supply voltage                                       | V <sub>SSA</sub> - 0.3 | V <sub>SSA</sub> + 3.63 | V    |
| V <sub>IN</sub>                    | Input voltage on 5V tolerant pin <sup>(3)</sup>                      | V <sub>SS</sub> - 0.3  | V <sub>DD</sub> + 3.63  | V    |
|                                    | Input voltage on other I/O                                           | V <sub>SS</sub> - 0.3  | 3.63                    | V    |
| ΔV <sub>DDx</sub>                  | Variations between different VDD power pins                          | —                      | 50                      | mV   |
| V <sub>SSx</sub> - V <sub>SS</sub> | Variations between different ground pins                             | —                      | 50                      | mV   |
| I <sub>IO</sub>                    | Maximum current for GPIO pins                                        | —                      | ±25                     | mA   |
| T <sub>A</sub>                     | Operating temperature range                                          | -40                    | +85                     | °C   |
| P <sub>D</sub>                     | Power dissipation at T <sub>A</sub> = 85°C of LQFP64 <sup>(5)</sup>  | —                      | 733                     | mW   |
|                                    | Power dissipation at T <sub>A</sub> = 85°C of LQFP48 <sup>(5)</sup>  | —                      | 574                     |      |
|                                    | Power dissipation at T <sub>A</sub> = 85°C of LQFP32 <sup>(5)</sup>  | —                      | 724                     |      |
|                                    | Power dissipation at T <sub>A</sub> = 85°C of QFN32 <sup>(5)</sup>   | —                      | 940                     |      |
|                                    | Power dissipation at T <sub>A</sub> = 105°C of LQFP64 <sup>(5)</sup> | —                      | 367                     |      |
| T <sub>STG</sub>                   | Storage temperature range                                            | -65                    | +150                    | °C   |
| T <sub>J</sub>                     | Maximum junction temperature                                         | —                      | +125                    | °C   |

(1) Guaranteed by design, not tested in production.

(2) All main power and ground pins should be connected to an external power source within the allowable range.

(3) V<sub>IN</sub> maximum value cannot exceed 5.5 V.

(4) It is recommended that V<sub>DD</sub> and V<sub>DDA</sub> are powered by the same source. The maximum difference between V<sub>DD</sub> and V<sub>DDA</sub> does not exceed 300 mV during power-up and operation.

(5) For grade 6 devices, the parameter of T<sub>A</sub>=85°C, For grade 7 devices, the parameter of T<sub>A</sub>=105°C.

### 4.2 Operating conditions characteristics

**Table 4-2. DC operating conditions**

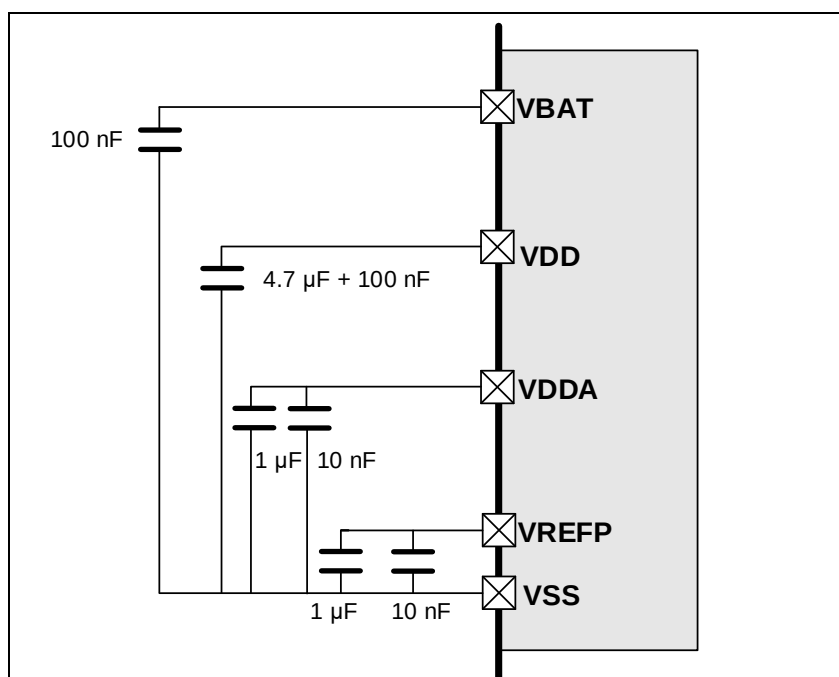
| Symbol           | Parameter              | Conditions | Min <sup>(1)</sup>  | Typ | Max <sup>(1)</sup> | Unit |
|------------------|------------------------|------------|---------------------|-----|--------------------|------|
| V <sub>DD</sub>  | Supply voltage         | —          | 1.71                | 3.3 | 3.63               | V    |
| V <sub>DDA</sub> | Analog supply voltage  | —          | 1.71                | 3.3 | 3.63               | V    |
| V <sub>BAT</sub> | Battery supply voltage | —          | 1.71 <sup>(2)</sup> | 3.3 | 3.63               | V    |

(1) Based on characterization, not tested in production.

(2) In the application which V<sub>BAT</sub> supply the backup domains, if the V<sub>BAT</sub> voltage drops below the minimum

value, when  $V_{DD}$  is powered on again, it is necessary to refresh the registers of backup domains and enable LXTAL again.

**Figure 4-1. Recommended power supply decoupling capacitors<sup>(1)</sup>**



(1) All decoupling capacitors need to be as close as possible to the pins on the PCB board. More details refer to **AN069 GD32L233 Hardware Development Guide**.

**Table 4-3. Clock frequency<sup>(1)</sup>**

| Symbol      | Parameter            | Conditions | Min | Max | Unit |
|-------------|----------------------|------------|-----|-----|------|
| $f_{HCLK1}$ | AHB1 clock frequency | —          | 0   | 64  | MHz  |
| $f_{HCLK2}$ | AHB2 clock frequency | —          | 0   | 64  | MHz  |
| $f_{APB1}$  | APB1 clock frequency | —          | 0   | 32  | MHz  |
| $f_{APB2}$  | APB2 clock frequency | —          | 0   | 64  | MHz  |

(1) Guaranteed by design, not tested in production.

**Table 4-4. Operating conditions at Power up/ Power down<sup>(1)</sup>**

| Symbol    | Parameter               | Conditions | Min | Max      | Unit |
|-----------|-------------------------|------------|-----|----------|------|
| $t_{VDD}$ | $V_{DD}$ rise time rate | —          | 0   | $\infty$ | us/v |
|           | $V_{DD}$ fall time rate |            | 50  | $\infty$ |      |

(1) Guaranteed by design, not tested in production.

**Table 4-5. Start-up timings of Operating conditions<sup>(1)</sup>**

| Symbol         | Parameter     | Conditions               | Typ  | Unit |
|----------------|---------------|--------------------------|------|------|
| $t_{start-up}$ | Start-up time | Clock source from HXTAL  | 1.24 | ms   |
|                |               | Clock source from IRC16M | 16.6 | us   |

(1) Based on characterization, not tested in production.

(2) After power-up, the start-up time is the time between the rising edge of NRST high and the first I/O instruction conversion in SystemInit function.

(3) PLL is off.

**Table 4-6. Power saving mode wakeup timings characteristics<sup>(1) (2)</sup>**

| Symbol                  | Parameter                                                          | Typ   | Unit |
|-------------------------|--------------------------------------------------------------------|-------|------|
| $t_{\text{Sleep}}$      | Wakeup from Sleep mode                                             | 1.29  | us   |
|                         | Wakeup from Sleep 1 mode (NPLDO=0.9V)                              | 1.30  |      |
|                         | Wakeup from Sleep 2 mode (NPLDO=0.9V and NPLDO in Low-driver mode) | 1.32  |      |
| $t_{\text{Deep-sleep}}$ | Wakeup from Deep-sleep mode (NPLDO in normal driver mode)          | 9.95  |      |
|                         | Wakeup from Deep-sleep mode (NPLDO in low driver mode)             | 9.93  |      |
|                         | Wakeup from Deep-sleep 1 mode                                      | 13.74 |      |
|                         | Wakeup from Deep-sleep 2 mode                                      | 15.50 |      |
| $t_{\text{Standby}}$    | Wakeup from Standby mode                                           | 20.92 |      |

(1) Based on characterization, not tested in production.

(2) The wakeup time is measured from the wakeup event to the point at which the application code reads the first instruction under the below conditions:  $V_{\text{DD}} = V_{\text{DDA}} = 3.3 \text{ V}$ , IRC16M = System clock = 16MHz.

## 4.3 Power consumption

The power measurements specified in the tables represent that code with data executing from on-chip Flash with the following specifications.

**Table 4-7. Power consumption characteristics<sup>(2)(3)(4)</sup>**

| Symbol                           | Parameter                 | Conditions                                                                                       | Min | Typ <sup>(1)</sup> | Max | Unit |
|----------------------------------|---------------------------|--------------------------------------------------------------------------------------------------|-----|--------------------|-----|------|
| $I_{\text{DD}} + I_{\text{DDA}}$ | Supply current (Run mode) | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 64 MHz, All peripherals enabled  | —   | 9.38               | —   | mA   |
|                                  |                           | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 64 MHz, All peripherals disabled | —   | 4.39               | —   |      |
|                                  |                           | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 48 MHz, All peripherals enabled  | —   | 7.23               | —   |      |
|                                  |                           | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 48 MHz, All peripherals disabled | —   | 3.46               | —   |      |
|                                  |                           | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 36 MHz, All peripherals enabled  | —   | 5.06               | —   |      |
|                                  |                           | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 36 MHz, All peripherals disabled | —   | 2.2                | —   |      |
|                                  |                           | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 24 MHz, All peripherals enabled  | —   | 4.47               | —   |      |
|                                  |                           | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 24 MHz, All peripherals disabled | —   | 2.54               | —   |      |
|                                  |                           | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 16 MHz, All peripherals enabled  | —   | 3.24               | —   |      |
|                                  |                           | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 16 MHz, All peripherals disabled | —   | 1.92               | —   |      |
|                                  |                           | $V_{\text{DD}} = 3.3 \text{ V}$ , HXTAL = 8 MHz, System clock = 8 MHz, All peripherals enabled   | —   | 1.98               | —   |      |

| Symbol | Parameter                      | Conditions                                                                                               | Min | Typ <sup>(1)</sup> | Max | Unit |
|--------|--------------------------------|----------------------------------------------------------------------------------------------------------|-----|--------------------|-----|------|
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, System clock = 8 MHz, All peripherals disabled                   | —   | 1.29               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, System clock = 4 MHz, All peripherals enabled                    | —   | 1.36               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, System clock = 4 MHz, All peripherals disabled                   | —   | 0.96               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, System clock = 2 MHz, All peripherals enabled                    | —   | 1.04               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, System clock = 2 MHz, All peripherals disabled                   | —   | 0.81               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 4 MHz, System clock = 1 MHz, All peripherals enabled                    | —   | 0.9                | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 4 MHz, System clock = 1 MHz, All peripherals disabled                   | —   | 0.75               | —   |      |
|        | Supply current<br>(Run 1 mode) | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 16 MHz, All peripherals enabled  | —   | 1.91               | —   | mA   |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 16 MHz, All peripherals disabled | —   | 0.77               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 8 MHz, All peripherals enabled   | —   | 1.29               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 8 MHz, All peripherals disabled  | —   | 0.69               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 4 MHz, All peripherals enabled   | —   | 0.8                | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 4 MHz, All peripherals disabled  | —   | 0.47               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 2 MHz, All peripherals enabled   | —   | 0.56               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 2 MHz, All peripherals disabled  | —   | 0.36               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 1 MHz, All peripherals enabled   | —   | 0.43               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 1 MHz, All peripherals disabled  | —   | 0.3                | —   |      |

| Symbol | Parameter                      | Conditions                                                                                              | Min | Typ <sup>(1)</sup> | Max | Unit |
|--------|--------------------------------|---------------------------------------------------------------------------------------------------------|-----|--------------------|-----|------|
|        | Supply current<br>(Run 2 mode) | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 2 MHz, All peripherals enabled  | —   | 0.45               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 2 MHz, All peripherals disabled | —   | 0.25               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 1 MHz, All peripherals enabled  | —   | 0.35               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, System clock = 16 MHz, HCLK = 1 MHz, All peripherals disabled | —   | 0.22               | —   |      |
|        | Supply current<br>(Sleep mode) | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 64 MHz, All peripherals enabled   | —   | 7.73               | —   | mA   |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 64 MHz, All peripherals disabled  | —   | 2.33               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 48 MHz, All peripherals enabled   | —   | 5.99               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 48 MHz, All peripherals disabled  | —   | 1.92               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 36 MHz, All peripherals enabled   | —   | 4.68               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 36 MHz, All peripherals disabled  | —   | 1.6                | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 24 MHz, All peripherals enabled   | —   | 3.38               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 24 MHz, All peripherals disabled  | —   | 1.29               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 16 MHz, All peripherals enabled   | —   | 2.51               | —   |      |
|        |                                | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 16 MHz, All peripherals disabled  | —   | 1.09               | —   |      |

| Symbol | Parameter                        | Conditions                                                                                            | Min | Typ <sup>(1)</sup> | Max | Unit |
|--------|----------------------------------|-------------------------------------------------------------------------------------------------------|-----|--------------------|-----|------|
|        |                                  | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 8 MHz, All peripherals enabled  | —   | 1.62               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 8 MHz, All peripherals disabled | —   | 0.86               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 4 MHz, All peripherals enabled  | —   | 1.18               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 4 MHz, All peripherals disabled | —   | 0.75               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 2 MHz, All peripherals enabled  | —   | 0.94               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, HXTAL = 8 MHz, CPU clock off, System clock = 2 MHz, All peripherals disabled | —   | 0.7                | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, HXTAL = 4 MHz, CPU clock off, System clock = 1 MHz, All peripherals enabled  | —   | 0.86               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, HXTAL = 4 MHz, CPU clock off, System clock = 1 MHz, All peripherals disabled | —   | 0.7                | —   |      |
|        | Supply current<br>(Sleep 1 mode) | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, CPU clock off, HCLK = 16 MHz, All peripherals enabled       | —   | 1.74               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, CPU clock off, HCLK = 16 MHz, All peripherals disabled      | —   | 0.52               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, CPU clock off, HCLK = 8 MHz, All peripherals enabled        | —   | 1.03               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, CPU clock off, HCLK = 8 MHz, All peripherals disabled       | —   | 0.38               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, CPU clock off, HCLK = 4 MHz, All peripherals enabled        | —   | 0.67               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, CPU clock off, HCLK = 4 MHz, All peripherals disabled       | —   | 0.31               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, CPU clock off, HCLK = 2 MHz, All peripherals enabled        | —   | 0.49               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, CPU clock off, HCLK = 2 MHz, All peripherals disabled       | —   | 0.28               | —   |      |
|        |                                  | V <sub>DD</sub> = 3.3 V, IRC16M = 16 MHz, CPU clock off, HCLK = 1 MHz, All peripherals enabled        | —   | 0.4                | —   |      |

| Symbol          | Parameter                             | Conditions                                                                                            | Min | Typ <sup>(1)</sup> | Max | Unit          |
|-----------------|---------------------------------------|-------------------------------------------------------------------------------------------------------|-----|--------------------|-----|---------------|
|                 |                                       | $V_{DD} = 3.3\text{ V}$ , IRC16M = 16 MHz, CPU clock off, HCLK = 1 MHz, All peripherals disabled      | —   | 0.26               | —   |               |
|                 | Supply current<br>(Sleep 2 mode)      | $V_{DD} = 3.3\text{ V}$ , IRC16M = 16 MHz, CPU clock off, HCLK = 2 MHz, All peripherals enabled       | —   | 0.43               | —   | mA            |
|                 |                                       | $V_{DD} = 3.3\text{ V}$ , IRC16M = 16 MHz, CPU clock off, HCLK = 2 MHz, All peripherals disabled      | —   | 0.22               | —   |               |
|                 |                                       | $V_{DD} = 3.3\text{ V}$ , IRC16M = 16 MHz, CPU clock off, HCLK = 1 MHz, All peripherals enabled       | —   | 0.34               | —   |               |
|                 |                                       | $V_{DD} = 3.3\text{ V}$ , IRC16M = 16 MHz, CPU clock off, HCLK = 1 MHz, All peripherals disabled      | —   | 0.21               | —   |               |
|                 | Supply current<br>(Deep-sleep mode)   | $V_{DD} = 3.3\text{ V}$ , NPLDO in Low driver mode, IRC32K off, RTC off                               | —   | 40.09              | —   | $\mu\text{A}$ |
|                 | Supply current<br>(Deep-sleep 1 mode) | $V_{DD} = 3.3\text{ V}$ , NPLDO off, LPLDO on, IRC32K off, RTC off                                    | —   | 3.144              | —   |               |
|                 | Supply current<br>(Deep-sleep 2 mode) | $V_{DD} = 3.3\text{ V}$ , NPLDO off, LPLDO on, COREOFF0/SRAM1/COREOFF1 off, IRC32K off, RTC off       | —   | 1.702              | —   |               |
|                 | Supply current<br>(Standby mode)      | $V_{DD} = 3.3\text{ V}$ , LXTAL off, IRC32K off, RTC off                                              | —   | 0.442              | —   |               |
| $I_{LXTAL+RTC}$ | LXTAL+RTC<br>current                  | $V_{DD}$ off, $V_{BAT} = 3.6\text{ V}$ , LXTAL on with external crystal, RTC on, Higher driving       | —   | 1.22               | —   | $\mu\text{A}$ |
|                 |                                       | $V_{DD}$ off, $V_{BAT} = 3.3\text{ V}$ , LXTAL on with external crystal, RTC on, Higher driving       | —   | 1.09               | —   | $\mu\text{A}$ |
|                 |                                       | $V_{DD}$ off, $V_{BAT} = 2.6\text{ V}$ , LXTAL on with external crystal, RTC on, Higher driving       | —   | 0.93               | —   | $\mu\text{A}$ |
|                 |                                       | $V_{DD}$ off, $V_{BAT} = 1.71\text{ V}$ , LXTAL on with external crystal, RTC on, Higher driving      | —   | 0.79               | —   | $\mu\text{A}$ |
|                 |                                       | $V_{DD}$ off, $V_{BAT} = 3.6\text{ V}$ , LXTAL on with external crystal, RTC on, Medium High driving  | —   | 1.09               | —   | $\mu\text{A}$ |
|                 |                                       | $V_{DD}$ off, $V_{BAT} = 3.3\text{ V}$ , LXTAL on with external crystal, RTC on, Medium High driving  | —   | 0.97               | —   | $\mu\text{A}$ |
|                 |                                       | $V_{DD}$ off, $V_{BAT} = 2.6\text{ V}$ , LXTAL on with external crystal, RTC on, Medium High driving  | —   | 0.8                | —   | $\mu\text{A}$ |
|                 |                                       | $V_{DD}$ off, $V_{BAT} = 1.71\text{ V}$ , LXTAL on with external crystal, RTC on, Medium High driving | —   | 0.66               | —   | $\mu\text{A}$ |
|                 |                                       | $V_{DD}$ off, $V_{BAT} = 3.6\text{ V}$ , LXTAL on with external crystal, RTC on, Medium Low driving   | —   | 0.92               | —   | $\mu\text{A}$ |
|                 |                                       |                                                                                                       |     |                    |     |               |

| Symbol | Parameter | Conditions                                                                                                | Min | Typ <sup>(1)</sup> | Max | Unit |
|--------|-----------|-----------------------------------------------------------------------------------------------------------|-----|--------------------|-----|------|
|        |           | V <sub>DD</sub> off, V <sub>BAT</sub> = 3.3V, LXTAL on with external crystal, RTC on, Medium Low driving  | —   | 0.79               | —   | μA   |
|        |           | V <sub>DD</sub> off, V <sub>BAT</sub> = 2.6V, LXTAL on with external crystal, RTC on, Medium Low driving  | —   | 0.63               | —   | μA   |
|        |           | V <sub>DD</sub> off, V <sub>BAT</sub> = 1.71V, LXTAL on with external crystal, RTC on, Medium Low driving | —   | 0.49               | —   | μA   |
|        |           | V <sub>DD</sub> off, V <sub>BAT</sub> = 3.6V, LXTAL on with external crystal, RTC on, Low driving         | —   | 0.87               | —   | μA   |
|        |           | V <sub>DD</sub> off, V <sub>BAT</sub> = 3.3V, LXTAL on with external crystal, RTC on, Low driving         | —   | 0.74               | —   | μA   |
|        |           | V <sub>DD</sub> off, V <sub>BAT</sub> = 2.6V, LXTAL on with external crystal, RTC on, Low driving         | —   | 0.57               | —   | μA   |
|        |           | V <sub>DD</sub> off, V <sub>BAT</sub> = 1.71V, LXTAL on with external crystal, RTC on, Low driving        | —   | 0.43               | —   | μA   |

- (1) Based on characterization, not tested in production.
- (2) When analog peripheral blocks such as ADCs, HXTAL, LXTAL, IRC8M, or IRC32K are ON, an additional power consumption should be considered.
- (3) The system clock 36MHZ (inclusive) to 64MHZ (inclusive) adopts FMC\_WAIT\_STATE\_1, the system clock 24MHZ (inclusive) to 1MHZ (inclusive) adopts FMC\_WAIT\_STATE\_0.
- (4) All GPIOs are configured as analog mode except standby mode.

**Figure 4-2. Typical supply current consumption in Run mode**

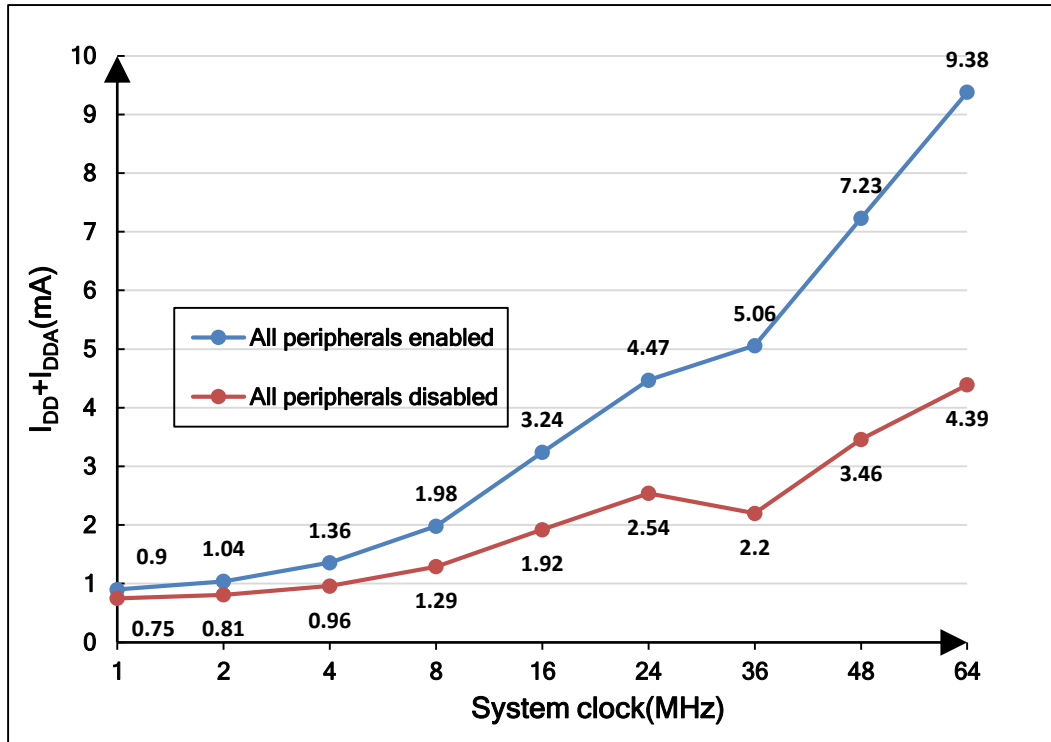
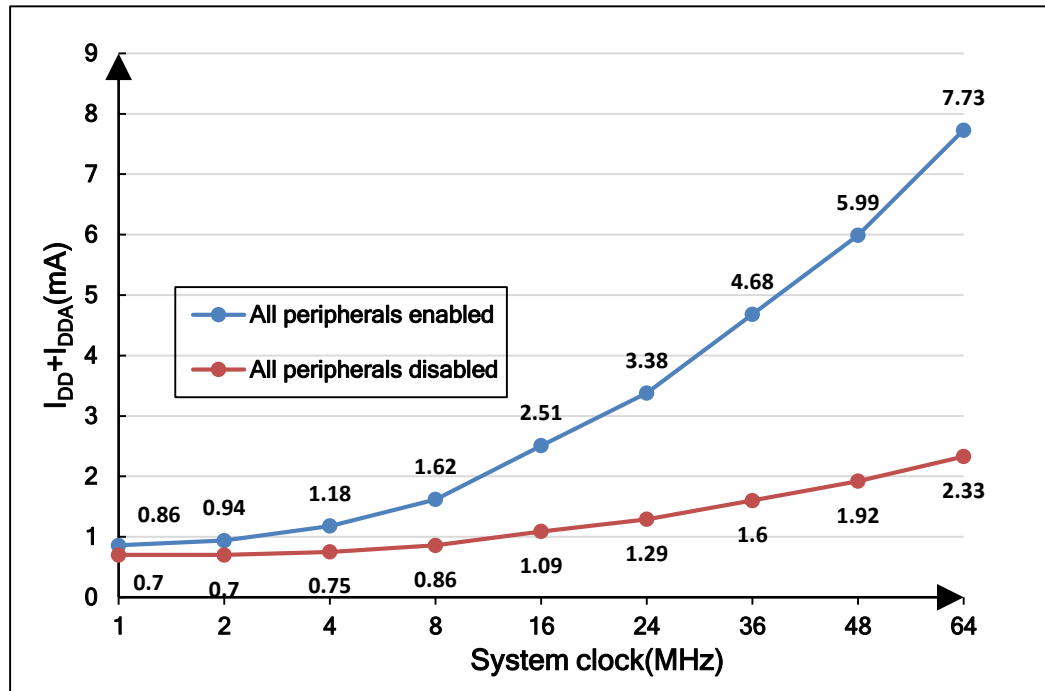


Figure 4-3. Typical supply current consumption in Sleep mode



## 4.4 EMC characteristics

EMS (electromagnetic susceptibility) includes ESD (Electrostatic discharge, positive and negative) and FTB (Burst of Fast Transient voltage, positive and negative) testing result is given in [Table 4-8. EMS characteristics](#), based on the EMS levels and classes compliant with IEC 61000 series standard.

**Table 4-8. EMS characteristics<sup>(1)</sup>**

| Symbol           | Parameter                                                                                                  | Conditions                                                                                  | Level/Class |
|------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------|
| V <sub>ESD</sub> | Voltage applied to all device pins to induce a functional disturbance                                      | V <sub>DD</sub> = 3.3 V, LQFP64,<br>f <sub>HCLK</sub> = 64 MHz<br>conforms to IEC 61000-4-2 | 3A          |
| V <sub>FTB</sub> | Fast transient voltage burst applied to induce a functional disturbance through 100 pF on VDD and VSS pins | V <sub>DD</sub> = 3.3 V, LQFP64,<br>f <sub>HCLK</sub> = 64 MHz<br>conforms to IEC 61000-4-4 | 4A          |

(1) Based on characterization, not tested in production.

EMI (Electromagnetic Interference) emission test result is given in the [Table 4-9. EMI characteristics](#)<sup>(1)</sup>, The electromagnetic field emitted by the device are monitored while an application, executing EEMBC code, is running. The test is compliant with SAE J1752-3:2017 standard which specifies the test board and the pin loading.

**Table 4-9. EMI characteristics<sup>(1)</sup>**

| Symbol           | Parameter  | Conditions                                                                                                         | Tested frequency band | Max vs. [f <sub>HXTAL</sub> /f <sub>HCLK</sub> ] | Unit |
|------------------|------------|--------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------|------|
|                  |            |                                                                                                                    |                       | 8/64 MHz                                         |      |
| S <sub>EMI</sub> | Peak level | V <sub>DD</sub> = 3.6 V, T <sub>A</sub> = +25 °C, LQFP64, f <sub>HCLK</sub> = 64 MHz, conforms to SAE J1752-3:2017 | 0.15 MHz to 30 MHz    | -11.89                                           | dBμV |
|                  |            |                                                                                                                    | 30 MHz to 130 MHz     | -1.54                                            |      |
|                  |            |                                                                                                                    | 130 MHz to 1 GHz      | 3.05                                             |      |

(1) Based on characterization, not tested in production.

## 4.5 Power supply supervisor characteristics

**Table 4-10. Power supply supervisor characteristics<sup>(1)</sup>**

| Symbol               | Parameter                      | Conditions                    | Min  | Typ  | Max  | Unit |
|----------------------|--------------------------------|-------------------------------|------|------|------|------|
| $V_{LVD}^{(1)}$      | Low Voltage Detector Threshold | LVDT[2:0] = 000, rising edge  | —    | 2.15 | —    | V    |
|                      |                                | LVDT[2:0] = 000, falling edge | —    | 2.05 | —    | V    |
|                      |                                | LVDT[2:0] = 001, rising edge  | —    | 2.30 | —    | V    |
|                      |                                | LVDT[2:0] = 001, falling edge | —    | 2.20 | —    | V    |
|                      |                                | LVDT[2:0] = 010, rising edge  | —    | 2.45 | —    | V    |
|                      |                                | LVDT[2:0] = 010, falling edge | —    | 2.35 | —    | V    |
|                      |                                | LVDT[2:0] = 011, rising edge  | —    | 2.60 | —    | V    |
|                      |                                | LVDT[2:0] = 011, falling edge | —    | 2.50 | —    | V    |
|                      |                                | LVDT[2:0] = 100, rising edge  | —    | 2.75 | —    | V    |
|                      |                                | LVDT[2:0] = 100, falling edge | —    | 2.65 | —    | V    |
|                      |                                | LVDT[2:0] = 101, rising edge  | —    | 2.90 | —    | V    |
|                      |                                | LVDT[2:0] = 101, falling edge | —    | 2.80 | —    | V    |
|                      |                                | LVDT[2:0] = 110, rising edge  | —    | 3.00 | —    | V    |
|                      |                                | LVDT[2:0] = 110, falling edge | —    | 2.90 | —    | V    |
| $V_{LVDhyst}^{(2)}$  | LVD hysteresis                 | —                             | —    | 100  | —    | mV   |
| $V_{BOR0}$           | Brown-out reset threshold 0    | rising edge                   | —    | 1.60 | —    | V    |
|                      |                                | falling edge                  | —    | 1.56 | —    |      |
| $V_{BOR1}$           | Brown-out reset threshold 1    | rising edge                   | —    | 2.10 | —    | V    |
|                      |                                | falling edge                  | —    | 2.00 | —    |      |
| $V_{BOR2}$           | Brown-out reset threshold 2    | rising edge                   | —    | 2.30 | —    | V    |
|                      |                                | falling edge                  | —    | 2.20 | —    |      |
| $V_{BOR3}$           | Brown-out reset threshold 3    | rising edge                   | —    | 2.60 | —    | V    |
|                      |                                | falling edge                  | —    | 2.50 | —    |      |
| $V_{BOR4}$           | Brown-out reset threshold 4    | rising edge                   | —    | 2.90 | —    | V    |
|                      |                                | falling edge                  | —    | 2.80 | —    |      |
| $V_{POR}^{(1)}$      | Power on reset threshold       | —                             | 1.56 | 1.60 | 1.63 | V    |
| $V_{PDR}^{(1)}$      | Power down reset threshold     |                               | 1.52 | 1.56 | 1.59 | V    |
| $V_{PDRhyst}^{(2)}$  | PDR hysteresis                 |                               | —    | 40   | —    | mV   |
| $t_{RSTTEMPO}^{(2)}$ | Reset temporization            |                               | —    | 550  | —    | us   |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

## 4.6 Electrical sensitivity

The device is strained in order to determine its performance in terms of electrical sensitivity. Electrostatic discharges (ESD) are applied directly to the pins of the sample. Static latch-up (LU) test is based on the two measurement methods.

**Table 4-11. ESD characteristics<sup>(1)</sup>**

| Symbol         | Parameter                                             | Conditions  | Min | Typ | Max  | Unit |
|----------------|-------------------------------------------------------|-------------|-----|-----|------|------|
| $V_{ESD(HBM)}$ | Electrostatic discharge voltage (human body model)    | JS-001-2017 | —   | —   | 2000 | V    |
| $V_{ESD(CDM)}$ | Electrostatic discharge voltage (charge device model) | JS-002-2018 | —   | —   | 500  | V    |

(1) Based on characterization, not tested in production.

**Table 4-12. Static latch-up characteristics<sup>(1)</sup>**

| Symbol | Parameter                 | Conditions | Min | Typ | Max  | Unit |
|--------|---------------------------|------------|-----|-----|------|------|
| LU     | I-test                    | JESD78     | —   | —   | ±200 | mA   |
|        | $V_{supply}$ over voltage |            | —   | —   | 5.4  | V    |

(1) Based on characterization, not tested in production.

## 4.7 External clock characteristics

**Table 4-13. High speed external clock (HXTAL) generated from a crystal/ceramic characteristics**

| Symbol                 | Parameter                                            | Conditions              | Min | Typ  | Max | Unit |
|------------------------|------------------------------------------------------|-------------------------|-----|------|-----|------|
| $f_{HXTAL}^{(1)}$      | Crystal or ceramic frequency                         | $V_{DD} = 3.3\text{ V}$ | 4   | 8    | 48  | MHz  |
| $R_F^{(2)}$            | Feedback resistor                                    | $V_{DD} = 3.3\text{ V}$ | —   | 400  | —   | kΩ   |
| $C_{HXTAL}^{(2)(3)}$   | Recommended matching capacitance on OSCIN and OSCOUT | —                       | —   | 20   | 30  | pF   |
| $D_{ucy(HXTAL)}^{(2)}$ | Crystal or ceramic duty cycle                        | —                       | 30  | 50   | 70  | %    |
| $g_m^{(2)}$            | Oscillator transconductance                          | Startup                 | —   | 20   | —   | mA/V |
| $I_{DD(HXTAL)}^{(1)}$  | Crystal or ceramic operating current                 | $V_{DD} = 3.3\text{ V}$ | —   | 0.32 | —   | mA   |
| $t_{SUHXTAL}^{(1)}$    | Crystal or ceramic startup time                      | $V_{DD} = 3.3\text{ V}$ | —   | 1.27 | —   | ms   |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

(3)  $C_{HXTAL1} = C_{HXTAL2} = 2 \times (C_{LOAD} - C_S)$ . For  $C_{HXTAL1}$  and  $C_{HXTAL2}$ , it is recommended matching capacitance on OSCIN and OSCOUT. For  $C_{LOAD}$ , it is crystal/ceramic load capacitance, provided by the crystal or ceramic manufacturer. For  $C_S$ , it is PCB and MCU pin stray capacitance.

**Table 4-14. High speed external user clock characteristics (HXTAL in bypass mode)**

| Symbol                               | Parameter                                     | Conditions                      | Min                 | Typ | Max                 | Unit |
|--------------------------------------|-----------------------------------------------|---------------------------------|---------------------|-----|---------------------|------|
| $f_{\text{HXTAL\_ext}}^{(1)}$        | External clock source or oscillator frequency | $V_{\text{DD}} = 3.3 \text{ V}$ | 1                   | 8   | 50                  | MHz  |
| $V_{\text{HXTALH}}^{(2)}$            | OSCIN input pin high level voltage            | $V_{\text{DD}} = 3.3 \text{ V}$ | $0.7 V_{\text{DD}}$ | —   | $V_{\text{DD}}$     | V    |
| $V_{\text{HXTALL}}^{(2)}$            | OSCIN input pin low level voltage             |                                 | $V_{\text{SS}}$     | —   | $0.3 V_{\text{DD}}$ |      |
| $t_{\text{H/L(HXTAL)}}^{(2)}$        | OSCIN high or low time                        | —                               | 5                   | —   | —                   | ns   |
| $t_{\text{R/F(HXTAL)}}^{(2)}$        | OSCIN rise or fall time                       | —                               | —                   | —   | 10                  |      |
| $C_{\text{IN}}^{(2)}$                | OSCIN input capacitance                       | —                               | —                   | 5   | —                   | pF   |
| $\text{Duty}_{(\text{HXTAL})}^{(2)}$ | Duty cycle                                    | —                               | 30                  | 50  | 70                  | %    |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

**Table 4-15. Low speed external clock (LXTAL) generated from a crystal/ceramic characteristics**

| Symbol                               | Parameter                                                | Conditions                      | Min | Typ    | Max | Unit            |
|--------------------------------------|----------------------------------------------------------|---------------------------------|-----|--------|-----|-----------------|
| $f_{\text{LXTAL}}^{(1)}$             | Crystal or ceramic frequency                             | $V_{\text{DD}} = 3.3 \text{ V}$ | —   | 32.768 | —   | kHz             |
| $C_{\text{LXTAL}}^{(2)(3)}$          | Recommended matching capacitance on OSC32IN and OSC32OUT | —                               | —   | 10     | —   | pF              |
| $\text{Duty}_{(\text{LXTAL})}^{(2)}$ | Crystal or ceramic duty cycle                            | —                               | 30  | —      | 70  | %               |
| $g_{\text{m}}^{(2)}$                 | Oscillator transconductance                              | Lower driving capability        | —   | 3.6    | —   | $\mu\text{A/V}$ |
|                                      |                                                          | Medium low driving capability   | —   | 4.8    | —   |                 |
|                                      |                                                          | Medium high driving capability  | —   | 8.4    | —   |                 |
|                                      |                                                          | Higher driving capability       | —   | 10.8   | —   |                 |
| $I_{\text{DDLXTAL}}^{(1)}$           | Crystal or ceramic operating current                     | Lower driving capability        | —   | 332    | —   | nA              |
|                                      |                                                          | Medium low driving capability   | —   | 392    | —   |                 |
|                                      |                                                          | Medium high driving capability  | —   | 562    | —   |                 |
|                                      |                                                          | Higher driving capability       | —   | 692    | —   |                 |
| $t_{\text{SULXTAL}}^{(1)(4)}$        | Crystal or ceramic startup time                          | $V_{\text{DD}} = 3.3 \text{ V}$ | —   | 0.32   | —   | s               |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

(3)  $C_{\text{LXTAL1}} = C_{\text{LXTAL2}} = 2 \times (C_{\text{LOAD}} - C_{\text{S}})$ , For  $C_{\text{LXTAL1}}$  and  $C_{\text{LXTAL2}}$ , it is recommended matching capacitance on OSC32IN and OSC32OUT. For  $C_{\text{LOAD}}$ , it is crystal/ceramic load capacitance, provided by the crystal or ceramic manufacturer. For  $C_{\text{S}}$ , it is PCB and MCU pin stray capacitance.

(4)  $t_{\text{SULXTAL}}$  is the startup time measured from the moment it is enabled (by software) to the 32.768 kHz oscillator stabilization flags is SET. This value varies significantly with the crystal manufacturer.

**Table 4-16. Low speed external user clock characteristics (LXTAL in bypass mode)**

| Symbol                 | Parameter                                     | Conditions              | Min          | Typ    | Max          | Unit |
|------------------------|-----------------------------------------------|-------------------------|--------------|--------|--------------|------|
| $f_{LXTAL\_ext}^{(1)}$ | External clock source or oscillator frequency | $V_{DD} = 3.3\text{ V}$ | —            | 32.768 | 1000         | kHz  |
| $V_{LXTALH}^{(2)}$     | OSC32IN input pin high level voltage          | $V_{DD} = 3.3\text{ V}$ | $0.7 V_{DD}$ | —      | $V_{DD}$     | V    |
| $V_{LXTALL}^{(2)}$     | OSC32IN input pin low level voltage           |                         | $V_{SS}$     | —      | $0.3 V_{DD}$ |      |
| $t_{H/L(LXTAL)}^{(2)}$ | OSC32IN high or low time                      | —                       | 250          | —      | —            | ns   |
| $t_{R/F(LXTAL)}^{(2)}$ | OSC32IN rise or fall time                     | —                       | —            | —      | 50           |      |
| $C_{IN}^{(2)}$         | OSC32IN input capacitance                     | —                       | —            | 5      | —            | pF   |
| $Duty_{(LXTAL)}^{(2)}$ | Duty cycle                                    | —                       | 30           | 50     | 70           | %    |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

## 4.8 Internal clock characteristics

**Table 4-17. High speed internal clock (IRC16M) characteristics**

| Symbol               | Parameter                                                               | Conditions                                                                                                                       | Min  | Typ                        | Max  | Unit          |
|----------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------|----------------------------|------|---------------|
| $f_{IRC16M}$         | High Speed Internal Oscillator (IRC16M) frequency                       | $V_{DD} = V_{DDA} = 3.3\text{ V}$                                                                                                | —    | 16                         | —    | MHz           |
| $ACC_{IRC16M}$       | IRC16M oscillator Frequency accuracy, Factory-trimmed                   | $V_{DD} = V_{DDA} = 3.3\text{ V}$ ,<br>$T_A = -40\text{ }^{\circ}\text{C} \sim +85\text{ }^{\circ}\text{C}$ for grade 6 devices  | —    | -1.5 to 1.5 <sup>(1)</sup> | —    | %             |
|                      |                                                                         | $V_{DD} = V_{DDA} = 3.3\text{ V}$ ,<br>$T_A = -40\text{ }^{\circ}\text{C} \sim +105\text{ }^{\circ}\text{C}$ for grade 7 devices | —    | -2 to 2 <sup>(1)</sup>     | —    | %             |
|                      |                                                                         | $V_{DD} = V_{DDA} = 3.3\text{ V}$                                                                                                | -1.0 | —                          | +1.0 | %             |
|                      | IRC16M oscillator Frequency accuracy, User trimming step <sup>(1)</sup> | —                                                                                                                                | —    | 0.3 <sup>(1)</sup>         | —    | %             |
| $D_{IRC16M}^{(2)}$   | IRC16M oscillator duty cycle                                            | $V_{DD} = V_{DDA} = 3.3\text{ V}$                                                                                                | 45   | 50                         | 55   | %             |
| $I_{DDIRC16M}^{(1)}$ | IRC16M oscillator operating current                                     | $V_{DD} = V_{DDA} = 3.3\text{ V}$ ,<br>$f_{IRC16M} = 16\text{ MHz}$                                                              | —    | 110                        | —    | $\mu\text{A}$ |
| $t_{SUIRC16M}^{(1)}$ | IRC16M oscillator startup time                                          | $V_{DD} = V_{DDA} = 3.3\text{ V}$ ,<br>$f_{IRC16M} = 16\text{ MHz}$                                                              | —    | 0.75                       | —    | $\mu\text{s}$ |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

**Table 4-18. Low speed internal clock (IRC32K) characteristics**

| Symbol                | Parameter                                        | Conditions                                                                                                    | Min | Typ          | Max | Unit          |
|-----------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----|--------------|-----|---------------|
| $f_{IRC32K}$          | Low Speed Internal oscillator (IRC32K) frequency | $V_{DD} = V_{DDA} = 3.3\text{ V}$ ,<br>$T_A = -40 \sim 85\text{ }^{\circ}\text{C}^{(2)}$ for grade 6 devices  | —   | 31.7 to 32.3 | —   | kHz           |
|                       |                                                  | $V_{DD} = V_{DDA} = 3.3\text{ V}$ ,<br>$T_A = -40 \sim 105\text{ }^{\circ}\text{C}^{(2)}$ for grade 7 devices | —   | 31.4 to 32.9 | —   |               |
|                       |                                                  | $V_{DD} = V_{DDA} = 3.3\text{ V}$                                                                             | 30  | —            | 35  |               |
| $I_{DDAIRC32K}^{(2)}$ | IRC32K oscillator operating current              | $V_{DD} = V_{DDA} = 3.3\text{ V}$                                                                             | —   | 160          | —   | nA            |
| $t_{SUIRC32K}^{(2)}$  | IRC32K oscillator startup time                   | $V_{DD} = V_{DDA} = 3.3\text{ V}$                                                                             | —   | 40           | —   | $\mu\text{s}$ |

(1) Guaranteed by design, not tested in production.

(2) Based on characterization, not tested in production.

**Table 4-19. High speed internal clock (IRC48M) characteristics**

| Symbol                | Parameter                                                | Conditions                                                                                                                       | Min  | Typ                          | Max  | Unit          |
|-----------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------|------------------------------|------|---------------|
| $f_{IRC48M}$          | High Speed Internal Oscillator (IRC48M) frequency        | $V_{DD} = V_{DDA} = 3.3\text{ V}$                                                                                                | —    | 48                           | —    | MHz           |
| $ACC_{IRC48M}$        | IRC48M oscillator Frequency accuracy, Factory-trimmed    | $V_{DD} = V_{DDA} = 3.3\text{ V}$ ,<br>$T_A = -40\text{ }^{\circ}\text{C} \sim +85\text{ }^{\circ}\text{C}$ for grade 6 devices  | —    | -3.3 to -0.25 <sup>(1)</sup> | —    | %             |
|                       |                                                          | $V_{DD} = V_{DDA} = 3.3\text{ V}$ ,<br>$T_A = -40\text{ }^{\circ}\text{C} \sim +105\text{ }^{\circ}\text{C}$ for grade 7 devices | —    | -3 to 3 <sup>(1)</sup>       | —    | %             |
|                       |                                                          | $V_{DD} = V_{DDA} = 3.3\text{ V}$                                                                                                | -2.0 | —                            | +2.0 | %             |
|                       | IRC48M oscillator Frequency accuracy, User trimming step | —                                                                                                                                | —    | 0.12                         | —    | %             |
| $D_{IRC48M}^{(2)}$    | IRC48M oscillator duty cycle                             | $V_{DD} = V_{DDA} = 3.3\text{ V}$                                                                                                | 45   | 50                           | 55   | %             |
| $I_{DDAIRC48M}^{(1)}$ | IRC48M oscillator operating current                      | $V_{DD} = V_{DDA} = 3.3\text{ V}$ ,<br>$f_{IRC48M} = 48\text{ MHz}$                                                              | —    | 327                          | —    | $\mu\text{A}$ |
| $t_{SUIRC48M}^{(1)}$  | IRC48M oscillator startup time                           | $V_{DD} = V_{DDA} = 3.3\text{ V}$ ,<br>$f_{IRC48M} = 48\text{ MHz}$                                                              | —    | 1.8                          | —    | $\mu\text{s}$ |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

## 4.9 PLL characteristics

Table 4-20. PLL characteristics

| Symbol               | Parameter                            | Conditions        | Min | Typ | Max | Unit |
|----------------------|--------------------------------------|-------------------|-----|-----|-----|------|
| $f_{PLLIN}^{(1)}$    | PLL input clock frequency            | —                 | 2   | —   | 16  | MHz  |
| $f_{PLLOUT}^{(2)}$   | PLL output clock frequency           | —                 | 16  | —   | 64  | MHz  |
| $f_{VCO}^{(2)}$      | PLL VCO output clock frequency       | —                 | —   | —   | 64  | MHz  |
| $t_{LOCK}^{(2)}$     | PLL lock time                        | —                 | —   | —   | 200 | μs   |
| $I_{DD}^{(1)}$       | Current consumption on $V_{DD}$      | VCO freq = 64 MHz | —   | 400 | —   | μA   |
| $Jitter_{PLL}^{(3)}$ | Cycle to cycle Jitter (rms)          | System clock      | —   | 120 | —   | ps   |
|                      | Cycle to cycle Jitter (peak to peak) |                   | —   | 900 | —   |      |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

(3) Value given with main PLL running.

## 4.10 Memory characteristics

Table 4-21. Flash memory characteristics

| Symbol             | Parameter                                                             | Conditions                         | Min | Typ  | Max | Unit    |
|--------------------|-----------------------------------------------------------------------|------------------------------------|-----|------|-----|---------|
| $PE_{CYC}^{(1)}$   | Number of guaranteed program /erase cycles before failure (Endurance) | $T_A$ range <sup>(3)</sup>         | 10  | —    | —   | kcycles |
| $t_{RET}^{(1)}$    | Data retention time                                                   | 10k cycles at $T_A$ <sup>(3)</sup> | 10  | —    | —   | years   |
| $t_{PROG}^{(2)}$   | Word programming time                                                 | $T_A$ range <sup>(3)</sup>         | —   | 37.5 | —   | μs      |
| $t_{ERASE}^{(2)}$  | Page erase time                                                       |                                    | —   | 11   | —   | ms      |
| $t_{MERASE}^{(2)}$ | Mass erase time                                                       |                                    | —   | 12   | —   | ms      |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

(3) For grade 6 devices,  $T_A$  range= -40° C ~ +85°C. For grade 7 devices,  $T_A$  range= -40° C ~ +105°C.

## 4.11 NRST pin characteristics

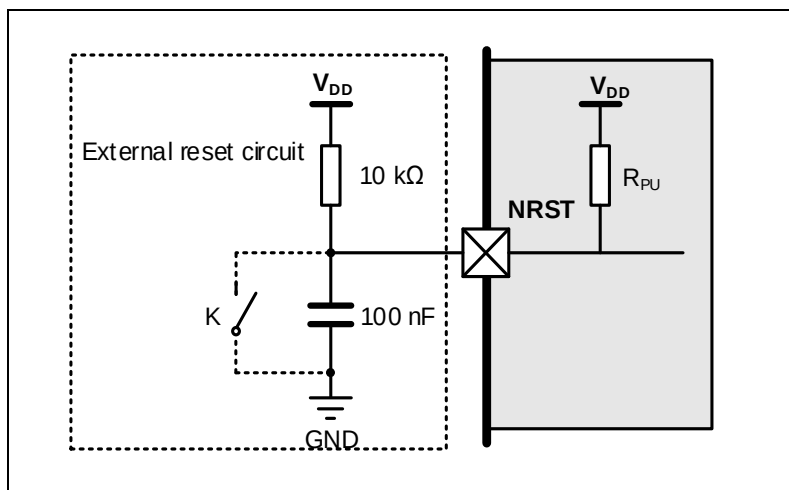
Table 4-22. NRST pin characteristics

| Symbol               | Parameter                          | Conditions                                               | Min           | Typ | Max            | Unit       |
|----------------------|------------------------------------|----------------------------------------------------------|---------------|-----|----------------|------------|
| $V_{IL(NRST)}^{(1)}$ | NRST Input low level voltage       | $1.71\text{ V} \leq V_{DD} = V_{DDA} \leq 3.63\text{ V}$ | -0.5          | —   | $0.35 V_{DD}$  | V          |
| $V_{IH(NRST)}^{(1)}$ | NRST Input high level voltage      |                                                          | $0.65 V_{DD}$ | —   | $V_{DD} + 0.5$ | V          |
| $V_{hyst}^{(1)}$     | Schmidt trigger Voltage hysteresis |                                                          | —             | 400 | —              | mV         |
| $R_{pu}^{(2)}$       | Pull-up equivalent resistor        | —                                                        | —             | 40  | —              | k $\Omega$ |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

**Figure 4-4. Recommended external NRST pin circuit<sup>(1)</sup>**



(1) Unless the voltage on NRST pin go below  $V_{IL(NRST)}$  level, the device would not generate a reliable reset.

## 4.12 VREF buffer characteristics

Table 4-23. VREF buffer characteristics

| Symbol              | Parameter                | Conditions                                                                                                                 | Min   | Typ  | Max   | Unit                      |
|---------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------|-------|------|-------|---------------------------|
| $V_{DDA}^{(1)}$     | Analog Supply Voltage    | —                                                                                                                          | 2.7   | 3.3  | 3.63  | V                         |
| $V_{REF}$           | Output Reference Voltage | $V_{DDA} = 3.3\text{ V}$                                                                                                   | 2.49  | 2.50 | 2.51  | V                         |
|                     |                          | $V_{DDA} = 3.3\text{ V}$ ,<br>$T_A = -40\text{ }^{\circ}\text{C} \sim +85\text{ }^{\circ}\text{C}$<br>for grade 6 devices  | 2.455 | 2.50 | 2.545 |                           |
|                     |                          | $V_{DDA} = 3.3\text{ V}$ ,<br>$T_A = -40\text{ }^{\circ}\text{C} \sim +105\text{ }^{\circ}\text{C}$<br>for grade 7 devices | 2.45  | 2.50 | 2.55  |                           |
| $PSRR^{(1)}$        | Power Supply Rejection   | DC ( $I_O = 0$ )                                                                                                           | —     | 57   | —     | dB                        |
|                     |                          | DC ( $I_O = 200\text{ }\mu\text{A}$ )                                                                                      | —     | 57   | —     |                           |
| $T_{SU}^{(1)}$      | Setup Time               | $C_L = 1\text{ }\mu\text{F} + 10\text{ nF}$                                                                                | —     | —    | 200   | $\mu\text{s}$             |
| $I_{LOAD\_R}^{(1)}$ | Load Regulation          | $I_{LOAD}$ from 0 to 200 $\mu\text{A}$                                                                                     | —     | 5    | —     | $\mu\text{V}/\mu\text{A}$ |
| $C_{LOAD}^{(1)}$    | Load Capacitor           | —                                                                                                                          | —     | 1    | —     | $\mu\text{F}$             |
| $TRIM^{(1)}$        | Trim Step                | —                                                                                                                          | —     | 3    | —     | mV                        |

(1). Guaranteed by design, not tested in production.

(2). Based on characterization, not tested in production.

## 4.13 GPIO characteristics

Table 4-24. I/O port DC characteristics<sup>(1)(3)</sup>

| Symbol                                      | Parameter                                                             | Conditions                                               | Min          | Typ  | Max          | Unit |
|---------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------|--------------|------|--------------|------|
| $V_{IL}$                                    | Standard IO Low level input voltage                                   | $1.71\text{ V} \leq V_{DD} = V_{DDA} \leq 3.63\text{ V}$ | —            | —    | $0.3 V_{DD}$ | V    |
|                                             | 5V-tolerant IO Low level input voltage                                | $1.71\text{ V} \leq V_{DD} = V_{DDA} \leq 3.63\text{ V}$ | —            | —    | $0.3 V_{DD}$ |      |
| $V_{IH}$                                    | Standard IO High level input voltage                                  | $1.71\text{ V} \leq V_{DD} = V_{DDA} \leq 3.63\text{ V}$ | $0.7 V_{DD}$ | —    | —            |      |
|                                             | 5 V-tolerant IO High level input voltage                              | $1.71\text{ V} \leq V_{DD} = V_{DDA} \leq 3.63\text{ V}$ | $0.7 V_{DD}$ | —    | —            |      |
| $V_{OL}$<br>( $I_{O\_speed}=50\text{MHz}$ ) | Low level output voltage for an IO Pin<br>( $I_{IO} = +8\text{ mA}$ ) | $V_{DD} = 1.71\text{ V}$                                 | —            | 0.26 | —            | V    |
|                                             |                                                                       | $V_{DD} = 3.3\text{ V}$                                  | —            | 0.13 | —            |      |
|                                             |                                                                       | $V_{DD} = 3.6\text{ V}$                                  | —            | 0.13 | —            |      |
|                                             |                                                                       | ( $I_{IO} = +10\text{ mA}$ )<br>$V_{DD} = 1.71\text{ V}$ | —            | 0.20 | —            |      |

| Symbol                                   | Parameter                                                            | Conditions               | Min | Typ  | Max | Unit       |
|------------------------------------------|----------------------------------------------------------------------|--------------------------|-----|------|-----|------------|
|                                          | Low level output voltage for an IO Pin ( $I_{IO} = +20\text{ mA}$ )  | $V_{DD} = 3.3\text{ V}$  | —   | 0.33 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.6\text{ V}$  | —   | 0.32 | —   |            |
| $V_{OH}$<br>( $IO\_speed=50\text{MHz}$ ) | High level output voltage for an IO Pin ( $I_{IO} = +8\text{ mA}$ )  | $V_{DD} = 1.71\text{ V}$ | —   | 1.46 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.3\text{ V}$  | —   | 3.15 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.6\text{ V}$  | —   | 3.45 | —   |            |
|                                          | High level output voltage for an IO Pin ( $I_{IO} = +20\text{ mA}$ ) | $V_{DD} = 1.71\text{ V}$ | —   | 1.38 | —   | V          |
|                                          |                                                                      | $V_{DD} = 3.3\text{ V}$  | —   | 2.91 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.6\text{ V}$  | —   | 3.22 | —   |            |
| $V_{OL}$<br>( $IO\_speed=10\text{MHz}$ ) | Low level output voltage for an IO Pin ( $I_{IO} = +8\text{ mA}$ )   | $V_{DD} = 1.71\text{ V}$ | —   | 0.31 | —   | V          |
|                                          |                                                                      | $V_{DD} = 3.3\text{ V}$  | —   | 0.36 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.6\text{ V}$  | —   | 0.35 | —   |            |
|                                          | Low level output voltage for an IO Pin ( $I_{IO} = +15\text{ mA}$ )  | $V_{DD} = 1.71\text{ V}$ | —   | —    | —   |            |
|                                          |                                                                      | $V_{DD} = 3.3\text{ V}$  | —   | 0.73 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.6\text{ V}$  | —   | 0.70 | —   |            |
| $V_{OH}$<br>( $IO\_speed=10\text{MHz}$ ) | High level output voltage for an IO Pin ( $I_{IO} = +8\text{ mA}$ )  | $V_{DD} = 1.71\text{ V}$ | —   | 1.33 | —   | V          |
|                                          |                                                                      | $V_{DD} = 3.3\text{ V}$  | —   | 2.87 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.6\text{ V}$  | —   | 3.19 | —   |            |
|                                          | High level output voltage for an IO Pin ( $I_{IO} = +15\text{ mA}$ ) | $V_{DD} = 1.71\text{ V}$ | —   | —    | —   |            |
|                                          |                                                                      | $V_{DD} = 3.3\text{ V}$  | —   | 2.42 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.6\text{ V}$  | —   | 2.78 | —   |            |
| $V_{OL}$<br>( $IO\_speed=2\text{MHz}$ )  | Low level output voltage for an IO Pin ( $I_{IO} = +4\text{ mA}$ )   | $V_{DD} = 1.71\text{ V}$ | —   | 0.32 | —   | V          |
|                                          |                                                                      | $V_{DD} = 3.3\text{ V}$  | —   | 0.55 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.6\text{ V}$  | —   | 0.53 | —   |            |
| $V_{OH}$<br>( $IO\_speed=2\text{MHz}$ )  | High level output voltage for an IO Pin ( $I_{IO} = +4\text{ mA}$ )  | $V_{DD} = 1.71\text{ V}$ | —   | 1.32 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.3\text{ V}$  | —   | 2.65 | —   |            |
|                                          |                                                                      | $V_{DD} = 3.6\text{ V}$  | —   | 2.99 | —   |            |
| $R_{PU}^{(2)}$                           | Internal pull-up resistor                                            | —                        | —   | 40   | —   | k $\Omega$ |
| $R_{PD}^{(2)}$                           | Internal pull-down resistor                                          | —                        | —   | 40   | —   | k $\Omega$ |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

(3) All pins except PC13 / PC14 / PC15. Since PC13 to PC15 are supplied through the Power Switch, which can only be obtained by a small current, the speed of GPIOs PC13 to PC15 should not exceed 2 MHz when they are in output mode(maximum load: 30 pF).

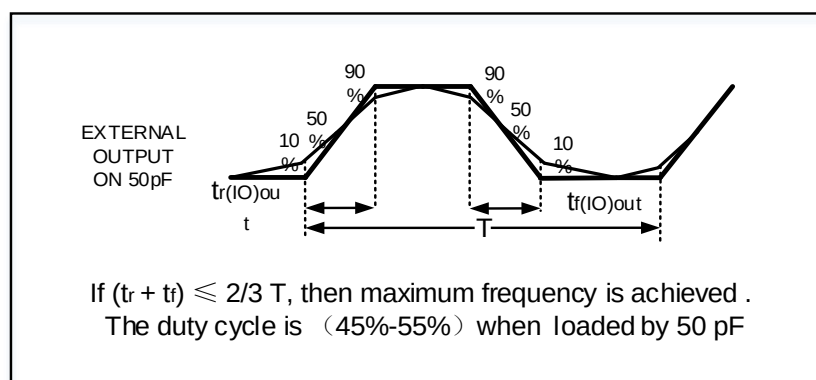
**Table 4-25. I/O port AC characteristics<sup>(1)</sup>**

| GPIOx_OSPD[1:0] bit value <sup>(2)</sup>           | Parameter         | Conditions                                                        | Max | Unit |
|----------------------------------------------------|-------------------|-------------------------------------------------------------------|-----|------|
| GPIOx_OSPD->OSPDy[1:0] = X0<br>(IO_Speed = 2 MHz)  | Maximum frequency | $1.71 \leq V_{DD} \leq 3.63 \text{ V}$ ,<br>$C_L = 10 \text{ pF}$ | 6   | MHz  |
|                                                    |                   | $1.71 \leq V_{DD} \leq 3.63 \text{ V}$ ,<br>$C_L = 30 \text{ pF}$ | 5   |      |
|                                                    |                   | $1.71 \leq V_{DD} \leq 3.63 \text{ V}$ , $C_L = 50 \text{ pF}$    | 4   |      |
| GPIOx_OSPD->OSPDy[1:0] = 01<br>(IO_Speed = 10 MHz) | Maximum frequency | $1.71 \leq V_{DD} \leq 3.63 \text{ V}$ ,<br>$C_L = 10 \text{ pF}$ | 17  | MHz  |
|                                                    |                   | $1.71 \leq V_{DD} \leq 3.63 \text{ V}$ ,<br>$C_L = 30 \text{ pF}$ | 14  |      |
|                                                    |                   | $1.71 \leq V_{DD} \leq 3.63 \text{ V}$ ,<br>$C_L = 50 \text{ pF}$ | 12  |      |
| GPIOx_OSPD->OSPDy[1:0] = 11<br>(IO_Speed = 50 MHz) | Maximum frequency | $1.71 \leq V_{DD} \leq 3.63 \text{ V}$ ,<br>$C_L = 10 \text{ pF}$ | 81  | MHz  |
|                                                    |                   | $1.71 \leq V_{DD} \leq 3.63 \text{ V}$ ,<br>$C_L = 30 \text{ pF}$ | 72  |      |
|                                                    |                   | $1.71 \leq V_{DD} \leq 3.63 \text{ V}$ ,<br>$C_L = 50 \text{ pF}$ | 60  |      |

(1) Based on characterization, not tested in production.

(2) The I/O speed is configured using the GPIOx\_OSPD->OSPDy [1:0] bits. Refer to the GD32L233 user manual which is selected to set the GPIO port output speed.

**Figure 4-5. I/O port AC characteristics definition**



## 4.14 ADC characteristics

**Table 4-26. ADC characteristics**

| Symbol           | Parameter               | Conditions | Min   | Typ | Max       | Unit |
|------------------|-------------------------|------------|-------|-----|-----------|------|
| $V_{DDA}^{(1)}$  | Operating voltage       | —          | 1.8   | 3.3 | 3.6       | V    |
| $V_{IN}^{(1)}$   | ADC input voltage range | —          | 0     | —   | $V_{REF}$ | V    |
| $V_{REFP}^{(2)}$ | Reference Voltage       | —          | 1.8   | 3.3 | $V_{DDA}$ | V    |
| $f_{ADC}^{(1)}$  | ADC clock               | —          | 0.125 | —   | 16        | MHz  |
| $f_s^{(1)}$      | Sampling rate           | 12-bit     | 0.008 | —   | 1.067     |      |

| Symbol                           | Parameter                                      | Conditions                      | Min   | Typ  | Max              | Unit                   |
|----------------------------------|------------------------------------------------|---------------------------------|-------|------|------------------|------------------------|
|                                  |                                                | 10-bit                          | 0.009 | —    | 1.23             | MSP<br>S               |
|                                  |                                                | 8-bit                           | 0.011 | —    | 1.45             |                        |
|                                  |                                                | 6-bit                           | 0.013 | —    | 1.78             |                        |
| V <sub>AIN</sub> <sup>(1)</sup>  | Analog input voltage                           | 16 external; 4 internal         | 0     | —    | V <sub>DDA</sub> | V                      |
| R <sub>AIN</sub> <sup>(2)</sup>  | External input impedance                       | See <a href="#">Equation 1</a>  | —     | —    | 513.6            | kΩ                     |
| R <sub>ADC</sub> <sup>(2)</sup>  | Input sampling switch resistance               | —                               | —     | —    | 0.5              | kΩ                     |
| C <sub>ADC</sub> <sup>(2)</sup>  | Input sampling capacitance                     | No pin/pad capacitance included | —     | —    | 3                | pF                     |
| t <sub>CAL</sub> <sup>(2)</sup>  | Calibration time                               | f <sub>ADC</sub> = 16 MHz       | —     | 13.4 | —                | μs                     |
| t <sub>s</sub> <sup>(2)</sup>    | Sampling time                                  | f <sub>ADC</sub> = 16 MHz       | 0.156 | —    | 14.97            | μs                     |
| t <sub>CONV</sub> <sup>(2)</sup> | Total conversion time(including sampling time) | 12-bit                          | —     | 15   | —                | 1/<br>f <sub>ADC</sub> |
|                                  |                                                | 10-bit                          | —     | 13   | —                |                        |
|                                  |                                                | 8-bit                           | —     | 11   | —                |                        |
|                                  |                                                | 6-bit                           | —     | 9    | —                |                        |
| I <sub>DDA(ADC)</sub>            | ADC consumption from V <sub>DDA</sub>          | f <sub>s</sub> = 1M             | —     | 133  | —                | μA                     |
|                                  |                                                | f <sub>s</sub> = 0.5M           | —     | 77   | —                |                        |
|                                  |                                                | f <sub>s</sub> = 10k            | —     | 17.5 | —                |                        |
| I <sub>DDV(ADC)</sub>            | ADC consumption from V <sub>REFP</sub>         | f <sub>s</sub> = 1M             | —     | 14.7 | —                | μA                     |
|                                  |                                                | f <sub>s</sub> = 0.5M           | —     | 7.6  | —                |                        |
|                                  |                                                | f <sub>s</sub> = 10k            | —     | 0.4  | —                |                        |
| t <sub>SU</sub> <sup>(2)</sup>   | Startup time                                   | —                               | —     | 5    | —                | μs                     |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

**Equation 1:** R<sub>AIN</sub> max formula 
$$R_{AIN} < \frac{T_s}{f_{ADC} \cdot C_{ADC} \cdot \ln(2^{N+2})} - R_{ADC}$$

The formula above ([Equation 1](#)) is used to determine the maximum external impedance allowed for an error below 1/4 of LSB. Here N = 12 (from 12-bit resolution).

**Table 4-27. ADC R<sub>AIN</sub> max for f<sub>ADC</sub> = 16 MHz<sup>(1)</sup>**

| T <sub>s</sub> (cycles) | t <sub>s</sub> (μs) | R <sub>AIN</sub> max (kΩ) |
|-------------------------|---------------------|---------------------------|
| 2.5                     | 0.16                | 4.8                       |
| 7.5                     | 0.47                | 15.6                      |
| 13.5                    | 0.85                | 28.4                      |
| 28.5                    | 1.79                | 60.6                      |
| 41.5                    | 2.60                | 88.5                      |
| 55.5                    | 3.47                | 118.6                     |
| 71.5                    | 4.47                | 153.0                     |
| 239.5                   | 14.97               | 513.6                     |

(1) Based on characterization, not tested in production.

**Table 4-28. ADC dynamic accuracy at  $f_{\text{ADC}} = 16 \text{ MHz}^{(1)}$**

| Symbol | Parameter                            | Test conditions                                    | Min  | Typ  | Max | Unit |
|--------|--------------------------------------|----------------------------------------------------|------|------|-----|------|
| ENOB   | Effective number of bits             | $f_{\text{ADC}} = 16 \text{ MHz}$                  | 10.8 | 11.2 | —   | bits |
| SNDR   | Signal-to-noise and distortion ratio | $V_{\text{DDA}} = V_{\text{REF+}} = 3.3 \text{ V}$ | 66.7 | 69.1 | —   | dB   |
| SNR    | Signal-to-noise ratio                | Input Frequency = 20 kHz                           | 66.9 | 69.3 | —   |      |
| THD    | Total harmonic distortion            |                                                    | —    | -82  | -78 |      |

(1) Based on characterization, not tested in production.

**Table 4-29. ADC static accuracy at  $f_{\text{ADC}} = 16 \text{ MHz}^{(1)}$**

| Symbol | Parameter                    | Test conditions                                                                         | Typ       | Max       | Unit |
|--------|------------------------------|-----------------------------------------------------------------------------------------|-----------|-----------|------|
| Offset | Offset error                 | $f_{\text{ADC}} = 16 \text{ MHz}$<br>$V_{\text{DDA}} = V_{\text{REF+}} = 3.3 \text{ V}$ | —         | —         | LSB  |
| DNL    | Differential linearity error |                                                                                         | $\pm 0.6$ | $\pm 1$   |      |
| INL    | Integral linearity error     |                                                                                         | $\pm 0.8$ | $\pm 1.5$ |      |

(1) Based on characterization, not tested in production.

**Table 4-30. ADC dynamic accuracy at  $f_{\text{ADC}} = 16 \text{ MHz}^{(1)}$**

| Symbol | Parameter                            | Test conditions                                                    | Min  | Typ  | Max | Unit |
|--------|--------------------------------------|--------------------------------------------------------------------|------|------|-----|------|
| ENOB   | Effective number of bits             | $f_{\text{ADC}} = 16 \text{ MHz}$                                  | 10.7 | 11.2 | —   | bits |
| SNDR   | Signal-to-noise and distortion ratio | $V_{\text{DDA}} = 3.3 \text{ V}$ $V_{\text{REF+}} = 2.5 \text{ V}$ | 66.2 | 69.1 | —   | dB   |
| SNR    | Signal-to-noise ratio                | Input Frequency = 20 kHz                                           | 66.4 | 69.3 | —   |      |
| THD    | Total harmonic distortion            |                                                                    | —    | -82  | -78 |      |

(1) Based on characterization, not tested in production.

**Table 4-31. ADC static accuracy at  $f_{ADC} = 16 \text{ MHz}^{(1)}$**

| Symbol | Parameter                    | Test conditions                                                                    | Typ       | Max       | Unit |
|--------|------------------------------|------------------------------------------------------------------------------------|-----------|-----------|------|
| Offset | Offset error                 | $f_{ADC} = 16 \text{ MHz}$ ,<br>$V_{DDA} = 3.3\text{V}$ $V_{REF+} = 2.5 \text{ V}$ | —         | —         | LSB  |
| DNL    | Differential linearity error |                                                                                    | $\pm 0.6$ | $\pm 1$   |      |
| INL    | Integral linearity error     |                                                                                    | $\pm 0.8$ | $\pm 1.5$ |      |

(1) Based on characterization, not tested in production.

**Table 4-32. ADC dynamic accuracy at  $f_{ADC} = 16 \text{ MHz}^{(1)}$**

| Symbol | Parameter                            | Test conditions                                                                                  | Min  | Typ  | Max | Unit |
|--------|--------------------------------------|--------------------------------------------------------------------------------------------------|------|------|-----|------|
| ENOB   | Effective number of bits             | $f_{ADC} = 16 \text{ MHz}$ ,<br>$V_{DDA} = V_{REF+} = 1.8 \text{ V}$<br>Input Frequency = 20 kHz | 10.5 | 10.8 | —   | bits |
| SNDR   | Signal-to-noise and distortion ratio |                                                                                                  | 64.9 | 66.7 | —   | dB   |
| SNR    | Signal-to-noise ratio                |                                                                                                  | 65.1 | 66.9 | —   |      |
| THD    | Total harmonic distortion            |                                                                                                  | —    | -71  | -68 |      |

(1) Based on characterization, not tested in production.

**Table 4-33. ADC static accuracy at  $f_{ADC} = 16 \text{ MHz}^{(1)}$**

| Symbol | Parameter                    | Test conditions                                                      | Typ       | Max       | Unit |
|--------|------------------------------|----------------------------------------------------------------------|-----------|-----------|------|
| Offset | Offset error                 | $f_{ADC} = 16 \text{ MHz}$ ,<br>$V_{DDA} = V_{REF+} = 1.8 \text{ V}$ | —         | —         | LSB  |
| DNL    | Differential linearity error |                                                                      | $\pm 0.8$ | $\pm 1$   |      |
| INL    | Integral linearity error     |                                                                      | $\pm 1$   | $\pm 1.5$ |      |

(1) Based on characterization, not tested in production.

## 4.15 DAC characteristics

**Table 4-34. DAC characteristics**

| Symbol                     | Parameter                                 | Conditions                                     | Min  | Typ       | Max                     | Unit          |
|----------------------------|-------------------------------------------|------------------------------------------------|------|-----------|-------------------------|---------------|
| $V_{DDA}^{(1)}$            | Operating voltage                         | —                                              | 1.71 | 3.3       | 3.63                    | V             |
| $V_{REFP}^{(2)}$           | Positive Reference Voltage                | —                                              | 1.71 | —         | $V_{DDA}$               | V             |
| $V_{REFN}^{(2)}$           | Negative Reference Voltage                | —                                              | —    | $V_{SSA}$ | —                       | V             |
| $R_{LOAD}^{(2)}$           | Resistive load                            | Resistive load with buffer ON                  | 5    | —         | —                       | k $\Omega$    |
| $R_o^{(2)}$                | Impedance output                          | Impedance output with buffer OFF               | —    | —         | 15                      | k $\Omega$    |
| $C_{LOAD}^{(2)}$           | Capacitive load                           | Capacitive load with buffer ON                 | —    | —         | 50                      | pF            |
| DAC_OUT min <sup>(2)</sup> | Lower DAC_OUT voltage                     | Lower DAC_OUT voltage with buffer ON           | 0.2  | —         | —                       | V             |
|                            |                                           | Lower DAC_OUT voltage with buffer OFF          | 0.5  | —         | —                       | mV            |
| DAC_OUT max <sup>(2)</sup> | Higher DAC_OUT voltage                    | Higher DAC_OUT voltage with buffer ON          | —    | —         | $V_{DDA} - 0.2$         | V             |
|                            |                                           | Higher DAC_OUT voltage with buffer OFF         | —    | —         | $V_{DDA} - 1\text{LSB}$ | V             |
| $I_{DDA}^{(1)}$            | DAC current consumption in quiescent mode | With no load, middle code(0x800) on the input, | —    | 400       | —                       | $\mu\text{A}$ |

| Symbol                     | Parameter                                                                     | Conditions                                                                    | Min | Typ       | Max       | Unit          |
|----------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----|-----------|-----------|---------------|
|                            |                                                                               | $V_{REFP} = 3.3\text{ V}$                                                     |     |           |           |               |
| $I_{DDVREF+}^{(1)}$        | DAC current consumption in quiescent mode                                     | With no load, middle code(0x800) on the input,<br>$V_{REFP} = 3.3\text{ V}$   | —   | 114       | —         | $\mu\text{A}$ |
| $DNL^{(1)}$                | Differential non linearity                                                    | 10-bit configuration                                                          | —   | —         | $\pm 0.5$ | LSB           |
|                            |                                                                               | 12-bit configuration                                                          | —   | —         | $\pm 2$   |               |
| $INL^{(1)}$                | Integral non linearity                                                        | 10-bit configuration                                                          | —   | —         | $\pm 1$   | LSB           |
|                            |                                                                               | 12-bit configuration                                                          | —   | —         | $\pm 4$   |               |
| Offset <sup>(1)</sup>      | Offset error                                                                  | DAC in 12-bit mode                                                            | —   | —         | $\pm 12$  | LSB           |
| GE <sup>(1)</sup>          | Gain error                                                                    | DAC in 12-bit mode                                                            | —   | $\pm 0.5$ | —         | %             |
| $T_{\text{setting}}^{(1)}$ | Settling time                                                                 | $C_{\text{LOAD}} \leq 50\text{ pF}$ , $R_{\text{LOAD}} \geq 5\text{ k}\Omega$ | —   | 0.5       | —         | $\mu\text{s}$ |
| $T_{\text{wakeup}}^{(2)}$  | Wakeup from off state                                                         | —                                                                             | —   | 5         | —         | $\mu\text{s}$ |
| Update rate <sup>(2)</sup> | Max frequency for a correct DAC_OUT change from code i to $i \pm 1\text{LSB}$ | $C_{\text{LOAD}} \leq 50\text{ pF}$ , $R_{\text{LOAD}} \geq 5\text{ k}\Omega$ | —   | —         | 4         | MS/s          |
| PSRR <sup>(2)</sup>        | Power supply rejection ratio(to $V_{\text{DDA}}$ )                            | No $R_{\text{Load}}$ , $C_{\text{LOAD}}=50\text{ pF}$                         | —   | -80       | —         | dB            |

(1) Based on characterization, not tested in production.

(2) Guaranteed by design, not tested in production.

## 4.16 Temperature sensor characteristics

**Table 4-35. Temperature sensor characteristics**

| Symbol                 | Parameter                                                           | Conditions                 | Min  | Typ         | Max | Unit                         |
|------------------------|---------------------------------------------------------------------|----------------------------|------|-------------|-----|------------------------------|
| $V_{\text{OFF}}$       | Uncalibrated Offset                                                 | $T_A = 30^\circ\text{C}$   | —    | 1022.8      | —   | mV                           |
| $E_{\text{OFF}}^{(1)}$ | Uncalibrated Offset Error                                           | $T_A = 30^\circ\text{C}$   | —    | 2           | —   | mV                           |
| M                      | Slope                                                               | —                          | —    | 3.3         | —   | mV/ $^\circ\text{C}$         |
| $E_M^{(1)}$            | Slope Error                                                         | —                          | —    | 30          | —   | $\mu\text{V}/^\circ\text{C}$ |
| LIN <sup>(3)</sup>     | Linearity                                                           | $T_A$ range <sup>(4)</sup> | —    | -0.4 to 1.2 | —   | $^\circ\text{C}$             |
| $t_{\text{ON}}$        | Turn-on Time                                                        | —                          | —    | —           | —   | $\mu\text{s}$                |
| ETOT <sup>(2)(3)</sup> | Temp Sensor Error Using Typical Slope and Factory-Calibrated Offset | $T_A$ range <sup>(4)</sup> | -3.5 | —           | 4.7 | $^\circ\text{C}$             |

(1) Represents one standard deviation from the mean.

(2) The factory-calibrated offset value is stored in the read-only area of flash in locations 0x1FFFF7F8.

(3) Based on characterization, not tested in production.

(4) For grade 6 devices,  $T_A$  range=  $-40^\circ\text{C} \sim +85^\circ\text{C}$ . For grade 7 devices,  $T_A$  range=  $-40^\circ\text{C} \sim +105^\circ\text{C}$ .

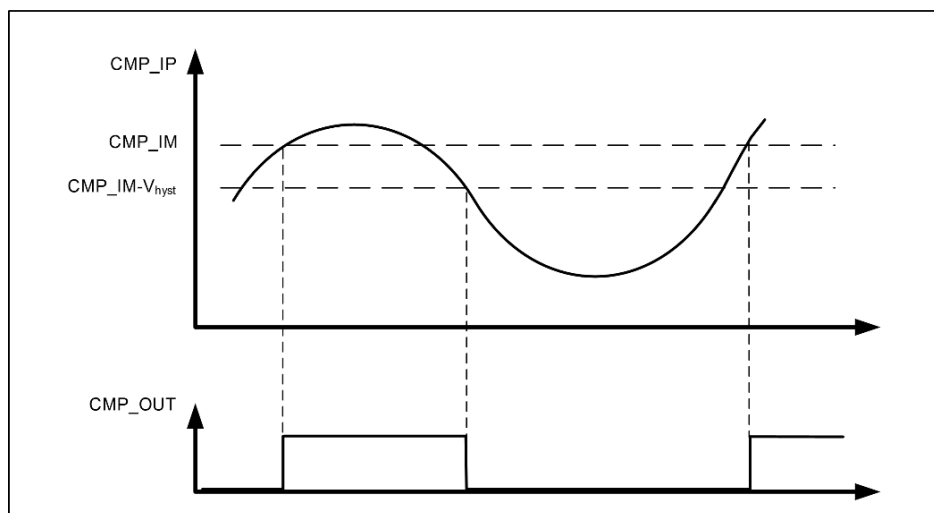
## 4.17 Comparators characteristics

**Table 4-36. CMP characteristics<sup>(1)</sup>**

| Symbol              | Parameter                                               | Conditions             | Min  | Typ     | Max       | Unit    |
|---------------------|---------------------------------------------------------|------------------------|------|---------|-----------|---------|
| $V_{DDA}$           | Operating voltage                                       | —                      | 1.71 | 3.3     | 3.63      | V       |
| $V_{IN}$            | Input voltage range                                     | —                      | 0    | —       | $V_{DDA}$ | V       |
| $V_{BG}$            | Scaler input voltage                                    | —                      | —    | 0.8     | —         | V       |
| $V_{SC}$            | Scaler offset voltage                                   | —                      | —    | $\pm 5$ | —         | mV      |
| $I_{DDA(SCALER)}$   | Scaler static consumption from $V_{DDA}$                | BEN=0 (bridge disable) | —    | 200     | —         | nA      |
|                     |                                                         | BEN=1 (bridge enable)  | —    | 0.8     | —         | $\mu A$ |
| $t_{START\_SCALER}$ | Scaler startup time                                     | —                      | —    | 100     | —         | $\mu s$ |
| $t_D$               | Propagation delay for 200 mV step with 100 mV overdrive | Ultra low power mode   | —    | 3.63    | —         | $\mu s$ |
|                     |                                                         | Medium power mode      | —    | 0.18    | —         | $\mu s$ |
|                     |                                                         | High speed power mode  | —    | 55      | —         | ns      |
| $I_{DD}$            | Current consumption                                     | Ultra low power mode   | —    | 0.5     | —         | $\mu A$ |
|                     |                                                         | Medium power mode      | —    | 4.7     | —         |         |
|                     |                                                         | High speed power mode  | —    | 47      | —         |         |
| $V_{offset}$        | Offset error                                            | —                      | —    | $\pm 5$ | —         | mV      |
| $V_{hyst}$          | Hysteresis Voltage                                      | No Hysteresis          | —    | 0       | —         | mV      |
|                     |                                                         | Low Hysteresis         | —    | 8       | —         |         |
|                     |                                                         | Medium Hysteresis      | —    | 16      | —         |         |
|                     |                                                         | High Hysteresis        | —    | 32      | —         |         |

(1) Based on characterization, not tested in production.

**Figure 4-6. CMP hysteresis**



## 4.18 TIMER characteristics

**Table 4-37. TIMER characteristics** <sup>(1)</sup>

| Symbol           | Parameter                                                   | Conditions                      | Min    | Max                  | Unit            |
|------------------|-------------------------------------------------------------|---------------------------------|--------|----------------------|-----------------|
| $t_{res}$        | Timer resolution time                                       | —                               | 1      | —                    | $t_{TIMERxCLK}$ |
|                  |                                                             | $f_{TIMERxCLK} = 64\text{ MHz}$ | 15.6   | —                    | ns              |
| $f_{EXT}$        | Timer external clock frequency                              | —                               | 0      | $f_{TIMERxCLK}/2$    | MHz             |
|                  |                                                             | $f_{TIMERxCLK} = 64\text{ MHz}$ | 0      | 32                   | MHz             |
| RES              | Timer resolution                                            | —                               | —      | 16                   | bit             |
| $t_{COUNTER}$    | 16-bit counter clock period when internal clock is selected | —                               | 1      | 65536                | $t_{TIMERxCLK}$ |
|                  |                                                             | $f_{TIMERxCLK} = 64\text{ MHz}$ | 0.0156 | 1024                 | $\mu\text{s}$   |
| $t_{MAX\_COUNT}$ | Maximum possible count                                      | —                               | —      | $65536 \times 65536$ | $t_{TIMERxCLK}$ |
|                  |                                                             | $f_{TIMERxCLK} = 64\text{ MHz}$ | —      | 67.11                | s               |

(1) Guaranteed by design, not tested in production.

## 4.19 SLCD controller characteristics

**Table 4-38. SLCD controller characteristics<sup>(1)</sup>**

| Symbol                           | Parameter                                                         | Conditions                                | Min | Typ                  | Max  | Unit |
|----------------------------------|-------------------------------------------------------------------|-------------------------------------------|-----|----------------------|------|------|
| V <sub>SLCD</sub>                | SLCD external voltage                                             |                                           | —   | —                    | 3.63 | V    |
| V <sub>SLCD0</sub>               | SLCD internal reference voltage 0                                 |                                           | —   | 2.65                 | —    |      |
| V <sub>SLCD1</sub>               | SLCD internal reference voltage 1                                 |                                           | —   | 2.80                 | —    |      |
| V <sub>SLCD2</sub>               | SLCD internal reference voltage 2                                 |                                           | —   | 2.92                 | —    |      |
| V <sub>SLCD3</sub>               | SLCD internal reference voltage 3                                 |                                           | —   | 3.08                 | —    |      |
| V <sub>SLCD4</sub>               | SLCD internal reference voltage 4                                 |                                           | —   | 3.23                 | —    |      |
| V <sub>SLCD5</sub>               | SLCD internal reference voltage 5                                 |                                           | —   | 3.37                 | —    |      |
| V <sub>SLCD6</sub>               | SLCD internal reference voltage 6                                 |                                           | —   | 3.52                 | —    |      |
| V <sub>SLCD7</sub>               | SLCD internal reference voltage 7                                 |                                           | —   | 3.67                 | —    |      |
| C <sub>ext</sub>                 | V <sub>SLCD</sub> external capacitance                            | Buffer OFF (VODEN=0 is SLCD_CTL register) | 0.2 | —                    | 2    | uF   |
|                                  |                                                                   | Buffer ON (VODEN=1 is SLCD_CTL register)  | 1   | —                    | 2    |      |
| I <sub>SLCD</sub> <sup>(2)</sup> | Supply current from VDD at V <sub>DD</sub> = 2.2 V                | Buffer OFF (VODEN=0, HDEN=0, PULSE=0)     | —   | 3.8                  | —    | uA   |
|                                  | Supply current from VDD at V <sub>DD</sub> = 3.0 V                | Buffer OFF (VODEN=0, HDEN=0, PULSE=0)     | —   | 4.0                  | —    |      |
| I <sub>VSLCD</sub>               | Supply current from V <sub>SLCD</sub> (V <sub>SLCD</sub> = 3.0 V) | Buffer OFF (VODEN = 0, HDEN=0, PULSE=0)   | —   | 0.55                 | —    | uA   |
|                                  |                                                                   | Buffer OFF (VODEN = 0, HDEN=1, PULSE=0)   | —   | 13                   | —    |      |
|                                  |                                                                   | Buffer ON (VODEN = 1, 1/2 Bias)           | —   | 0.68                 | —    |      |
|                                  |                                                                   | Buffer ON (VODEN = 1, 1/3 Bias)           | —   | 0.95                 | —    |      |
|                                  |                                                                   | Buffer ON (VODEN = 1, 1/4 Bias)           | —   | 0.95                 | —    |      |
| R <sub>HN</sub>                  | Total High Resistor value for Low drive resistive network         |                                           | —   | 6                    | —    | MΩ   |
| R <sub>LN</sub>                  | Total Low Resistor value for High drive resistive network         |                                           | —   | 230                  | —    | kΩ   |
| V <sub>44</sub>                  | Segment/Common highest level voltage                              |                                           | —   | V <sub>SLCD</sub>    | —    | V    |
| V <sub>34</sub>                  | Segment/Common 3/4 level voltage                                  |                                           | —   | 3/4V <sub>SLCD</sub> | —    |      |
| V <sub>23</sub>                  | Segment/Common 2/3 level voltage                                  |                                           | —   | 2/3V <sub>SLCD</sub> | —    |      |
| V <sub>12</sub>                  | Segment/Common 1/2 level voltage                                  |                                           | —   | 1/2V <sub>SLCD</sub> | —    |      |
| V <sub>13</sub>                  | Segment/Common 1/3 level voltage                                  |                                           | —   | 1/3V <sub>SLCD</sub> | —    |      |
| V <sub>14</sub>                  | Segment/Common 1/4 level voltage                                  |                                           | —   | 1/4V <sub>SLCD</sub> | —    |      |
| V <sub>0</sub>                   | Segment/Common lowest level voltage                               |                                           | —   | 0                    | —    |      |

- (1) Guaranteed by design, not tested in production.
- (2) SLCD enabled with 3V internal step-up active, 1/8 duty, 1/4 bias, ps=1, div=0, flcdclk=32.768KHz, all pixels active, no SLCD, after VSLCD setup.

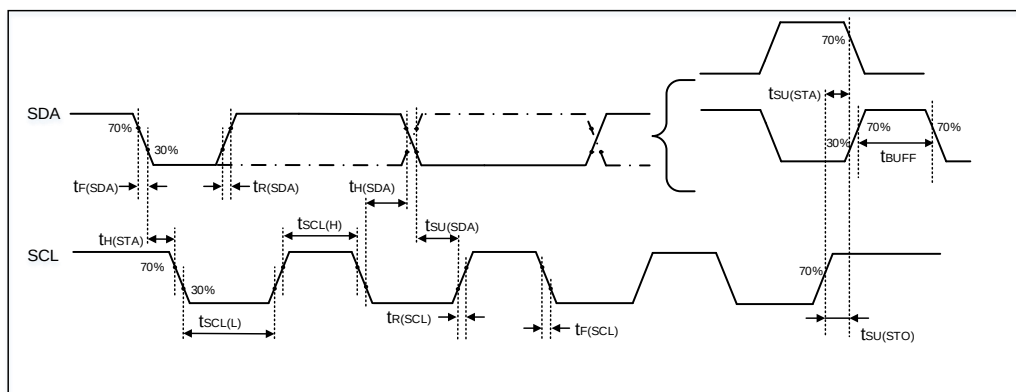
## 4.20 I2C characteristics

**Table 4-39. I2C characteristics<sup>(1)(2)</sup>**

| Symbol           | Parameter                               | Condi<br>tions | Standard<br>mode |      | Fast mode |     | Fast mode<br>plus |     | Unit    |
|------------------|-----------------------------------------|----------------|------------------|------|-----------|-----|-------------------|-----|---------|
|                  |                                         |                | Min              | Max  | Min       | Max | Min               | Max |         |
| $t_{SCL(H)}$     | SCL clock high time                     | —              | 4.0              | —    | 0.6       | —   | 0.2               | —   | $\mu s$ |
| $t_{SCL(L)}$     | SCL clock low time                      | —              | 4.7              | —    | 1.3       | —   | 0.5               | —   | $\mu s$ |
| $t_{SU(SDA)}$    | SDA setup time                          | —              | 250              | —    | 100       | —   | 50                | —   | ns      |
| $t_{H(SDA)}$     | SDA data hold time                      | —              | 0 <sup>(3)</sup> | 3450 | 0         | 900 | 0                 | 450 | ns      |
| $t_{R(SDA/SCL)}$ | SDA and SCL rise time                   | —              | —                | 1000 | —         | 300 | —                 | 120 | ns      |
| $t_{F(SDA/SCL)}$ | SDA and SCL fall time                   | —              | —                | 300  | —         | 300 | —                 | 120 | ns      |
| $t_{H(STA)}$     | Start condition hold time               | —              | 4.0              | —    | 0.6       | —   | 0.26              | —   | $\mu s$ |
| $t_{SU(STA)}$    | Repeated Start condition setup time     | —              | 4.7              | —    | 0.6       | —   | 0.26              | —   | $\mu s$ |
| $t_{SU(STO)}$    | Stop condition setup time               | —              | 4.0              | —    | 0.6       | —   | 0.26              | —   | $\mu s$ |
| $t_{BUFF}$       | Stop to Start condition time (bus free) | —              | 4.7              | —    | 1.3       | —   | 0.5               | —   | $\mu s$ |

- (1) Guaranteed by design, not tested in production.
- (2) To ensure the standard mode I2C frequency,  $f_{PCLK1}$  must be at least 2 MHz. To ensure the fast mode I2C frequency,  $f_{PCLK1}$  must be at least 4 MHz. To ensure the fast mode plus I2C frequency,  $f_{PCLK1}$  must be at least a multiple of 10 MHz.
- (3) The device should provide a data hold time of 300 ns at least in order to bridge the undefined region of the falling edge of SCL.

Figure 4-7. I2C bus timing diagram



## 4.21 SPI characteristics

Table 4-40. Standard SPI characteristics<sup>(1)</sup>

| Symbol                 | Parameter                | Conditions                                      | Min | Typ | Max | Unit |
|------------------------|--------------------------|-------------------------------------------------|-----|-----|-----|------|
| $f_{SCK}$              | SCK clock frequency      | —                                               | —   | —   | 16  | MHz  |
| $t_{SCK(H)}$           | SCK clock high time      | Master mode, $f_{PCLKx} = 64$ MHz,<br>presc = 4 | —   | 20  | —   | ns   |
| $t_{SCK(L)}$           | SCK clock low time       | Master mode, $f_{PCLKx} = 64$ MHz,<br>presc = 4 | —   | 20  | —   | ns   |
| <b>SPI master mode</b> |                          |                                                 |     |     |     |      |
| $t_{V(MO)}$            | Data output valid time   | —                                               | —   | —   | 10  | ns   |
| $t_{SU(MI)}$           | Data input setup time    | —                                               | 1   | —   | —   | ns   |
| $t_{H(MI)}$            | Data input hold time     | —                                               | 0   | —   | —   | ns   |
| <b>SPI slave mode</b>  |                          |                                                 |     |     |     |      |
| $t_{SU(NSS)}$          | NSS enable setup time    | —                                               | 0   | —   | —   | ns   |
| $t_{H(NSS)}$           | NSS enable hold time     | —                                               | 1   | —   | —   | ns   |
| $t_{A(SO)}$            | Data output access time  | —                                               | —   | 8   | —   | ns   |
| $t_{DIS(SO)}$          | Data output disable time | —                                               | —   | 9   | —   | ns   |
| $t_{V(SO)}$            | Data output valid time   | —                                               | —   | 9   | —   | ns   |
| $t_{SU(SI)}$           | Data input setup time    | —                                               | 0   | —   | —   | ns   |
| $t_{H(SI)}$            | Data input hold time     | —                                               | 1   | —   | —   | ns   |

(1) Based on characterization, not tested in production.

Figure 4-8. SPI timing diagram - master mode

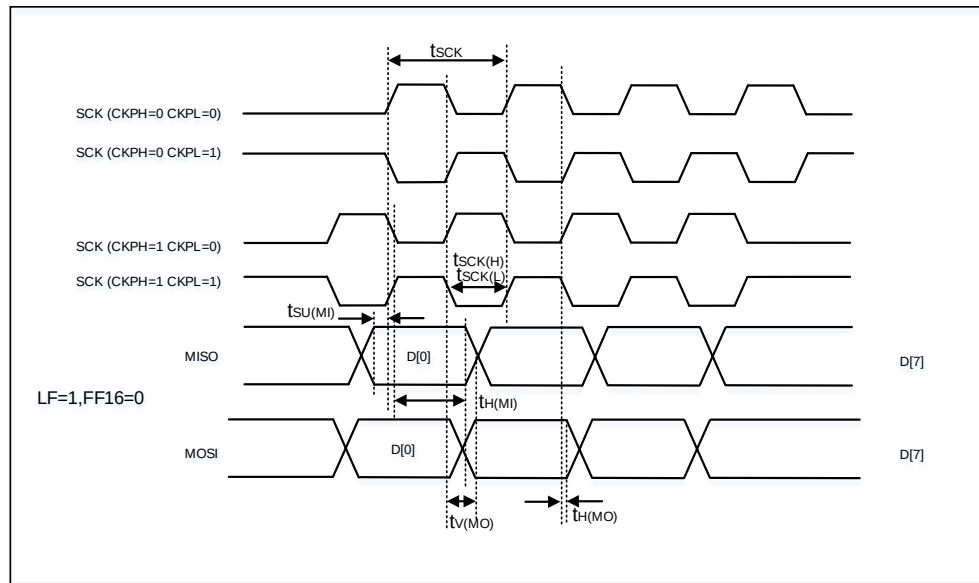
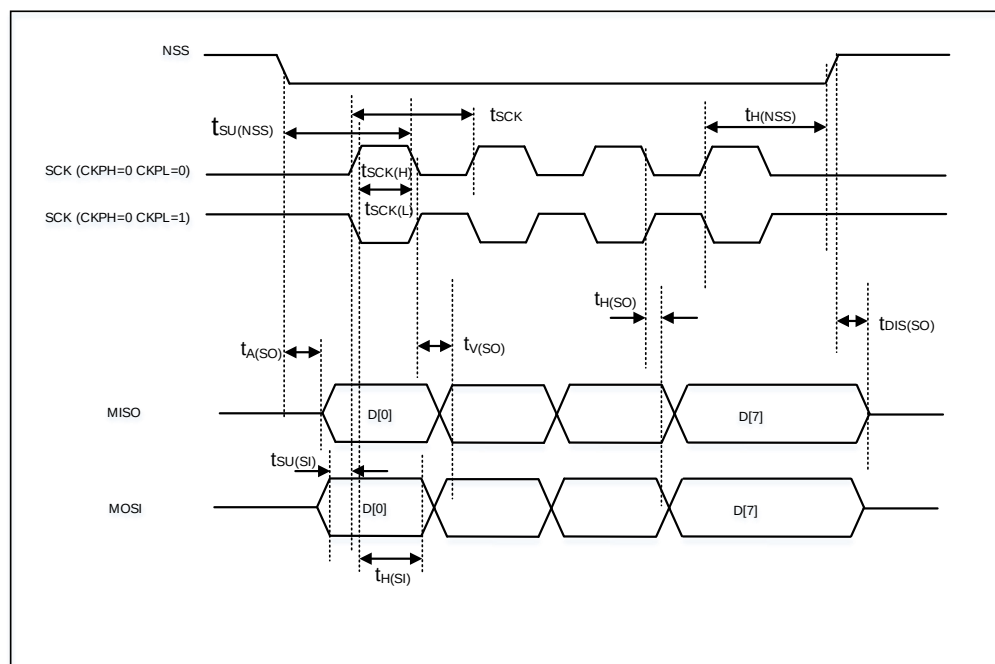


Figure 4-9. SPI timing diagram - slave mode



## 4.22 I2S characteristics

**Table 4-41. I2S characteristics<sup>(1)</sup>**

| Symbol           | Parameter                        | Conditions                                               | Min | Typ  | Max  | Unit |
|------------------|----------------------------------|----------------------------------------------------------|-----|------|------|------|
| $f_{CK}$         | Clock frequency                  | Master mode (data: 16 bits,<br>Audio frequency = 96 kHz) | —   | 6.25 | —    | MHz  |
|                  |                                  | Slave mode                                               | —   | —    | 12.5 |      |
| $t_H$            | Clock high time                  | —                                                        | —   | 80   | —    | ns   |
| $t_L$            | Clock low time                   |                                                          | —   | 80   | —    | ns   |
| $t_{V(WS)}$      | WS valid time                    | Master mode                                              | —   | 3    | —    | ns   |
| $t_{H(WS)}$      | WS hold time                     | Master mode                                              | —   | 3    | —    | ns   |
| $t_{SU(WS)}$     | WS setup time                    | Slave mode                                               | 0   | —    | —    | ns   |
| $t_{H(WS)}$      | WS hold time                     | Slave mode                                               | 3   | —    | —    | ns   |
| $Duty_{(SCK)}$   | I2S slave input clock duty cycle | Slave mode                                               | —   | 50   | —    | %    |
| $t_{SU(SD\_MR)}$ | Data input setup time            | Master mode                                              | 1   | —    | —    | ns   |
| $t_{SU(SD\_SR)}$ | Data input setup time            | Slave mode                                               | 0   | —    | —    | ns   |
| $t_{H(SD\_MR)}$  | Data input hold time             | Master receiver                                          | 0   | —    | —    | ns   |
| $t_{H(SD\_SR)}$  |                                  | Slave receiver                                           | 1   | —    | —    | ns   |
| $t_{V(SD\_ST)}$  | Data output valid time           | Slave transmitter<br>(after enable edge)                 | —   | —    | 10   | ns   |
| $t_{H(SD\_ST)}$  | Data output hold time            | Slave transmitter<br>(after enable edge)                 | 3   | —    | —    | ns   |
| $t_{V(SD\_MT)}$  | Data output valid time           | Master transmitter<br>(after enable edge)                | —   | —    | 10   | ns   |
| $t_{H(SD\_MT)}$  | Data output hold time            | Master transmitter<br>(after enable edge)                | 0   | —    | —    | ns   |

(1) Based on characterization, not tested in production.

Figure 4-10. I2S timing diagram - master mode

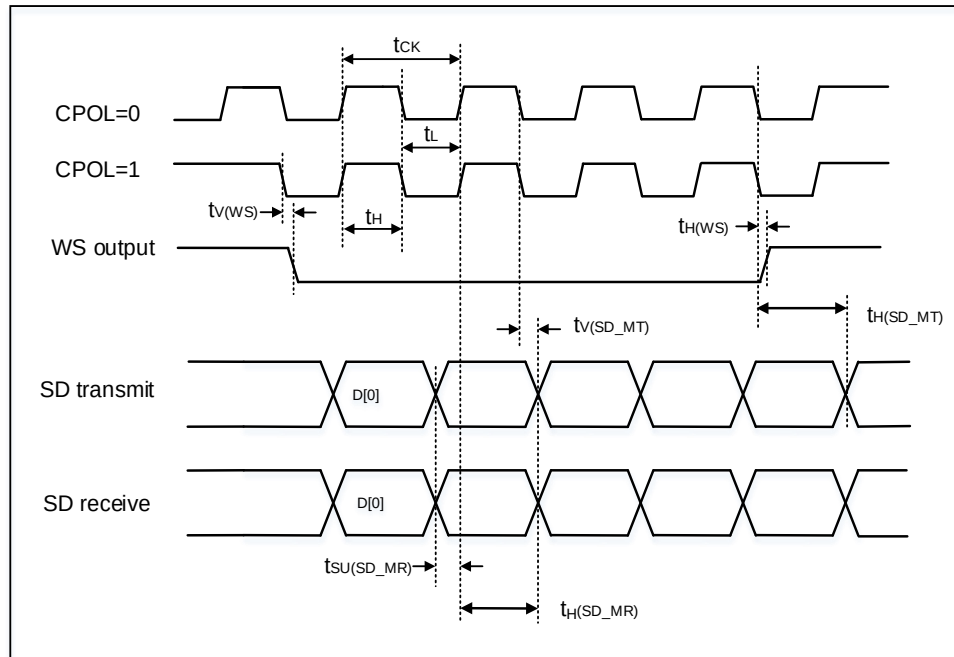
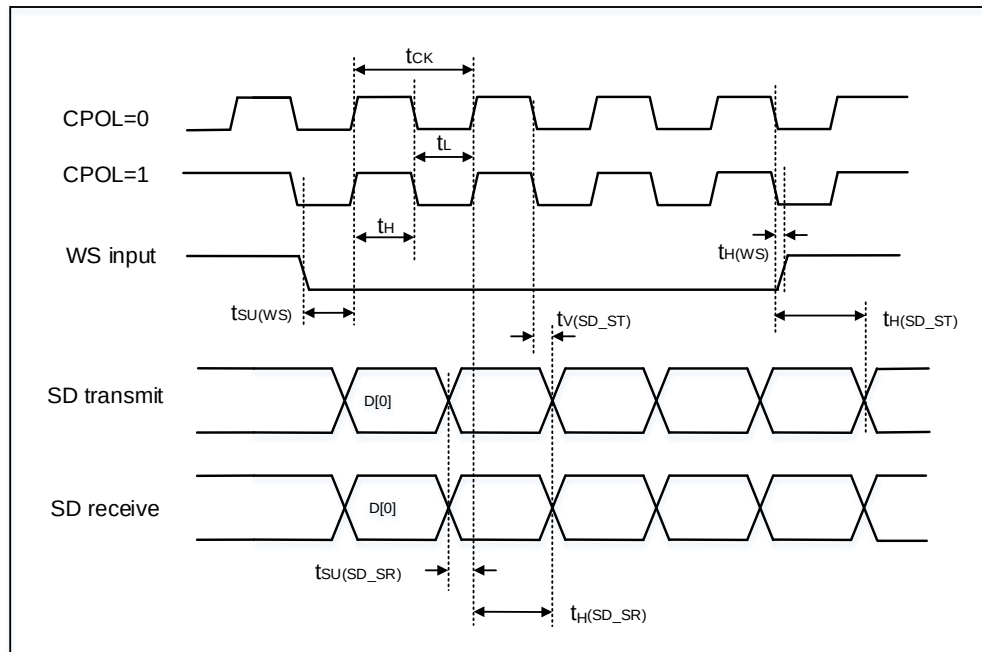


Figure 4-11. I2S timing diagram - slave mode



## 4.23 USART/LPUART characteristics

**Table 4-42. USART/LPUART characteristics<sup>(1)</sup>**

| Symbol              | Parameter           | Conditions                  | Min    | Typ | Max | Unit |
|---------------------|---------------------|-----------------------------|--------|-----|-----|------|
| f <sub>SCK</sub>    | SCK clock frequency | f <sub>PCLKx</sub> = 64 MHz | —      | —   | 32  | MHz  |
| t <sub>SCK(H)</sub> | SCK clock high time | f <sub>PCLKx</sub> = 64 MHz | 15.625 | —   | —   | ns   |
| t <sub>SCK(L)</sub> | SCK clock low time  | f <sub>PCLKx</sub> = 64 MHz | 15.625 | —   | —   | ns   |

(1) Guaranteed by design, not tested in production.

## 4.24 USB2 characteristics

**Table 4-43. USB2 startup time**

| Symbol                              | Parameter         | Max | Unit |
|-------------------------------------|-------------------|-----|------|
| t <sub>STARTUP</sub> <sup>(1)</sup> | USB2 startup time | 1   | μs   |

(1) Guaranteed by design, not tested in production.

**Table 4-44. USB2 DC electrical characteristics**

| Symbol                         | Parameter       | Conditions                        | Min                                        | Typ | Max | Unit |
|--------------------------------|-----------------|-----------------------------------|--------------------------------------------|-----|-----|------|
| Input levels <sup>(1)</sup>    | V <sub>DD</sub> | USB2 operating voltage            | —                                          | 3.0 | —   | 3.63 |
|                                | V <sub>DI</sub> | Differential input sensitivity    | —                                          | 0.2 | —   | —    |
|                                | V <sub>CM</sub> | Differential common mode range    | Includes V <sub>DI</sub> range             | 0.8 | —   | 2.5  |
|                                | V <sub>SE</sub> | Single ended receiver threshold   | —                                          | 0.8 | —   | 2.0  |
| Output levels <sup>(2)</sup>   | V <sub>OL</sub> | Static output level low           | R <sub>L</sub> of 1.0 kΩ to 3.63 V         | —   | —   | 0.3  |
|                                | V <sub>OH</sub> | Static output level high          | R <sub>L</sub> of 15 kΩ to V <sub>SS</sub> | 2.8 | 3.3 | 3.63 |
| R <sub>PU</sub> <sup>(2)</sup> | USB2P           | V <sub>IN</sub> = V <sub>SS</sub> | 1.2                                        | 1.5 | 1.8 | KΩ   |

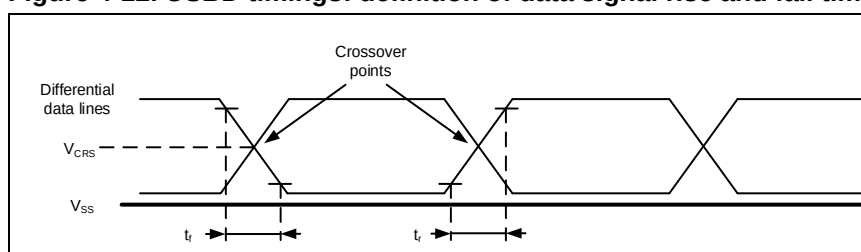
(1) Guaranteed by design, not tested in production.

(2) Based on characterization, not tested in production.

**Table 4-45. USB2 full speed-electrical characteristics<sup>(1)</sup>**

| Symbol           | Parameter                       | Conditions                      | Min  | Typ | Max | Unit |
|------------------|---------------------------------|---------------------------------|------|-----|-----|------|
| t <sub>R</sub>   | Rise time                       | C <sub>L</sub> = 50 pF          | 4    | 5   | 20  | ns   |
| t <sub>F</sub>   | Fall time                       | C <sub>L</sub> = 50 pF          | 4    | 5   | 20  | ns   |
| t <sub>RFM</sub> | Rise/ fall time matching        | t <sub>R</sub> / t <sub>F</sub> | 90   | —   | 111 | %    |
| V <sub>CRS</sub> | Output signal crossover voltage | —                               | 1.09 | —   | 2.0 | V    |

(1) Guaranteed by design, not tested in production.

**Figure 4-12. USB2 timings: definition of data signal rise and fall time**


## 4.25 WDGT characteristics

**Table 4-46. FWDGT min/max timeout period at 32 kHz (IRC32K)<sup>(1)</sup>**

| Prescaler divider | PSC[2:0] bits | Min timeout RLD[11:0]=<br>0x000 | Max timeout RLD[11:0]=<br>0xFFFF | Unit |
|-------------------|---------------|---------------------------------|----------------------------------|------|
| 1/4               | 000           | 0.03125                         | 511.90625                        | ms   |
| 1/8               | 001           | 0.03125                         | 1023.78125                       |      |
| 1/16              | 010           | 0.03125                         | 2047.53125                       |      |
| 1/32              | 011           | 0.03125                         | 4095.03125                       |      |
| 1/64              | 100           | 0.03125                         | 8190.03125                       |      |
| 1/128             | 101           | 0.03125                         | 16380.03125                      |      |
| 1/256             | 110 or 111    | 0.03125                         | 32760.03125                      |      |

(1) Guaranteed by design, not tested in production.

**Table 4-47. WWDGT min-max timeout value at 32 MHz ( $f_{PCLK1}$ )<sup>(1)</sup>**

| Prescaler divider | PSC[3:0] | Min timeout value<br>CNT[6:0] = 0x40 | Unit    | Max timeout value<br>CNT[6:0] = 0x7F | Unit |
|-------------------|----------|--------------------------------------|---------|--------------------------------------|------|
| 1/1               | 0000     | 128                                  | $\mu$ s | 8.192                                | ms   |
| 1/2               | 0001     | 256                                  |         | 16.384                               |      |
| 1/4               | 0010     | 512                                  |         | 32.768                               |      |
| 1/8               | 0011     | 1.024                                | ms      | 65.536                               |      |
| 1/16              | 0100     | 2.048                                |         | 131.072                              |      |
| 1/32              | 0101     | 4.096                                |         | 262.144                              |      |
| 1/64              | 0110     | 8.192                                |         | 524.288                              |      |
| 1/128             | 0111     | 16.384                               |         | 1048.576                             |      |
| 1/256             | 1000     | 32.768                               |         | 2097.152                             |      |
| 1/512             | 1001     | 65.536                               |         | 4194.304                             |      |
| 1/1024            | 1010     | 131.072                              |         | 8388.608                             |      |
| 1/2048            | 1011     | 262.144                              |         | 16777.216                            |      |
| 1/4096            | 1100     | 524.288                              |         | 33554.432                            |      |
| 1/8192            | 1101     | 1048.576                             |         | 67108.864                            |      |
| 1/1               | 1110     | 128                                  | $\mu$ s | 4.096                                |      |
| 1/1               | 1111     | 128                                  |         | 4.096                                |      |

(1) Guaranteed by design, not tested in production.

## 4.26 Parameter conditions

Unless otherwise specified, all values given for  $V_{DD} = V_{DDA} = 3.3$  V,  $T_A = 25$  °C.

## 5 Package information

### 5.1 LQFP64 package outline dimensions

Figure 5-1. LQFP64 package outline

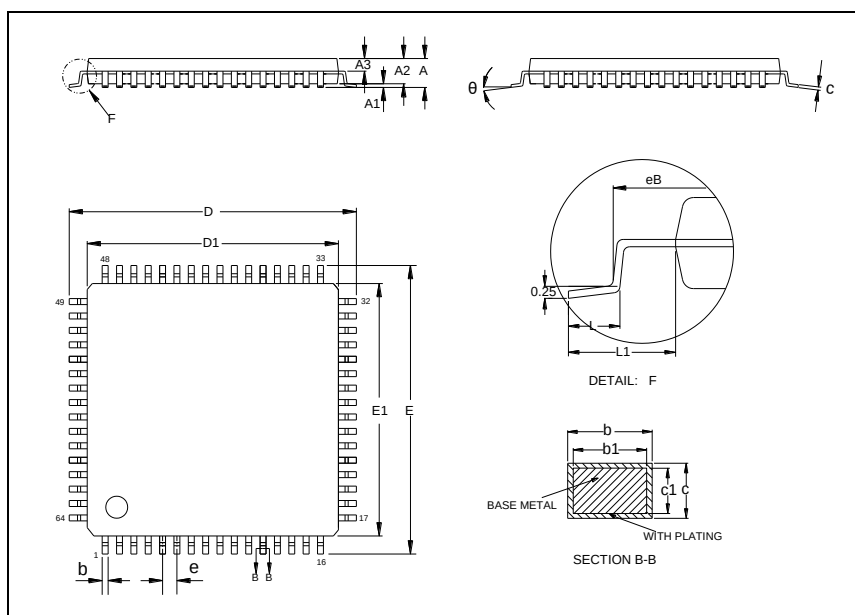
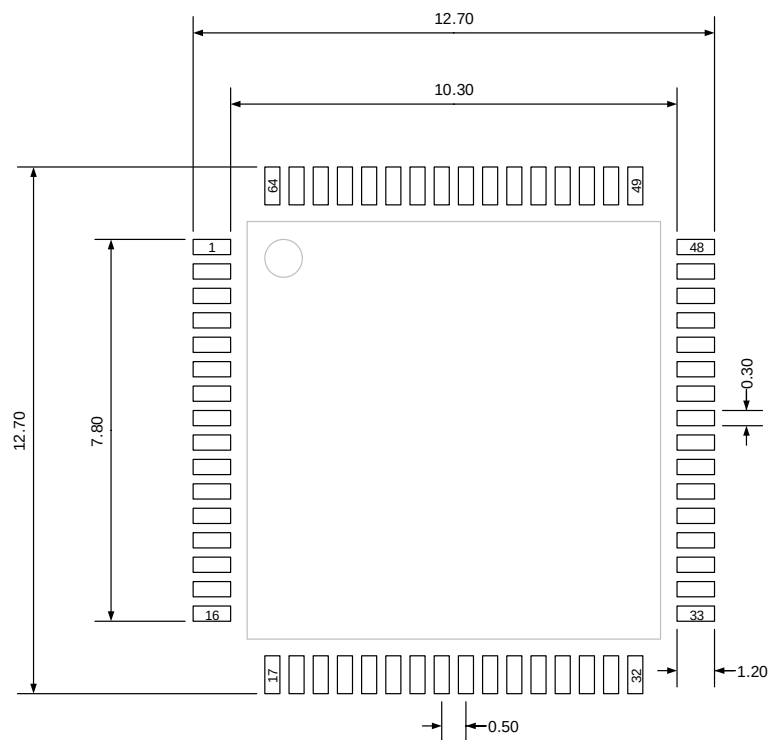


Table 5-1. LQFP64 package dimensions

| Symbol   | Min   | Typ   | Max   |
|----------|-------|-------|-------|
| A        | —     | —     | 1.60  |
| A1       | 0.05  | —     | 0.15  |
| A2       | 1.35  | 1.40  | 1.45  |
| A3       | 0.59  | 0.64  | 0.69  |
| b        | 0.18  | —     | 0.26  |
| b1       | 0.17  | 0.20  | 0.23  |
| c        | 0.13  | —     | 0.17  |
| c1       | 0.12  | 0.13  | 0.14  |
| D        | 11.80 | 12.00 | 12.20 |
| D1       | 9.90  | 10.00 | 10.10 |
| E        | 11.80 | 12.00 | 12.20 |
| E1       | 9.90  | 10.00 | 10.10 |
| e        | —     | 0.50  | —     |
| eB       | 11.25 | —     | 11.45 |
| L        | 0.45  | —     | 0.75  |
| L1       | —     | 1.00  | —     |
| $\theta$ | 0°    | —     | 7°    |

(Original dimensions are in millimeters)

Figure 5-2. LQFP64 recommended footprint



(Original dimensions are in millimeters)

## 5.2 LQFP48 package outline dimensions

Figure 5-3. LQFP48 package outline

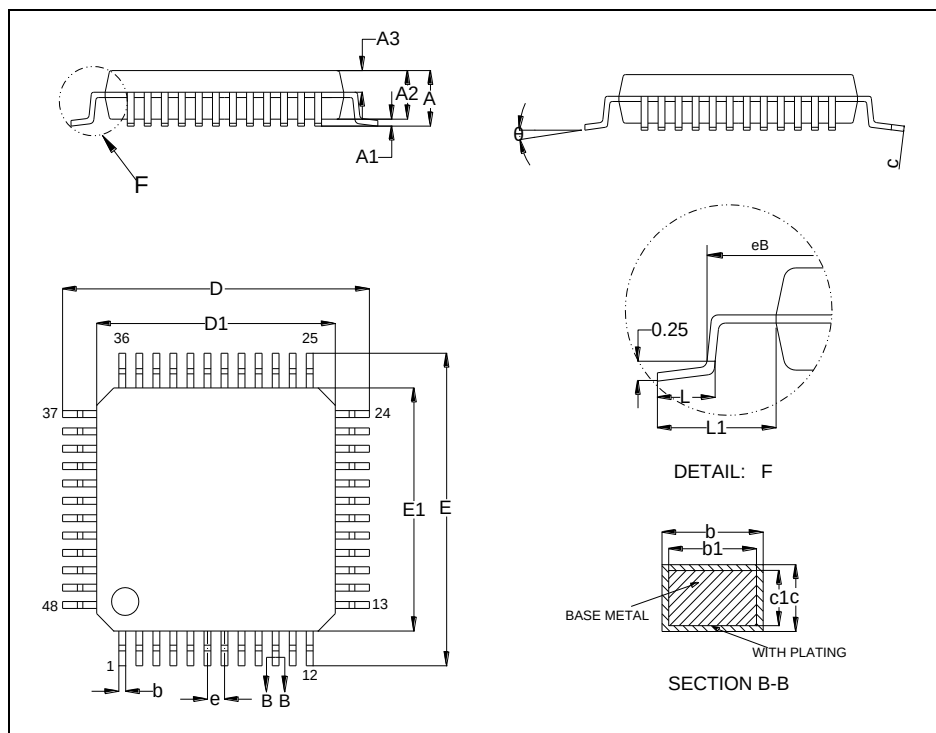
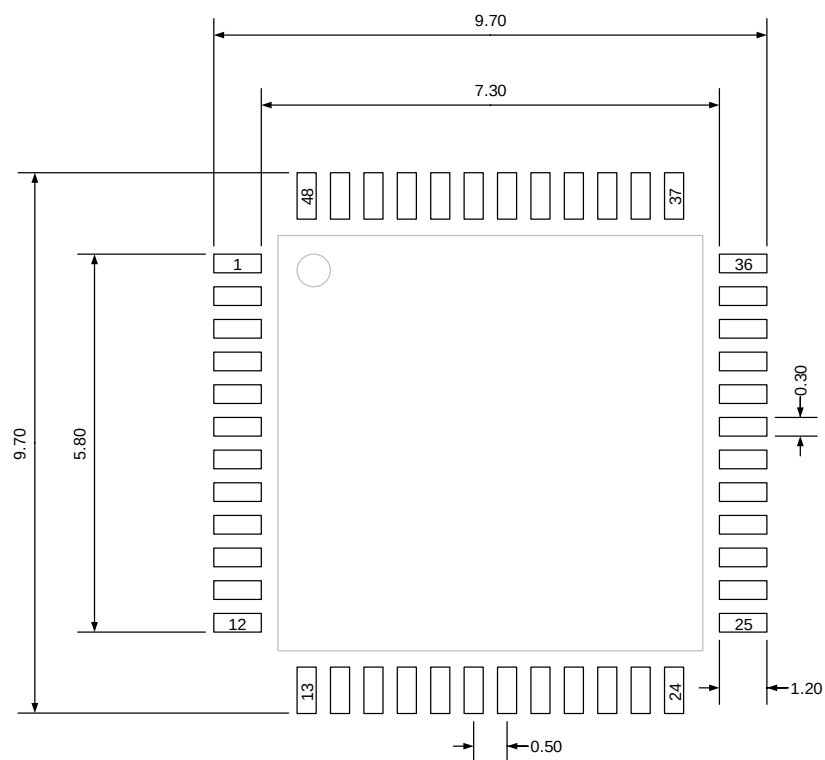


Table 5-2. LQFP48 package dimensions

| Symbol | Min  | Typ  | Max  |
|--------|------|------|------|
| A      | —    | —    | 1.60 |
| A1     | 0.05 | —    | 0.15 |
| A2     | 1.35 | 1.40 | 1.45 |
| A3     | 0.59 | 0.64 | 0.69 |
| b      | 0.18 | —    | 0.26 |
| b1     | 0.17 | 0.20 | 0.23 |
| c      | 0.13 | —    | 0.17 |
| c1     | 0.12 | 0.13 | 0.14 |
| D      | 8.80 | 9.00 | 9.20 |
| D1     | 6.90 | 7.00 | 7.10 |
| E      | 8.80 | 9.00 | 9.20 |
| E1     | 6.90 | 7.00 | 7.10 |
| e      | —    | 0.50 | —    |
| eB     | 8.10 | —    | 8.25 |
| L      | 0.45 | —    | 0.75 |
| L1     | —    | 1.00 | —    |
| θ      | 0°   | —    | 7°   |

(Original dimensions are in millimeters)

Figure 5-4. LQFP48 recommended footprint



(Original dimensions are in millimeters)

## 5.3 LQFP32 package outline dimensions

Figure 5-5. LQFP32 package outline

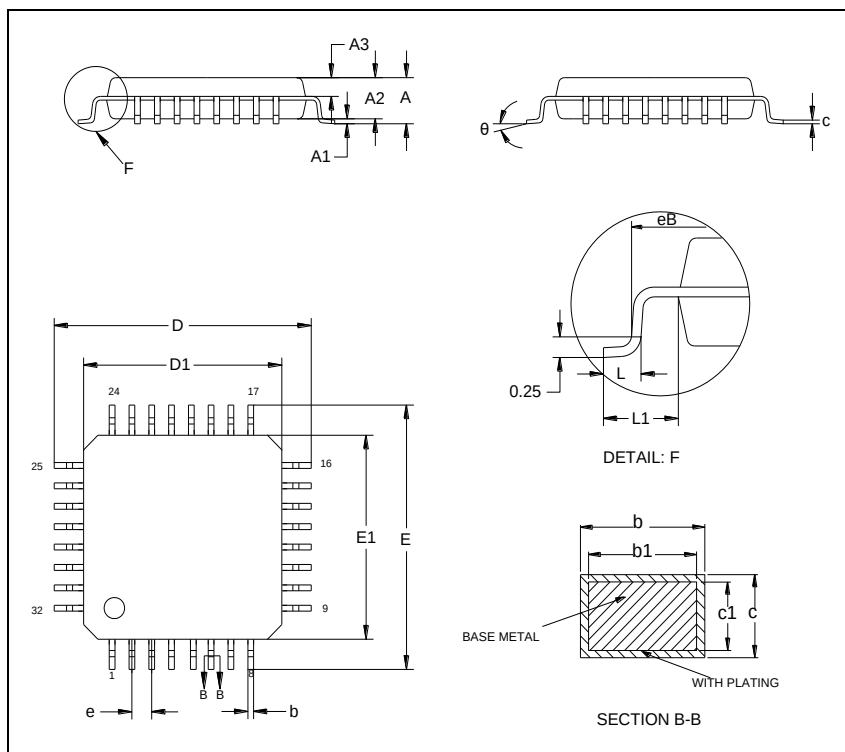
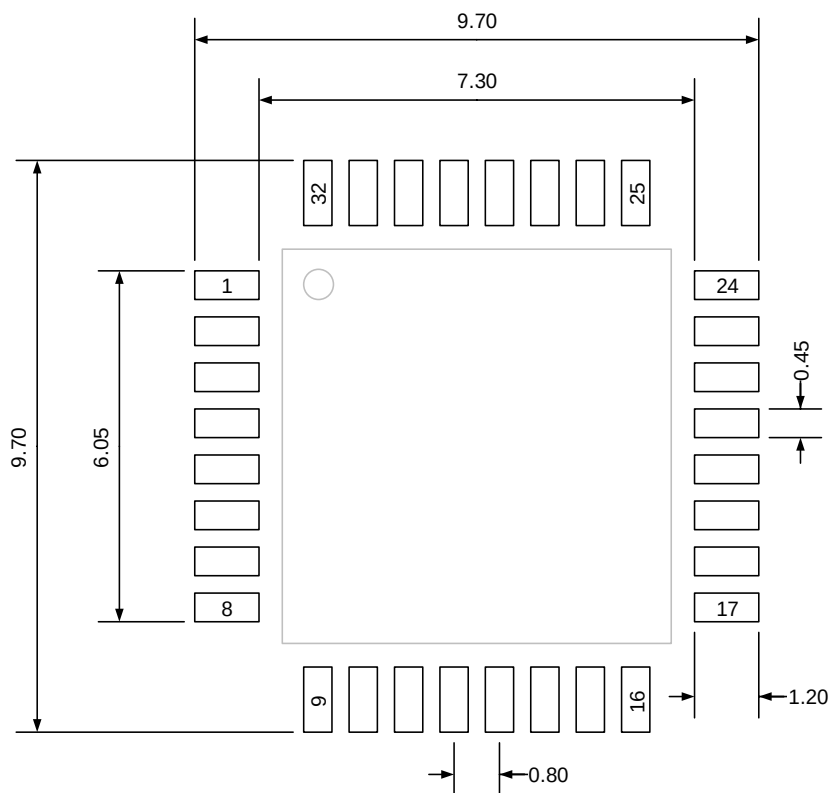


Table 5-3. LQFP32 package dimensions

| Symbol   | Min  | Typ  | Max  |
|----------|------|------|------|
| A        | —    | —    | 1.60 |
| A1       | 0.05 | —    | 0.15 |
| A2       | 1.35 | 1.40 | 1.45 |
| A3       | 0.59 | 0.64 | 0.69 |
| b        | 0.33 | —    | 0.41 |
| b1       | 0.32 | 0.35 | 0.38 |
| c        | 0.13 | —    | 0.17 |
| c1       | 0.12 | 0.13 | 0.14 |
| D        | 8.80 | 9.00 | 9.20 |
| D1       | 6.90 | 7.00 | 7.10 |
| E        | 8.80 | 9.00 | 9.20 |
| E1       | 6.90 | 7.00 | 7.10 |
| e        | —    | 0.80 | —    |
| eB       | 8.10 | —    | 8.25 |
| L        | 0.45 | —    | 0.75 |
| L1       | —    | 1.00 | —    |
| $\theta$ | 0°   | —    | 7°   |

(Original dimensions are in millimeters)

**Figure 5-6. LQFP32 recommended footprint**



(Original dimensions are in millimeters)

## 5.4 QFN32 package outline dimensions

Figure 5-7. QFN32 package outline

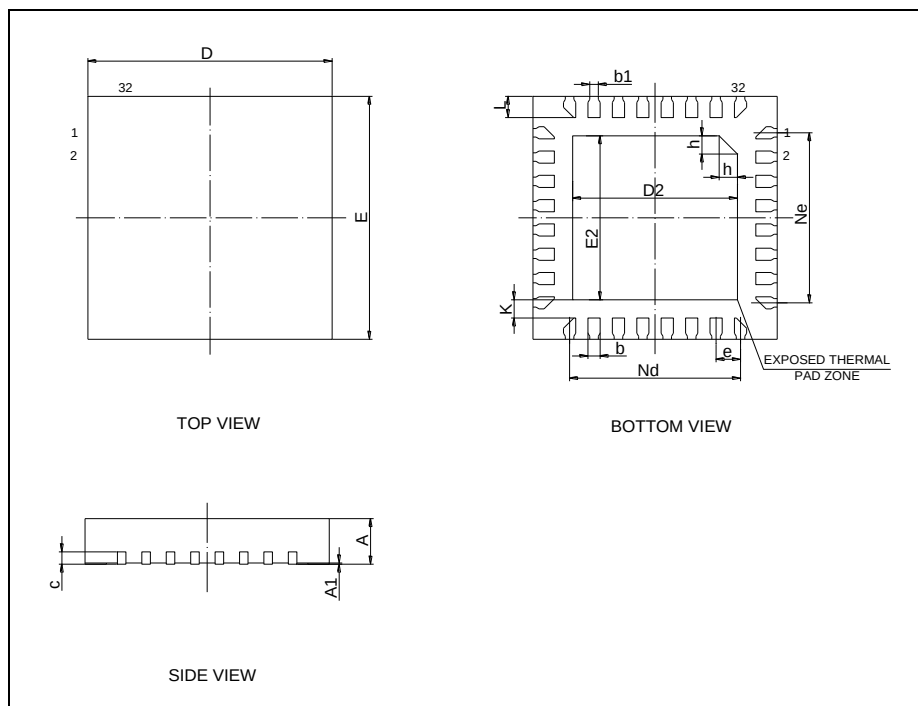
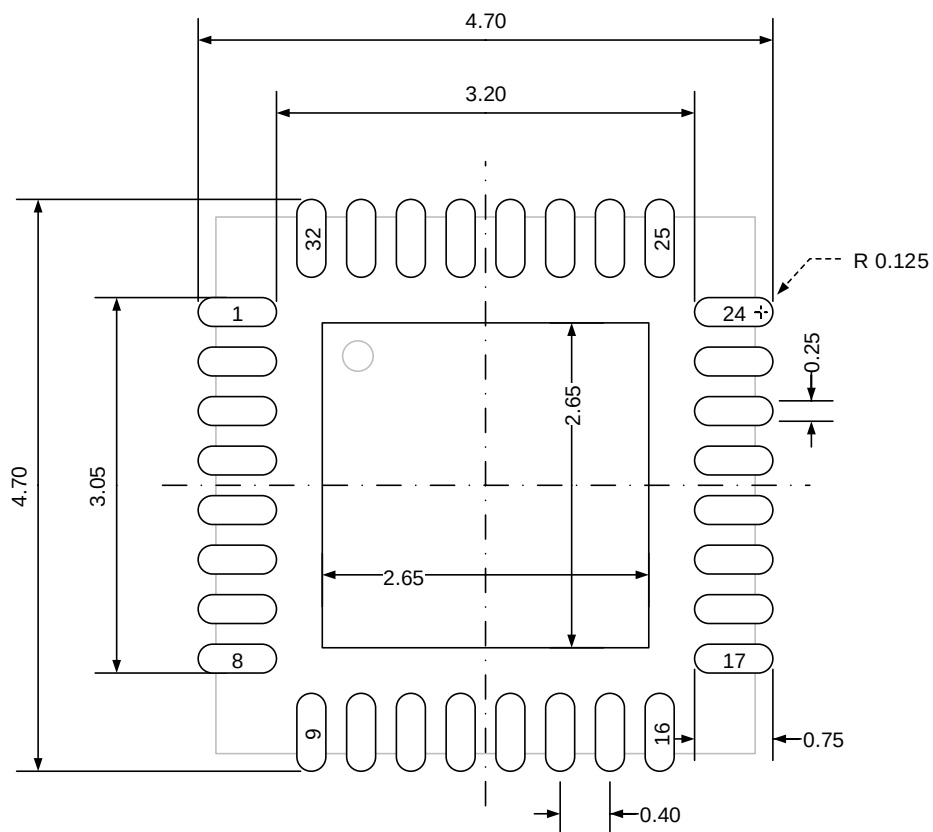


Table 5-4. QFN32 package dimensions

| Symbol | Min  | Typ  | Max  |
|--------|------|------|------|
| A      | 0.70 | 0.75 | 0.80 |
| A1     | 0    | 0.02 | 0.05 |
| b      | 0.15 | 0.20 | 0.25 |
| b1     | —    | 0.14 | —    |
| c      | —    | 0.20 | —    |
| D      | 3.90 | 4.00 | 4.10 |
| D2     | 2.60 | 2.70 | 2.80 |
| E      | 3.90 | 4.00 | 4.10 |
| E2     | 2.60 | 2.70 | 2.80 |
| e      | —    | 0.40 | —    |
| h      | 0.25 | 0.30 | 0.35 |
| K      | —    | 0.30 | —    |
| L      | 0.30 | 0.35 | 0.40 |
| Nd     | —    | 2.80 | —    |
| Ne     | —    | 2.80 | —    |

(Original dimensions are in millimeters)

Figure 5-8. QFN32 recommended footprint



(Original dimensions are in millimeters)

## 5.5 Thermal characteristics

Thermal resistance is used to characterize the thermal performance of the package device, which is represented by the Greek letter “ $\theta$ ”. For semiconductor devices, thermal resistance represents the steady-state temperature rise of the chip junction due to the heat dissipated on the chip surface.

$\theta_{JA}$ : Thermal resistance, junction-to-ambient.

$\theta_{JB}$ : Thermal resistance, junction-to-board.

$\theta_{JC}$ : Thermal resistance, junction-to-case.

$\Psi_{JB}$ : Thermal characterization parameter, junction-to-board.

$\Psi_{JT}$ : Thermal characterization parameter, junction-to-top center.

$$\theta_{JA} = (T_J - T_A) / P_D \quad (5-1)$$

$$\theta_{JB} = (T_J - T_B) / P_D \quad (5-2)$$

$$\theta_{JC} = (T_J - T_C) / P_D \quad (5-3)$$

Where,  $T_J$  = Junction temperature.

$T_A$  = Ambient temperature

$T_B$  = Board temperature

$T_C$  = Case temperature which is monitoring on package surface

$P_D$  = Total power dissipation

$\theta_{JA}$  represents the resistance of the heat flows from the heating junction to ambient air. It is an indicator of package heat dissipation capability. Lower  $\theta_{JA}$  can be considerate as better overall thermal performance.  $\theta_{JA}$  is generally used to estimate junction temperature.

$\theta_{JB}$  is used to measure the heat flow resistance between the chip surface and the PCB board.

$\theta_{JC}$  represents the thermal resistance between the chip surface and the package top case.  $\theta_{JC}$  is mainly used to estimate the heat dissipation of the system (using heat sink or other heat dissipation methods outside the device package).

**Table 5-5. Package thermal characteristics<sup>(1)</sup>**

| Symbol        | Condition                    | Package | Value | Unit |
|---------------|------------------------------|---------|-------|------|
| $\theta_{JA}$ | Natural convection, 2S2P PCB | LQFP64  | 54.57 | °C/W |
|               |                              | LQFP48  | 69.64 |      |
|               |                              | LQFP32  | 55.26 |      |
|               |                              | QFN32   | 42.57 |      |
| $\theta_{JB}$ | Cold plate, 2S2P PCB         | LQFP64  | 35.08 | °C/W |
|               |                              | LQFP48  | 43.16 |      |

| Symbol        | Condition                    | Package | Value | Unit                 |
|---------------|------------------------------|---------|-------|----------------------|
|               |                              | LQFP32  | 26.24 |                      |
|               |                              | QFN32   | 19.21 |                      |
| $\theta_{JC}$ | Cold plate, 2S2P PCB         | LQFP64  | 18.11 | $^{\circ}\text{C/W}$ |
|               |                              | LQFP48  | 25.36 |                      |
|               |                              | LQFP32  | 25.23 |                      |
|               |                              | QFN32   | 19.10 |                      |
| $\Psi_{JB}$   | Natural convection, 2S2P PCB | LQFP64  | 35.41 | $^{\circ}\text{C/W}$ |
|               |                              | LQFP48  | 47.75 |                      |
|               |                              | LQFP32  | 32.03 |                      |
|               |                              | QFN32   | 19.18 |                      |
| $\Psi_{JT}$   | Natural convection, 2S2P PCB | LQFP64  | 1.10  | $^{\circ}\text{C/W}$ |
|               |                              | LQFP48  | 2.45  |                      |
|               |                              | LQFP32  | 2.06  |                      |
|               |                              | QFN32   | 0.62  |                      |

(1) Thermal characteristics are based on simulation, and meet JEDEC specification.

## 6 Ordering information

Table 6-1. Part ordering code for GD32L233xx devices

| Ordering code | Flash (KB) | Package | Package type | Temperature operating range   |
|---------------|------------|---------|--------------|-------------------------------|
| GD32L233RCT6  | 256        | LQFP64  | Green        | Industrial<br>-40°C to +85°C  |
| GD32L233RCT7  | 256        | LQFP64  | Green        | Industrial<br>-40°C to +105°C |
| GD32L233RBT6  | 128        | LQFP64  | Green        | Industrial<br>-40°C to +85°C  |
| GD32L233R8T6  | 64         | LQFP64  | Green        | Industrial<br>-40°C to +85°C  |
| GD32L233CCT6  | 256        | LQFP48  | Green        | Industrial<br>-40°C to +85°C  |
| GD32L233CBT6  | 128        | LQFP48  | Green        | Industrial<br>-40°C to +85°C  |
| GD32L233C8T6  | 64         | LQFP48  | Green        | Industrial<br>-40°C to +85°C  |
| GD32L233KBT6  | 128        | LQFP32  | Green        | Industrial<br>-40°C to +85°C  |
| GD32L233K8T6  | 64         | LQFP32  | Green        | Industrial<br>-40°C to +85°C  |
| GD32L233KBQ6  | 128        | QFN32   | Green        | Industrial<br>-40°C to +85°C  |
| GD32L233K8Q6  | 64         | QFN32   | Green        | Industrial<br>-40°C to +85°C  |

## 7 Revision history

Table 7-1. Revision history

| Revision No. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Date          |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1.0          | Initial Release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Oct.19, 2021  |
| 1.1          | <ol style="list-style-type: none"> <li>1. Modify description in <u><b>Debug mode.</b></u></li> <li>2. Modify description in <u><b>Embedded memory.</b></u></li> <li>3. Add EMI parameter in <u><b>Table 4-9. EMI characteristics(1).</b></u></li> <li>4. Modify <u><b>table 4-17</b></u> and <u><b>table 4-19.</b></u></li> <li>5. Modify power consumption value in <u><b>Table 4-7. Power consumption characteristics(2)(3)</b></u></li> <li>6. Update QFN32 recommended footprint figure.</li> </ol>             | Apr.13, 2022  |
| 1.2          | <ol style="list-style-type: none"> <li>1. Modify USART pin function description from USARTx_RTS to USARTx_RTS/USARTx_DE in <u><b>Pin definitions.</b></u></li> <li>2. Fixed the description of Flash memory and SRAM waiting state in <u><b>Embedded memory.</b></u></li> <li>3. Add note of <u><b>Figure 4-4. Recommended external NRST pin circuit.</b></u></li> <li>4. Update <u><b>4.12 VREF buffer characteristics.</b></u></li> <li>5. Update <u><b>5.1 LQFP64 package outline dimensions.</b></u></li> </ol> | Jul. 27, 2022 |
| 1.3          | <ol style="list-style-type: none"> <li>1. Unify the pin names of the full text.</li> <li>2. Update <u><b>Figure 4-7. I2C bus timing diagram.</b></u></li> </ol>                                                                                                                                                                                                                                                                                                                                                     | Dec.20, 2022  |
| 1.4          | <ol style="list-style-type: none"> <li>1. Add information about GD32L233RCT7.</li> <li>2. Update <u><b>Table 4-28. SLCD controller characteristics.</b></u></li> <li>3. Update comment of <u><b>Table 4-7. Power consumption characteristics.</b></u></li> </ol>                                                                                                                                                                                                                                                    | Jun.5, 2023   |
| 1.5          | <ol style="list-style-type: none"> <li>1. Modify the pin definitions of PB1//PB2 in <u><b>Pin definitions.</b></u></li> <li>2. Update the pin definitions table header, add chip type and package in <u><b>Pin definitions.</b></u></li> <li>3. Modify the description of CMP in <u><b>3.23. Comparators (CMP).</b></u></li> </ol>                                                                                                                                                                                  | Jul.7, 2023   |

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